Firm Pay, Amenities, and Inequality*

Sydnee Caldwell — UC Berkeley & NBER

Ingrid Haegele — LMU & IAB

Jörg Heining — IAB

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Abstract

We estimate firm-specific amenity valuations using discrete choice experiments embedded in a large-scale survey of German workers linked to administrative records. Workers rank hypothetical offers from real firms they would consider joining, with randomized wages identifying money-metric valuations. Valuations vary substantially across firms and demographic groups, yet a single index performs surprisingly well. Valuations are approximately orthogonal to firm wage premia; they therefore do not offset between-firm wage inequality. However, male-female differences in amenity valuations explain part of the gender wage gap.

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1 Introduction

Firms vary in both pay and non-wage dimensions (Mas, 2025). A large literature has examined firm-level wage heterogeneity and its implications for inequality. Far less work has asked how non-wage job characteristics vary across firms and how workers value them (Card, 2022).

Identifying the value of firm-provided amenities is empirically challenging. Regressing wages on non-wage characteristics does not identify the value of those characteristics, both because workers sort across firms and because such regressions conflate preferences with sorting and with unobserved firm characteristics. A long literature on compensating differentials has emphasized that these approaches often yield "wrong-signed" estimates, and that richer data on job attributes alone cannot resolve this problem (Lucas, 1977; Brown, 1980; Lavetti, 2023). Workers' transitions between firms can yield estimates of the overall value provided by a firm—given assumptions on offer arrival rates and wage dynamics—but these methods recover only the component of amenities orthogonal to wages (Sorkin, 2018). More recent work has turned to stated preference methods to estimate willingness to pay for specific job characteristics (see, e.g., Mas and Pallais, 2017; Wiswall and Zafar, 2018; Folke and Rickne, 2022; Maestas et al., 2023), but these studies typically examine specific job bundles and do not yield firm-level estimates of amenity value.

This paper uses discrete choice experiments we embedded in a large-scale survey of German workers, fielded by a group within the German Employment Agency, to estimate firm-specific amenity valuations and examine their role in job choice, sorting, and inequality. We asked workers to rank hypothetical job offers from real firms they themselves identified as likely application targets. We randomize raises across offers, allowing us to recover money-metric valuations of firm amenities, which are not confounded by equilibrium wage differences and which do not rely on assumptions about offer arrival rates. We link the survey to administrative Social Security records and to firm-level data from Kununu, a major employer review platform. These data allow us to study how amenity valuations relate to worker mobility, firm characteristics, and between-firm inequality.

Preferences over non-wage attributes shape job choice. Workers often select lower-wage offers in the discrete choice experiment, implying substantial trade-offs between pay and non-wage attributes even among the firms they are marginal to. Valuations of these non-wage attributes ("amenity valuations") exhibit substantial heterogeneity across workers, even when those workers are marginal to the same firms. For example, the Pearson correlation in amenity valuations between men and women is 0.239, and the correlation between college-educated and non-college workers is 0.336. Formal tests confirm that male and female valuations differ statistically.

Most models of the labor market assume a uniform ranking of firms: a single-index model (Card, Cardoso and Kline, 2016; Berger, Herkenhoff and Mongey, 2022; Mas, 2025). The between-group disagreement in valuations would seem to challenge this assumption. Yet a common valuation index has substantial predictive power for aggregate patterns. Conditional on their wage, firms with higher amenity values (measured using a single index) are larger and more favorably reviewed by employees—consistent with structural models that infer firm desirability from firm size or retention (see, e.g., Volpe, 2024). Valuations of non-wage attributes also correlate positively with firm rankings from external datasets—a pattern inconsistent with pure idiosyncratic match effects and consistent with firms differing in amenity quality along dimensions most workers agree on.

While amenity values vary widely across firms, they are approximately orthogonal to employment-weighted firm-specific wage premia. Amenities therefore do not offset wage inequality through compensating differentials: a typical worker at a high-wage firm does not enjoy systematically better or worse working conditions than a typical worker at a low-wage firm. Converting to money-metric units using the structural wage coefficient, the variance of amenity values is about one-third that of wage premia.

Gender differences in amenity valuations explain part of why women work at lower-wage firms. Among full-time workers, women work at firms that offer lower pay premia. Gender differences in firm-specific valuations, however, mean that women do not work at firms that offer them lower overall value. Consistent with prior work, women in our survey are significantly more likely than men to say that specific non-wage amenities (such as reduced hours or the availability of remote work) are important in their job choice.

This paper contributes to two main strands of literature. First, our work builds on the large literature on amenities and compensating differentials (Rosen, 1986; Mas and Pallais, 2017; Sorkin, 2018; Volpe, 2024), which examines how workers trade off wages for non-wage job attributes. Much of this literature infers the value of amenities from observational wage and mobility patterns, though important recent work has introduced experimental methods—notably Mas and Pallais (2017), who use a field experiment to value specific job attributes. Our approach extends this experimental tradition to estimate workers' valuations of firm-specific amenities. By randomizing wages across hypothetical offers in discrete choice experiments, we identify the money-metric value of firm amenities, avoiding both the endogeneity of hedonic wage regressions and the structural assumptions required by mobility-based approaches. Our results show that the bundle of non-wage characteristics associated with a firm is not, taken as a whole, a compensating differential: high-wage firms are not systematically worse on non-wage dimensions.

Our results are complementary to those in Maestas et al. (2023), who use a similar stated preference approach to estimate willingness to pay for a broad set of specific job amenities in a representative US sample. Their design prices individual amenities but does not yield firm-level estimates, and their data are not linked to administrative employment or mobility records. We do not estimate valuations of specific job attributes. Instead, we recover the total non-wage value of specific firms and use the linkage with administrative records to examine whether firms that pay more also provide better amenities, and whether the gender gap in firm-specific pay premia partly reflects a gender gap in firm-specific non-wage values.

Second, our paper contributes to research on firm wage premia and worker sorting (see, e.g., Card et al., 2018; Card, Heining and Kline, 2013; Song et al., 2019; Lamadon, Mogstad and Setzler, 2022). Prior work has demonstrated that firms differ systematically in the wages they pay, even after controlling for worker characteristics, and that these wage premia play a central role in shaping labor market inequality, including the gender wage gap (Card, Cardoso and Kline, 2016; Sorkin, 2017; Palladino et al., 2025). However, much less is known about the role of firm-specific amenities in driving worker mobility and sorting patterns. Our findings show that firm amenities are important in job choice, and can explain some of the sorting of women into lower-wage firms.[1] However, because they are approximately orthogonal to wage premia, prior estimates of between-firm wage inequality are not biased by the omission of amenities.

The paper proceeds as follows: Section 2 describes the data. Section 3 presents the conceptual framework and estimation strategy. Section 4 presents our main estimates. Section 5 examines implications for mobility and inequality. Section 6 concludes.

2 Empirical Setting and Data

Our analysis combines discrete choice experiments embedded in a large-scale survey of workers, linked to administrative Social Security records and to information on specific amenities provided by firms.

2.1 Data

We obtain data from two primary sources.

CHH Worker Survey. First, we use data from a large-scale survey of German workers to identify workers' valuations of firm-provided amenities. We fielded this survey to full-time German workers between the ages of 25 and 50 (Caldwell, Haegele and Heining, 2025*a*). We fielded the main wave of the survey in fall 2022 with a follow-up in spring 2024; we pool data from both waves in most of our analysis.[2] In previous papers we analyzed modules focused on wage bargaining and on workers' beliefs about pay (Caldwell, Haegele and Heining, 2025*a*,*b*). An online survey appendix provides details on implementation, as well as the full survey instruments; we

[1]Lavetti and Schmutte (2023) also explore whether gender differences in the valuation of amenities can explain the sorting of women into low-wage firms. They find that women sort on physical but not financial risk. Our findings are complementary and focus on the overall non-wage value of a firm, rather than the value of specific job attributes.

[2]Both waves were fielded well after the acute phase of the COVID-19 pandemic. To the extent that the pandemic permanently changed firms' amenity provision—for example, by expanding remote work options—this would drive a wedge between our survey-based amenity valuations and pre-pandemic measures used in some of our external validations.

include some of this information in Appendix E.[3]

To elicit firm-specific valuations, we designed a survey module in which workers reported where they would apply if they decided to switch firms. This approach allows us to focus on firms that workers actively consider as potential employers. After identifying relevant firms, we asked workers to rank hypothetical job offers from these firms under randomly assigned wage changes. We standardized and cleaned the free-text firm names provided by respondents and matched them to establishment identifiers in the German Social Security records housed at the IAB using establishment addresses.

In some robustness exercises we also use data from a survey module in which we provided workers with the names of specific firms. In this module we asked workers whether they would consider applying to each of seven firms; we then asked them to rank hypothetical job offers from three of these firms (chosen at random) under randomly assigned wage changes.

A natural concern with survey elicitations is that workers may respond inattentively or strategically. In a previous paper that used the bargaining modules embedded in this survey, we showed that workers' responses to questions on their bargaining behavior correlate with employer-side responses from a separate survey, fielded by a different organization one year prior (Caldwell, Haegele and Heining, 2025*a*). These and other validations reported in the appendices of that paper and in Caldwell, Haegele and Heining (2025*b*) suggest that workers generally provided truthful, accurate responses in the survey. We validate the specific modules and questions designed for this paper in Section 3.4 and Appendix D.5.

IEB Administrative Records. We use the administrative Social Security records compiled by the IAB to compare amenity valuations to observed pay policies and to link valuations to job-to-job mobility and inequality. These data provide comprehensive employment histories, including job start and end dates, wages, and establishment identifiers. They also include information on demographics and education.[4] We match the CHH worker survey to these records in two ways. First,

[3] See here for more information: https://sydneec.github.io/Website/CHH_Survey_Appendix.pdf.

[4] Throughout the text we say a worker is college educated if they have a university degree (BA or more).

we match each respondent to their own Social Security records, giving us their complete employment history. Second, we use the crosswalk we generated between firm names and establishment identifiers to match each respondent to data on the firms they named. Appendix B.2 provides more information on how we cleaned the provided strings and generated the crosswalk.

Beyond the data on surveyed workers, we use two extracts from these records. First, we use a 2% random sample of workers to describe the broader labor force and to examine the relationship between firm valuations and the gender gap; these data cover 2010 to 2019. Second, we use data on all workers who switched firms between 2010 and 2019 to study mobility. In all samples, we focus on full-time workers between the ages of 25 and 50, for consistency with our survey sample. Appendix Section B.1 provides more information on these extracts.

We obtain additional firm-level data from the IAB, including estimates of firm wage premia and workforce composition. The firm wage premia capture systematic differences in pay across firms, controlling for worker characteristics, and allow us to examine how firm-provided amenities relate to wage-setting (Bellmann et al., 2020). While we prefer to use these estimates for comparability with the literature, in some robustness checks we use our own estimates of the wage premia, calculated from the mobility sample. We describe the estimation procedure, as well as the comparison with estimates in Bellmann et al. (2020), in Appendix B.4. We use the Betriebs-Historik Panel (BHP) to assign establishments to sectors, and to identify the number of total employees at each establishment. Throughout the paper, we use "firm" to refer to the economic unit, which may span multiple establishments; we construct firm-level measures (for example, wage premia) by taking employment-weighted averages across establishments.

Other Data. We also link our survey and administrative data to employer reviews from Kununu, a leading employer rating platform in Germany. Like Glassdoor, Kununu allows current and former employees to evaluate their workplaces across multiple dimensions, including compensation, work-life balance, career opportunities, and organizational culture. The primary measure we use from Kununu is the overall firm rating, which summarizes the information that reviewers provide about a firm.[5] Appendix B.6 describes these data.

2.2 Descriptive Statistics and Sample Selection

We estimate firm-specific valuations using data from the more than 2,000 workers who named at least one firm in either the initial or follow-up survey wave.[6] Table 1 compares our surveyed workers (Columns 1–2) to two benchmark samples: a 2% random sample of full-time workers ages 25–50 (Columns 3–4) and the universe of full-time job movers between 2010 and 2019 (Columns 5–6). Relative to the full workforce, our sample is younger, more educated, and higher-earning—partly because it overrepresents manufacturing workers. We oversampled workers at firms that participated in a previous survey we fielded on bargaining behavior; many of these firms were in the manufacturing sector (Caldwell, Haegele and Heining, 2025*a*). Compared to a random sample of workers, job movers are somewhat younger and lower-earning. We describe the firms for which we have amenity estimates in Section 4.

Because our survey sample differs somewhat from the German workforce, we take two approaches to ensure our results reflect population-relevant patterns. First, in firm-level analyses we weight firms by employment by default, so that estimates reflect the experience of a typical worker rather than a typical firm in our sample. We depart from this default when employment weighting is inappropriate—for example, when firm size is itself the dependent variable—and in some specifications we present unweighted results before the employment-weighted ones. We show robustness to alternative weighting schemes throughout. Second, in worker-level analyses that combine our survey with the 2% population sample—such as the gender gap decomposition—we probe the robustness of our results by reweighting so that our sample matches the demographic composition of the broader workforce. We do so using inverse probability weights estimated from a logit of

[5] In unreported results we found this to be highly correlated with the first principal component obtained from using amenity-specific measures.

[6] We cannot estimate valuations using data for workers who declined to provide the names of specific outside firms. For workers who did not provide this information in either wave, we asked about their reasons for non-response. Appendix Table A2 shows that the majority either had no interest in switching firms or could not identify specific companies. The first group of inframarginal workers is unlikely to be relevant for firms' wage–amenity setting decisions, while the second group would not have provided usable responses in the choice experiment.

survey inclusion on age, education, earnings, and sector.

3 Valuing Firms' Amenities

This section presents a conceptual framework for workers' choices over firms, outlines our empirical strategy for estimating money-metric valuations of firm-specific amenities using discrete choice experiments (DCEs), and compares our approach to alternative estimation methods.

3.1 Conceptual Framework

Workers derive utility from wages and non-wage amenities. The utility that worker i receives from working at firm j is

$$u_{ij} = \beta \log w_{ij} + a_j + \epsilon_{ij} \tag{1}$$

where β is the marginal utility of log wages, w_{ij} is the wage worker i earns at firm j, a_j is the value of the firm's non-wage attributes (in utility units), and ϵ_{ij} captures idiosyncratic preferences. Following a large literature, we refer to a_j as "amenities," though a_j captures the full bundle of non-wage factors that attract workers to a firm—working conditions, career prospects, job security, and culture. This is the standard quantity in models of compensating differentials and job ladders: the total non-wage value a firm provides, not any single non-wage attribute.

A worker prefers firm k to firm j if:

$$\beta \log w_{ij} + a_j + \epsilon_{ij} < \beta \log w_{ik} + a_k + \epsilon_{ik}.$$

Assuming the ϵ_{ij} are independent and identically distributed and follow a Type I extreme value (logit) distribution, the probability a worker prefers firm k over firm j, denoted P_{kj}, is:

$$\begin{aligned} P_{kj} &= \Pr[\epsilon_{ij} - \epsilon_{ik} < \beta(\log w_{ik} - \log w_{ij}) + (a_k - a_j)] \\ &= \mathbb{E}_{j,k}\left[\frac{\exp(\beta \log w_{ik} + a_k)}{\exp(\beta \log w_{ik} + a_k) + \exp(\beta \log w_{ij} + a_j)}\right], \end{aligned}$$

where the expectation is taken over the distribution of randomized wage offers. The probability workers prefer firm k to firm j depends on both the gap in log wages $(\log w_{ik} - \log w_{ij})$ and the gap in amenities $(a_k - a_j)$.

Group-Specific Valuations and the Common Index. Equation (1) treats a_j as a single firm-level amenity value. In practice, different workers may value the same firm differently, so we also estimate group-specific valuations a_j^g (e.g., for men and women separately) in Section 5.3. These group-specific valuations are the relevant object for questions about disagreement across groups and for exercises, such as the gender-gap analysis, in which workers' own valuations matter directly. At the same time, heterogeneity across groups does not imply that a common firm component is absent. Rather, amenity valuations may contain both a common component and group-specific or match-specific deviations.

This parallels a central question in the wage-setting literature: whether wages contain a common firm component or are driven primarily by match-specific factors. The common index a_j, which can be interpreted as the amenity value for the employment-weighted average of marginal applicants to firm j, is the relevant object for questions about firm-level inequality, mobility, and the relationship between amenities and firm wage premia. We provide direct evidence for the importance of this common component in Section 4.4, where we show that the common index predicts firm size, employee satisfaction, and worker retention. The headline results on wage–amenity orthogonality (Section 5.1) and mobility (Section 5.2) therefore use the common index a_j. Both sets of results use split-sample instrumental variables—instrumenting one random half-sample's amenity estimate with the other—and thus isolate the common component of amenity value. Idiosyncratic match effects are independent across the two half-samples, so they are uncorrelated with the instrument and do not bias the IV estimates. Section 4.3 characterizes the extent of this heterogeneity, and Section 5.3 examines its implications for the gender wage gap.

3.2 Recovering Amenity Valuations Through Discrete Choice Experiments

We estimate firm-specific amenity valuations using DCEs embedded in our worker survey. Randomizing wage offers across firms isolates preferences over amenities, enabling direct identification without assumptions about offer rates, search frictions, or endogenous wage-setting.

To construct realistic choice sets, we first asked each worker to name three firms they would consider applying to if they were to leave their current employer. We then presented these three firms—alongside the worker's current job—in a personalized DCE. Each outside firm was paired with a randomly assigned wage change, ranging from a 5% pay cut to a 20% raise in 5-percentage-point increments.[7] Workers were asked to rank the four job options from most to least preferred. The instructions read:

> "Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take."

Appendix Table A3 confirms that the randomized wages assigned to different firms are uncorrelated with firm characteristics, including both perceived and actual wage differences, Kununu pay scores, and brand recognition.[8]

We estimate amenity valuations using workers' preferences over the three outside firms. For each pair of firms j and k, we calculate the share of workers who choose k over j; this yields an empirical matrix of choice probabilities. As we show in Appendix C, because we randomized raises across firms, the amenity parameters a_k are identified up to a normalization. We can recover

[7] We did not design the survey to have power to test whether pay cuts matter differently from pay increases. The initial wave randomized raises of 5% to 20% for outside firms; the follow-up wave extended the range to include a 5% pay cut. Because we did not provide pay cuts in the initial wave, robustness checks using data from only this wave provide a joint test that the valuations are stable between waves and that the valuations are not affected by the range of wages provided.

[8] A remaining concern is that workers' prior beliefs about firm pay could contaminate the amenity estimates. Two features of the design address this. First, each offer is framed as a raise on the worker's *current* pay, not as a level of pay at the outside firm; prior beliefs about firm pay therefore do not enter the choice scenario directly. Second, within-firm wage dispersion is large (Caldwell, Haegele and Heining, 2025*a*). Raises of 5–20% are within the range workers observe at their own firm, making the offers credible regardless of beliefs about the firm's average pay.

a term proportional to $\exp(a_j)$ by finding the principal eigenvector of the matrix implied by the empirical matrix of choice probabilities; this identifies a_j up to a constant (Sorkin, 2018). The DCE maps directly into this revealed-preference framework: we use pairwise choice probabilities from the experiment to construct the pairwise comparison matrix analogous to the job-to-job transition matrix in Sorkin (2018), but because wages are randomized, the eigenvector recovers amenity values alone rather than the combined value of wages and amenities. This approach also allows us to rank firms that are not themselves explicitly ranked and to handle firm pairs where workers unanimously prefer one firm.

Our approach contrasts with revealed preference methods, which recover only the combined utility value of firm wages and amenities. Our strategy shares three limitations with both those approaches and with the estimation of firm wage effects. First, as in Abowd, Kramarz and Margolis (1999) and Sorkin (2018), we can only estimate values for firms in the strongly connected set. In our setting, this requires that at least one firm named by a worker is connected, directly or indirectly, to others through observed comparisons. Each DCE yields three pairwise comparisons among the three outside firms, excluding comparisons involving the current employer. Second, we recover amenity values only up to a normalization (see Appendix C). We normalize to an arbitrary firm with an amenity value near the median. Finally, our strategy assumes there is a single ranking of firms; we test the validity of this assumption in Section 4.3.

In some exercises, we combine amenity valuations with observed wage premia, which are the firm fixed effects from wage regressions that control for worker education, experience, and time-invariant worker and firm characteristics (Bellmann et al., 2020). We use ψ_j to denote the wage premium associated with firm j. In robustness specifications that use our own estimates of these premia, we can also quantify measurement error. Appendix B.4 describes how we produced these estimates.

3.3 Comparison with Alternative Estimation Approaches

Stated choice data allow us to directly estimate workers' firm-specific valuations of amenities: by randomizing wages, we isolate amenity preferences from wage effects. In standard compensating differentials frameworks, amenities are treated as non-monetary compensation that offsets wage differences. However, amenities may also reinforce—not offset—between-firm wage variation. Revealed preference methods cannot distinguish these cases: they identify only the component of utility orthogonal to wages (Sorkin, 2018).

Stated choice data avoid a key limitation of revealed preference designs: reliance on assumptions about firm offer rates. Observed job-to-job transitions reflect both preferences and the frequency with which firms extend offers. The fact that a worker does not transition from firm j to firm k does not reveal that the worker prefers firm j; the worker may simply not have received an offer from firm k. A common proxy for a firm's offer rate is the share of non-employed workers a firm hires. This proxy assumes that (1) non-employed workers accept all jobs they receive and (2) the probability a firm offers a worker a job does not depend on the worker's employment status.[9] But if firms target different types of job seekers, this proxy may be invalid (Kroft, Lange and Notowidigdo, 2013; Caldwell, Haegele and Heining, 2025*b*).[10] If high-amenity firms have queues of applicants, revealed preference estimates will understate workers' willingness to pay for those amenities. In contrast, discrete choice experiments present workers with hypothetical offers that are not constrained by firm hiring behavior.[11]

Interpretation of Estimates. Our design identifies ex-ante firm-specific non-wage value for workers who are marginal to a firm: the value workers assign to joining a firm before they experience the match, among workers who view that firm as a realistic outside option. This object

[9] Both assumptions can be relaxed up to a constant.

[10] Kroft, Lange and Notowidigdo (2013) use an audit experiment to document that employed jobseekers receive callbacks at a different rate than unemployed jobseekers. Caldwell, Haegele and Heining (2025*b*) show that, among employed workers with the same occupation, education, and experience, the productivity of search is increasing in the firm pay premium of a worker's current firm. This is true even after controlling for workers' individual-specific wage premia, a common proxy for unobserved skill.

[11] The firms that workers provide and apply to may be shaped by beliefs regarding offer probability. This is not a concern for our design because these are still firms the worker is marginal to.

may differ from ex-post valuations because workers accumulate firm-specific human capital, form workplace-specific social connections, and become attached to particular teams, managers, or organizational environments over time. The ex-ante valuation among marginal workers is the relevant object in standard models of firm wage and amenity setting (Rosen, 1986).

Because our experiments randomize wages, holding all other firm characteristics constant, these estimates capture the entire bundle of non-wage factors offered by a firm. This includes tangible benefits—such as flexible work arrangements, career development opportunities, and job security—as well as harder-to-quantify attributes, including workplace culture and managerial quality. Disentangling the value of each specific component would require cross-randomizing both wages and amenities at the firm level; supplementary firm-level data on amenities cannot address this identification problem. While we present correlations between our valuations and amenity provision (in Appendix Section D.3), this is simply a descriptive analysis.

A key advantage of our eigenvector approach is that the randomization of wages causes the wage coefficient to approximately cancel in expectation, so the estimated amenity index does not depend on *estimating* the coefficient on log wages (see Appendix C.2). The eigenvector recovers the amenity values a_j that enter the utility function—that is, amenity values in utility units. If β is the coefficient on log wages in the utility function, the money-metric (log-wage equivalent) amenity value is a_j/β. We estimate β from the structural logit coefficient on the experimental wage difference in Table A4 and use it to convert utility values to money-metric units when comparing amenity and wage dispersions (Section 5.1).

Amenity valuations may capture dynamic considerations. For instance, workers may incorporate expectations about promotions, wage growth, or firm-specific human capital when evaluating prospective employers. To address this, we asked a random subset of workers to re-rank the same job offers under the assumption that all outside firms provided identical promotion and wage-growth trajectories. This allows us to isolate static amenity preferences and assess how much of the total valuation reflects differences in future earnings potential. We refer to these as "no-growth" valuations where relevant.

Accounting for Measurement Error. Our approach generates amenity valuations that are noisy, especially for firms that are ranked by only a few workers. We address this in several ways: weighting firm-level regressions by employment, which is positively correlated with the number of survey mentions; presenting estimates that explicitly reweight by the number of mentions; and using a split-sample approach that naturally accounts for measurement error. Our results are robust across these approaches. Measurement error in amenity estimates would, if anything, push us toward over-interpreting common preferences as reflecting idiosyncratic worker–firm match effects, since noise attenuates the correlation across split samples.

Several of our exercises use estimated amenity values as data in subsequent regressions. In these cases, standard errors do not account for first-stage estimation uncertainty in the amenity values. The remedy depends on the exercise. The split-sample variance decomposition (Section 5.1) is designed to remove estimation noise: the covariance between independent half-sample estimates is a consistent estimator of the signal variance regardless of the first-stage error distribution. The IV binscatters in Section 5.2 similarly use one split-half as an instrument for the other, so that first-stage noise attenuates neither the slope nor the covariance. In the remaining exercises—the size and quit-rate regressions (Section 4.4) and the Kununu correlations (Appendix D.3)—estimated amenity values enter as regressors, and first-stage noise attenuates coefficients toward zero; these exercises therefore provide conservative estimates of the true relationships. In the gender gap regressions (Section 5.3), the amenity valuation is the dependent variable; measurement error inflates standard errors but does not attenuate the point estimates. When exercises treat estimated amenities as data, estimation noise will tend to attenuate the sharpness of the patterns we document.

3.4 Design Validity and Sample Selection

Our design required three choices: what method to use, which firms to evaluate, and which workers to survey.

What Method. We use discrete choice experiments—the standard experimental approach in the literature for valuing non-wage job attributes—because randomizing wages allows clean identification of amenity values without assumptions about offer rates and wage dynamics that revealed-preference methods require. A natural alternative would have been to ask workers directly how much they would be willing to forgo to leave their current firm or how much they would require to join a specific firm. In theory, such an approach would yield each worker's exact willingness to pay for a particular firm, instead of the bound revealed by their choice at a given wage differential. However, stated willingness-to-pay measures are unreliable, as responses can be sensitive to framing effects, reference points, and hypothetical bias (Johnston et al., 2017; see also the discussion in Mas, 2025). Other work has shown that stated willingness-to-pay measures can yield estimates that diverge from experimental estimates by orders of magnitude (Rodemeier, 2023). Prior research shows that DCEs produce amenity values comparable to revealed preference approaches, provided workers evaluate familiar job bundles (see, e.g., Hainmueller, Hangartner and Yamamoto, 2015; Maestas et al., 2023; Mas and Pallais, 2017; Wiswall and Zafar, 2018). For this reason, researchers use DCEs to estimate the value of specific amenities; firms use them to design benefit plans and schedules in real-stakes environments (DeBellis, 2018).

Which Firms. Three considerations shaped our choice of firms for workers to evaluate.

First, when preferences over firms are heterogeneous, estimated valuations depend on which workers evaluate each firm. The workers most relevant for firm decisions—those on the margin of accepting or rejecting an offer—are also the most responsive to changes in amenities (Rosen, 1986).[12] We asked workers to compare job offers from firms they had previously identified as likely application targets if they were to switch employers. The positive relationship between the share of workers from a given origin firm who provide a destination firm and the share of workers from that origin firm who have transitioned to that firm (Appendix Figure A2) supports the idea

[12]The trade-off with focusing on worker-provided firms is that we do not observe the full distribution of potential firms, only those workers consider applying to. As we show in Section 5.1, this selection does not generate a spurious wage–amenity correlation.

we successfully identified these workers.[13]

Second, if preferences evolve over the course of an employment spell, workers' valuations may differ depending on whether they have worked at a firm. We chose to focus on firms that workers identified as potential destinations—not firms that workers currently or previously worked for. By asking workers to evaluate outside firms rather than their current or former employer, our estimates reflect ex-ante valuations—what workers expect a firm to be like—rather than ex-post assessments shaped by tenure. Preferences may evolve during tenure through the accumulation of firm-specific capital or the formation of social ties (Caldwell, Haegele and Heining, 2025*b*; Humlum, Rasmussen and Rose, 2025; see also Pakes et al., 2021).[14] We discuss the distinction between ex-ante and ex-post valuations further in Section 5.1.

Finally, a common concern in survey-based approaches is that respondents may struggle to evaluate unfamiliar options. By focusing on firms workers specified they would apply to, we mitigate this concern. Appendix D.5 confirms that the characteristics of these firms closely resemble those of firms that workers from the same employer have historically moved to, further supporting the validity of this design.

Which Workers. We fielded the survey to a broad sample of workers rather than recent job-to-job movers. We targeted workers on the margin of moving; conditioning on movers can generate a spurious negative correlation between wages and amenities even when the population-level correlation is positive (Bonhomme and Jolivet, 2009; Lavetti, 2023; Lavetti and Schmutte, 2023). Such bias would mechanically lead us to conclude that amenities dampen overall between-firm inequality. We discuss this issue and present direct evidence of the bias in Section 5.2.

[13]Because we use job transitions from 2010–2019 for this comparison, and our survey was fielded in 2022, this correlation is not mechanically driven by workers providing information on their future transitions.

[14]A separate concern is that workers' valuations of a firm may change because they have left it if, for instance, they burn bridges with coworkers when they leave or if they believe their previous firm will treat them differently if they rejoin (e.g., because they perceive the worker to be less loyal).

4 Firm Wages, Amenities, and Valuations

We begin by describing the set of firms for which we have estimates of non-wage values. We then show that the distribution of these values is more dispersed than that of wages. While there is considerable heterogeneity in valuations, a single-index model still has predictive power.

4.1 Characteristics of Valued Firms

Our estimation strategy rests on the assumption that workers perceive a trade-off between wages and non-wage characteristics when choosing among firms. Because we randomized raises across outside firms, any systematic preference for one firm over another reveals a difference in amenity value. If, on average, workers prefer a lower-paying firm, this implies that they perceive it as offering superior non-wage attributes.

Because we ask workers to choose between offers from firms they said they would apply to, the firms they compare are likely more similar on non-wage attributes than randomly chosen firms in their labor market. Yet even among this set, pay is not the only determinant of workers' choices. On average, 35% of workers selected the lower-wage firm. This is consistent with observational evidence that a large share of job-to-job transitions involve wage cuts—a pattern we confirm in Section 5.2—and reinforces the interpretation that workers value a broader bundle of job characteristics beyond pay alone. It also suggests that variation in amenity valuations is large, even among the firms workers plan to apply to.

Table A4 reports the coefficient on the experimental wage difference from an OLS regression of the binary choice indicator. The pooled estimate is 2.86 (SE 0.11): a one log-point increase in the wage offered by firm i relative to firm j raises the probability of choosing firm i by 2.86 percentage points. We cannot reject equality of this coefficient across men and women ($p = 0.67$) or across workers with and without a college degree ($p = 0.49$), consistent with a common response to wages across demographic groups. The structural logit coefficient on the wage difference—the preference parameter β that enters the utility function—is 12.09 (SE 0.50). As with the linear probability model, we cannot reject equality of this coefficient across men and women ($p = 0.69$) or across workers with and without a college degree ($p = 0.50$). Because the eigenvector recovers amenity values in utility units (Section 3.2), converting to log-wage equivalents requires dividing by β. We use the logit estimate $\hat{\beta}$ for this conversion in Section 5.1.[15]

The surveyed workers named 3,899 unique firms, described in Table 2. Like the workers in our sample, these firms span all parts of the wage distribution and all major sectors. Firms in the strongly connected set have similar wage premia and sectoral composition to the full set. This holds regardless of the weighting scheme. Of the 3,899 firms workers named, 951 are in the connected set.[16] These firms account for 69.7% of total employment of all named firms, indicating that the connected set covers the bulk of the labor market. Because workers can only sort across firms they know, the consideration set is the relevant object for studying wage–amenity trade-offs. Firms outside this set are typically mentioned by only one worker, making it impossible to compare them to other firms. Firms in the connected set are mentioned a median of 3 times (mean 13.4; interquartile range 2–7), providing sufficient pairwise comparisons for estimation.

4.2 Distribution of Valuations

Amenity values are highly dispersed across firms. Figure 2 plots this distribution, weighted by the size of the corresponding firm. Recall that we normalize the valuation of an arbitrary firm to zero. The figure shows that amenity values are highly dispersed: the spread between the highest- and lowest-valued firms is approximately four utility units. Our estimates do not simply reflect noise. Weighting by firm size, the correlation between estimates from two random halves of the data is 0.5.

[15] As a calibration check, the same discrete choice experiments yield separation elasticities of 2–4 when used to estimate the responsiveness of job-to-job transitions to wage differentials (Caldwell, Haegele and Heining, 2025*b*), consistent with the range reported in the meta-analysis of Sokolova and Sorensen (2018).

[16] This limitation is similar to that in the AKM literature where firm wage premia can only be estimated within a connected set of firms.

Static and Dynamic Considerations. In many models of job choice, workers trade off current wages for expected future wage growth. If firms offering high amenities also provide steeper wage-growth profiles, part of what we measure may reflect dynamic wage considerations rather than static job attributes. To test this, we use an additional discrete choice experiment in which workers re-ranked firms under the assumption that all outside firms offered promotion and wage-growth trajectories identical to those of the incumbent firm. Figure 3 shows that the two sets of estimates are highly correlated and that the dispersion of amenity values net of wage growth closely matches that of the baseline.

Correlates of Firm-Specific Amenities. Because we did not cross-randomize provision of specific amenities, we cannot "price" individual amenities. Without exogenous variation in the provision of individual amenities, regressions of valuations on characteristics conflate the direct value of those characteristics with the value of correlated but potentially omitted workplace attributes.[17] With this caveat in mind, we examine how our estimated valuations correlate with workers' ratings of specific firm amenities from Kununu. Appendix D.3 shows that our valuations are positively correlated with a range of firm characteristics and policies.

4.3 Substantial Between-Group Heterogeneity in Valuations

These valuations come from the preferences of workers who are marginal to a firm. However, as noted in Section 3.2, our model assumes a uniform ranking of firms (apart from idiosyncratic valuations). To probe the realism of this assumption, we re-estimate firm-specific amenity values separately for women, men, college-educated workers, and workers without a college degree. Table 3 reports pairwise correlations between these subgroup-specific amenity valuations.

Between-group heterogeneity in amenity valuations is large. The Pearson correlation between

[17] For example, suppose a firm's dog-friendliness happens to correlate perfectly with other workplace attributes that workers do value, such as flexible hours. A regression of our valuations on dog-friendliness would find a strong positive coefficient even if workers place no value on the policy itself—or even view it as a disamenity. More generally, a positive correlation between a characteristic and our valuations tells us only that the characteristic tends to co-occur with valued workplace attributes, not that the characteristic itself is valued.

amenity valuations for women and men is 0.239, indicating that the types of firms women and men view as high-amenity differ markedly. This low correlation is striking given that the estimates come from workers who are marginal to the same firms. We find similar divergence by education: the correlation between valuations of college-educated and non-college workers is 0.336. Spearman rank correlations are higher (0.437 and 0.389, respectively): groups agree more on the ordering of firms than on the magnitude of money-metric valuations.

This divergence contrasts with the literature on firm wage premia. In a broad set of countries including Germany and the United States, firm wage premia for different demographic groups (including men and women and college and non-college workers) are strikingly similar (Card, Cardoso and Kline, 2016; Sorkin, 2017; Palladino et al., 2025). Similarly, workers' perceived wage premia do not differ substantially by gender or education (Caldwell, Haegele and Heining, 2025*b*). Workers agree much less on the ranking of firms by non-wage attributes.

Robustness. A natural concern is that measurement error could attenuate between-group correlations, making groups appear to disagree more than they actually do. If amenity estimates are noisy, the noise in male- and female-specific estimates may be independent even when the true values are similar, mechanically reducing the correlation. However, if noise in rarely mentioned firms drives the low correlations, then upweighting well-measured firms should increase them. Appendix D.1 shows that weighting by the number of times a firm appears in the data yields similarly low correlations. Restricting the sample to firms mentioned at least 5 or 10 times—which substantially reduces noise—also has little effect. Paired t-tests confirm these patterns: we reject equality of male and female valuations ($p = 0.032$) but cannot reject equality by education ($p = 0.687$).

To address measurement error more directly, we turn to a separate module of the survey in which we selected a set of large firms and asked all respondents to evaluate them, regardless of whether they had named these firms as application targets. Because many workers evaluated the same researcher-provided firms, the resulting amenity estimates are substantially more precise than those from the worker-provided module. Using all valuations from this module—normalized relative to a common base firm (manufacturing)—we strongly reject equality of male and female amenity values ($p < 0.001$, Appendix Table D5, Panel A). Panel B restricts to firms workers named as application targets. With the smaller sample we no longer reject equality ($p = 0.758$), but between-group correlations remain low.

4.4 Predictive Power of the Common Index

The between-group heterogeneity in Section 4.3 poses a challenge for models that represent amenities as a single firm-level index. Many structural models of the labor market assume such an index (Lamadon, Mogstad and Setzler, 2022; Berger, Herkenhoff and Mongey, 2022; Volpe, 2024; Mas, 2025), and for many policy questions—such as whether to mandate amenity disclosure or how to evaluate firm-level regulations—a common index is the relevant object. Our inequality question also requires one: between-firm inequality is about firm effects ψ_j, not match quality.

This challenge parallels a central question in the wage-setting literature: whether firm wage premia reflect a common component or are dominated by worker–firm match effects (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013; Bonhomme, Lamadon and Manresa, 2019; Song et al., 2019). In the wage context, various approaches have established that a meaningful common firm component exists (Kline, Saggio and Sølvsten, 2020). The analogous question for amenities—do firms offer a common non-wage value, or are amenity valuations primarily match-specific?—has received far less attention.

Several forces could weaken the link between our valuations and observed labor market outcomes. First, measurement error in amenity estimates—particularly for firms mentioned by few workers—could attenuate any relationship. Second, our valuations are elicited from workers considering hypothetical job changes; if workers' ex-ante expectations of a firm's amenities diverge substantially from their ex-post experience, valuations may not predict realized outcomes such as retention or firm size. Third, if idiosyncratic worker–firm match effects dominate the common component of amenity value, firm-level averages of our estimates could be uninformative about the actual amenity environment. Our split-sample correlation for amenity valuations (approximately 0.5) bounds this concern directly. The common component accounts for half the cross-firm variance in amenity estimates.[18] All three forces—measurement error, ex-ante/ex-post divergence, and match effects—bias against finding a relationship; any positive association we find therefore understates the true predictive power.

Table 4 presents two tests of the predictive power of the common index. For each test we use three alternative wage controls: the observed wage premium (Panel A), log median pay (Panel B), and log mean pay (Panel C). Within each panel, the first row shows that amenity valuations positively predict firm size (as measured in the administrative records), conditional on the wage premium, with an elasticity of 0.34 ($p < 0.01$). This supports a key assumption in the structural labor literature. In many models, amenity values are derived from the assumption that firms which are large relative to their wage must be providing non-wage value that attracts and retains workers (Lamadon, Mogstad and Setzler, 2022; Volpe, 2024).[19] Our estimates validate this logic independently: firm-specific amenity values recovered from a discrete choice experiment predict administrative employment, even though employment played no role in estimation.

The second row within each panel shows that amenity valuations also positively predict the share of employees who would recommend the firm on Kununu, again conditional on the wage. Kununu measures ex-post employee satisfaction, independently of our survey. The correlation with our ex-ante valuations suggests that workers anticipate what they will experience after joining.

If our estimates were dominated by idiosyncratic match effects, they would not systematically predict firm size or employee satisfaction conditional on wages. That they do rules out a pure match-effects interpretation and confirms that firms differ in the non-wage value they provide. Appendix Section D.7 provides additional evidence on this point. This appendix shows that firms with higher amenity valuations also have lower quit rates, conditional on wages.

[18] Match effects are idiosyncratic to each worker–firm pair and therefore independent across the two random half-samples; only the common component a_j generates covariance between the split-sample estimates. A reliability ratio of 0.5 implies that the common component accounts for half the cross-firm variance in amenity estimates; match effects and measurement error together account for the other half. Since experimental noise from the discrete choice design also contributes to the residual, the match-effect share is strictly less than one-half.

[19] A common critique of such approaches is that firms may also vary in recruiting standards or intensity. A low-wage firm may be large simply because it is willing to hire a large share of workers, not because it provides a good working environment.

5 Amenities and Inequality

In this section we document four facts about between-firm inequality: (1) the dispersion of amenities is about one-third that of pay; (2) non-wage valuations are approximately orthogonal to wage premia; (3) wage cuts in the data are often paired with amenity gains; and (4) gender-specific amenity valuations close or reverse the gender gap in firm-provided value.

5.1 Firm Wage Premia and Inequality in Utility

Standard compensating differentials models predict a slope of -1 when amenity values are expressed in log-wage equivalent units: a one-unit increase in the wage premium is fully offset by a one-unit decrease in amenity value. Our estimates show instead that wages and amenities are approximately orthogonal. Because inequality in firm wage premia is a statement about the common firm component ψ_j, the relevant test requires an amenity measure that also isolates the common component—not one that conflates firm attributes with idiosyncratic match quality.

Figure 5 plots firm-specific amenity valuations against firm wage premia, using the split-sample approach described in Section 3.2 to account for measurement error. Panel A shows that, with each firm weighted equally, there is a mildly positive relationship: firms with higher pay premia typically offer weakly better amenities. Panel B shows that this correlation shrinks substantially toward zero when we weight by firm size, which captures the relationship experienced by the average worker.

That weighting by firm size pushes the correlation toward zero rules out one obvious concern: that measurement error is attenuating a strong positive true correlation. Firm size is positively correlated with the number of times a firm is mentioned in our survey, which directly indexes the precision of our amenity estimates. If the positive unweighted correlation were genuine but attenuated by measurement error, weighting by firm size (or by mentions) should strengthen it, since larger and more frequently mentioned firms have less noisy estimates.[20] Instead, weighting

[20]This is true both for our valuations and for the firm wage premia.

pushes the correlation toward zero—the opposite of what measurement error correction would produce.

Because workers name only firms they view as realistic outside options, our orthogonality result is identified among firms that compete for the same workers rather than across the full universe of firms. We view this as the relevant object for firm wage and amenity setting. In standard models, firms choose wage–amenity bundles to attract workers on the margin of accepting or rejecting an offer, so the relevant competitors are firms workers plausibly consider, not firms outside their consideration set. More broadly, some form of selection is unavoidable in any study of wage–amenity trade-offs: realized mobility studies condition on accepted offers, and cross-sectional studies condition on observed matches. Our design makes this selection explicit and targets the margin which is theoretically relevant for firm competition.

The variance of amenity valuations within consideration sets is likely smaller than across all firms in the economy. Workers will likely exclude firms where obvious disamenities—such as long commuting distances—dominate. Further, compressed variance is the expected result when studying firms that compete for the same workers. This selection does not lead to a mechanical positive correlation between wages and amenities. Rather, if workers direct their search on the basis of the total value of a firm, they may name a firm either due to high wages or high amenities. Conditioning on being named would, if anything, induce a negative correlation between wages and amenities. Our near-zero estimate is therefore conservative. Appendix Figure D3 shows, consistent with this reasoning, that firms named most frequently—those most selected on desirability—do not show a stronger relationship in either direction.

Variance Decomposition. We bias-correct using a split-sample approach: the covariance between estimates from two independent random halves of survey respondents provides a consistent estimate of the true variance, analogous to leave-out estimators for AKM firm effects (Kline, Saggio and Sølvsten, 2020).[21] Converting variances to money-metric (log-wage equivalent) units by

[21]The reliability ratio for firm wage premia (0.85) is the split-sample correlation from Card, Heining and Kline (2013). Our own split-sample correlation for amenity valuations is approximately 0.5, implying a signal-to-total

dividing by $\hat{\beta}^2$, where $\hat{\beta} = 12.09$ is the structural logit estimate from Table A4, the signal variance of amenity valuations is about one-third that of firm wage premia. Amenities are dispersed in utility units but compressed in log-wage equivalents.

The orthogonality holds when we decompose by demographic group: $\text{Cov}(a_j^M, \psi_j)$ and $\text{Cov}(a_j^F, \psi_j)$ are each approximately zero, ruling out the possibility that the result is an artifact of aggregating groups with offsetting correlations.

Robustness. The analyses in this section link amenity valuations elicited in 2022 and 2024 to administrative employment records spanning 2010–2019. We do not have direct evidence that firms' relative amenity positions are stable over this gap. The predictive relationships in Section 4.4, however, bound the degree of mismatch: substantial drift would attenuate the size and quit-rate relationships toward zero, yet we find sizable associations with both.

We find similar results when we use the alternative amenity estimates that hold career growth expectations constant (Appendix Figure D11), and when we restrict to Wave 1 of the survey, in which randomized raises ranged from 5% to 20% with no pay cuts offered (Appendix Figure D5). The latter rules out the possibility that the introduction of pay cuts in the follow-up wave drives our findings.

Relationship with Other Work. We find orthogonality, whereas studies based on movers find a negative wage–amenity correlation (Humlum, Rasmussen and Rose, 2025). Two features of our design plausibly contribute to this divergence. Bonhomme and Jolivet (2009) and Lavetti and Schmutte (2023) demonstrate that analyses focusing on job-to-job movers can generate a spurious negative correlation even when the true population correlation is zero or positive.[22] Workers who move to lower-wage firms disproportionately gain in amenities, mechanically inducing a negative correlation in the movers sample. A related selection issue arises with reservation-wage designs that elicit valuations from movers: a worker who accepts a lower-paying job must value the desti-

variance ratio of one-half.

[22]We confirm this bias in simulations reported in Appendix C.5.

nation's amenities enough to make the move worthwhile. The resulting sample can overrepresent transitions in which wages and amenities move in opposite directions.[23] Because our amenity estimates come from a design fielded to a broad sample of workers—not from the transitions of job changers—our estimates are not subject to this selection by construction. Our focus on outside firms also lets us avoid taking a stand on how valuations change once a worker joins; previous work has shown that ex-ante and ex-post valuations can diverge substantially (Caldwell, Haegele and Heining, 2025*b*; Humlum, Rasmussen and Rose, 2025). That ex-ante valuations predict firm size conditional on the wage confirms that these measures capture firm-level conditions relevant for labor market equilibrium.

Our finding of orthogonality between firm wage premia and amenity values is consistent with previous work showing that higher-wage workers report better working conditions (Maestas et al., 2023). That worker-level finding reflects sorting: higher-skilled workers earn more and work at firms with better amenities. Our estimates isolate the firm-level relationship and show that, conditional on the firm, higher wages do not come at the cost of worse amenities.

5.2 Amenities and Mobility

Because our estimates of amenity values are not derived from worker flows, we can ask how job-to-job transitions move workers up or down the firm wage and amenity ladders. We link administrative records on all job-to-job moves in Germany to our survey-based amenity valuations, classifying each move by whether the worker's wage increased, decreased, or remained similar (within $\pm 2\%$), and separately by whether the worker moved to a firm with higher, lower, or similar amenity value.

Transition Matrices. Table 5 presents transition matrices that categorize job-to-job movers based on changes in both wages and estimated firm amenity values. Panels A and B use our baseline amenity valuations, while Panels C and D use valuations estimated under the assumption that

[23] Humlum, Rasmussen and Rose (2025) use survey data on a large number of job movers in Denmark. To address the selection issue, they use a structural model.

wage growth at the firm remains constant. Within each pair, the first panel examines changes in workers' own wages and the second considers changes in the firm wage premium.

A substantial share of workers who take wage cuts move to firms with similar or better amenities, rather than experiencing losses in both dimensions. Focusing on Panel A, which tracks changes in a worker's own wage, 22.3% of job movers take a wage cut of more than 2%. Among these downward wage moves, only 1.1% involve a simultaneous decline in amenities, while 19.4% of wage-cut movers transition to firms with similar amenities, and 1.8% move to firms with at least a 2% increase in amenities. For the vast majority of downward wage moves, workers maintain or improve their non-wage attributes. Workers who take cuts in the firm wage premium (Panel B) show the same pattern: only a small fraction experience a simultaneous decline in amenities.

Workers are more likely to move up the wage ladder than up the amenity ladder. About 41.9% of workers move to firms that offer them higher own wages (>2% increase) and 33.1% move to firms with higher firm wage premia (>2% increase). By contrast, only 9.2% of transitions are to firms with substantially higher amenities (>2% increase). Panels C and D repeat the analysis using amenity estimates that hold wage-growth expectations constant. The pattern is similar: 38.7% of movers experience an own-wage increase, while only 7.1% move to firms with higher amenities. Wage gains are the primary driver of job-to-job mobility; however, 7.1–9.2% of workers move primarily to improve their working conditions rather than their pay.

Selection into Mobility: The Apparent Wage–Amenity Trade-off. Although wages and amenities are orthogonal in the cross-section of firms (Figure 5), a negative relationship emerges among movers—a pattern consistent with selection rather than the underlying firm-level relationship. To quantify the magnitude of this selection-induced slope, we regress changes in sex-specific amenity valuations on changes in the firm wage premium, allowing the slope to vary by gender. OLS estimates yield a negative slope: workers who move to higher-paying firms tend to experience a decline in amenities. Instrumental variables binscatters—instrumenting the change in amenity valuations from one random split-half with the change from the other—corroborate this pattern:

workers who move to firms with higher predicted amenities accept lower wages and lower firm premia (Appendix Figure D6).

Robustness. These patterns are robust to several alternative specifications. First, we repeat the transition-matrix analysis using our alternative measure of amenities that equalizes wage-growth expectations across firms; the results remain broadly similar, suggesting that the observed trade-offs are not merely a reflection of forward-looking wage considerations. Second, we restrict the sample to likely voluntary movers—workers who transition directly from one employer to another without an intervening unemployment spell—and find similar patterns (Appendix Table D3). Third, we require that both the origin and destination firms have at least five (or ten) survey respondents, ensuring that the amenity estimates used in the classification are precisely estimated; the results are again robust. Fourth, we repeat the analysis using a 10% threshold to define large changes, with similar conclusions (Appendix Table D4).

Implications for Estimating the Wage–Amenity Correlation. These transition patterns illuminate why analyses that condition on movers may find a negative wage–amenity correlation even when the population-level correlation is approximately zero. Among downward wage movers in our data, 95% move to firms with similar or higher firm-level amenity valuations—they accept lower wages while moving to firms that offer, on average, better non-wage attributes. Among upward wage movers, the amenity change is approximately zero on average. Conditioning on the sample of all job-to-job movers mechanically generates a negative slope between wage changes and amenity changes, even though the underlying firm-level objects are uncorrelated. This is the selection mechanism formalized by Bonhomme and Jolivet (2009) and emphasized in Lavetti and Schmutte (2023), and our data provide a direct empirical demonstration: the same amenity estimates that are orthogonal to wages in the cross-section of firms (Figure 5) produce the appearance of compensating differentials when viewed through the lens of realized transitions. Approaches that estimate the wage–amenity relationship from movers will capture this selection in addition to the underlying firm-level relationship; recovering the latter requires either an explicit selection correction or a design (like ours) that does not condition on moving.

Gender Differences in Mobility Patterns. Figure 7 presents transition classifications separately by sex, using sex-specific amenity valuations and changes in the firm wage premium. The patterns differ sharply. Among men, amenity increases dominate amenity decreases regardless of the direction of the wage change: men who move to higher-premium firms tend to gain amenities, as do men who move to lower-premium firms. Among women, the pattern reverses with wage direction. Women who move to lower-premium firms disproportionately gain amenities, while women who move to higher-premium firms disproportionately lose them. This asymmetry is consistent with women facing a steeper wage–amenity trade-off in their actual job transitions.

5.3 Amenities and the Gender Wage Gap

Many papers document that, across countries and institutional settings, women sort into firms that pay all workers less (Card, Cardoso and Kline, 2016; Sorkin, 2017; Palladino et al., 2025; Datta Gupta et al., 2025). A leading explanation is that women place greater weight on non-wage amenities when choosing where to work. Prior work confirms that women value specific amenities—such as flexible scheduling and reduced hours—more than men (Wiswall and Zafar, 2018; Mas and Pallais, 2017). Our own survey data are consistent with this: women are significantly more likely than men to rate options for home office, reduced hours, and predictable schedules as "very" or "extremely" important (Appendix Figure A3). We use our firm-level valuations, linked to the 2% IEB sample described in Section 2, to measure how accounting for amenities changes the gender gap in job quality.

We estimate the gender gap in firm-level outcomes by regressing each outcome on a female indicator and progressively richer controls:

$$y_{ij} = \theta_0 + \theta_1 \,\text{Female}_i + X_i'\gamma + \eta_{ij}, \tag{2}$$

where y_{ij} is the outcome (firm wage premium ψ_j or amenity valuation a_j) for worker i at firm j, X_i includes Mincer controls and fixed effects depending on the specification, and standard errors are clustered at the firm level.

Baseline Results. Table 6 presents the analysis. Panel A documents the wage gap: the raw gender gap in the firm wage premium is -0.014 (Column 1), and it persists after adding Mincer controls (Column 2, -0.010) and occupation fixed effects (Column 3, -0.003). Panel B asks whether amenities reinforce or offset this wage gap. Using our baseline amenity index—a single ranking applied to all workers—women work at firms with amenity values -0.003 to -0.003 log-wage equivalents lower than men's (Columns 1–2). Under a uniform ranking, women appear to receive lower value from their employers on both dimensions.

This conclusion depends on the assumption that men and women value the same firms equally. Figure 4 shows they do not: the Pearson correlation between male-specific and female-specific amenity valuations is only 0.239 (Table 3). Columns 4–6 re-estimate the gender gap using sex-specific valuations. The coefficient on the female indicator in Panel B changes sign and is significantly positive in our most stringent specification: with Mincer controls and occupation fixed effects (Column 6), women work at firms whose amenity value is 0.014 log-wage equivalents *higher* than men's ($p < 0.05$). On the same sample, the wage gap is -0.013 (Panel A, Column 6)—similar in magnitude but opposite in sign. Accounting for sex-specific amenity valuations therefore roughly closes the gender gap in total firm-provided value: the wage and amenity coefficients are offsetting, and the implied gap is statistically indistinguishable from zero.

Robustness. Of course, when we consider the sex-specific valuations, the comparison across sexes—how much higher women's amenity values are—does depend on the normalization. The estimated gender gap at firm k reflects

$$(\hat{a}_k^F - \hat{a}_k^M) = (a_k^F - a_k^M) - (c^F - c^M), \tag{3}$$

where $c^F - c^M$ is set by the choice of anchor. We normalize by setting the employment-weighted mean amenity valuation in services to zero within each sex. This amounts to assuming that men and women derive, on average, the same amenity value from firms in this sector. Services is a conservative anchor: women are overrepresented in the sector, so if services firms are in fact more attractive to women, this normalization will understate the female amenity advantage.

Table A6 examines the sensitivity of these results. The sign of the amenity coefficient is positive across all specifications, though the magnitude varies. Adding sector fixed effects (Column 1) yields 0.011 ($p < 0.05$). Using no-growth valuations, which removes differential expectations about wage growth, gives 0.040 ($p < 0.01$, Column 3)—the larger coefficient implies that women's firms offer worse wage-growth prospects, which partially offsets their amenity advantage in the baseline. Column 2 reweights observations to match the full population sample (0.008). Columns 4–6 vary the normalization anchor: the overall employment-weighted mean (0.003), the manufacturing sector (0.006), and the median firm (0.004). Under every normalization we consider, the coefficient remains positive: the sign of the female amenity advantage does not depend on the choice of anchor.

Discussion. Even if women sort into firms they value more highly on non-wage dimensions, this does not imply that the gender wage gap is welfare-neutral. Our estimates reflect preferences over the existing distribution of jobs. Some of the amenities women value at their current employers—such as the absence of sexual harassment, a known determinant of women's labor supply (Folke and Rickne, 2022), or a less male-dominated work environment—are likely shaped by the equilibrium itself. Changing a firm's gender composition could alter its culture and non-wage attributes, potentially making higher-wage firms more attractive to women. But moving a single woman to a higher-wage firm, holding the firm's characteristics fixed, would not necessarily enhance her welfare: it would separate her from the non-wage attributes she currently values without changing the environment at the destination firm.

Our work is related to Maestas et al. (2023), who find that accounting for the incidence and valuation of nine specific amenities narrows the gender wage gap. We examine the same question at the firm level, recovering the total non-wage value of specific firms—inclusive of amenities that are difficult to enumerate, such as workplace culture, social environment, and managerial quality. The gender gap in total firm-provided value is statistically indistinguishable from zero in our preferred specification (Column 6 of Table 6).

6 Conclusion

This paper provides new evidence on how workers value firm-specific amenities and how those valuations shape mobility, sorting, and inequality, including the gender wage gap. Using discrete choice experiments embedded in a large-scale survey of German workers, we estimate the money-metric value of firm amenities as perceived by marginal applicants, workers who would consider moving to a given firm. By randomizing wages across hypothetical job offers, we recover firm-specific valuations that are independent of observed wage premia, offer rates, or hiring frictions.

Non-wage attributes are an important driver of job choice: workers frequently choose lower-paying offers. Amenity valuations are highly dispersed across firms and approximately orthogonal to wages, so amenities do not offset between-firm pay differences. In money-metric units, the signal variance of amenities is about one-third that of wage premia. Conditional on the wage, high-amenity firms tend to be larger, have lower quit rates, and are more favorably reviewed by employees.

Amenity preferences vary across demographic groups. Men and women do not value the same firms equally: the correlation between their firm-specific valuations is 0.239. Women work at firms that pay less. They also work at firms that offer them higher amenity value. Using gender-specific valuations, women do not work at firms that offer them lower overall value. In some specifications, they work at firms that offer more.

Our results motivate several directions for future work. First, while we document substantial heterogeneity in amenities, we do not take a stand on which specific attributes drive worker preferences. Experimental variation in individual amenities at the firm level would allow precise pricing. Second, we abstract from the firm side of the market. Understanding how and why firms choose particular wage–amenity bundles, and how these decisions respond to worker preferences, is a natural next step. Third, our analysis is restricted to full-time workers; extending it to the part-time margin, where trade-offs among hourly wages, hours, and workplace amenities are simultaneous, would be particularly valuable for understanding the gender gap in job quality. Fourth, our analysis is restricted to the private sector; understanding how amenities shape sorting and inequality in the public sector is a natural extension.

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7 Tables and Figures

Table 1: Descriptive Statistics

	Survey Sample		2% Sample		Job-to-Job Movers	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
	(1)	(2)	(3)	(4)	(5)	(6)
Demographics						
Female	0.32	(0.47)	0.35	(0.48)	0.34	(0.47)
Age	32.96	(5.93)	39.39	(11.80)	35.80	(7.62)
German Citizen	0.91	(0.29)	0.95	(0.22)	0.92	(0.27)
College Degree	0.62	(0.48)	0.15	(0.36)	0.26	(0.44)
Apprenticeship	0.30	(0.46)	0.75	(0.44)	0.65	(0.48)
Earnings						
Daily Pay	173.11	(56.77)	107.58	(62.49)	103.54	(55.19)
Censored Pay	0.23	(0.42)	0.05	(0.23)	0.05	(0.21)
AKM Firm Effect	0.46	(0.15)	0.36	(0.21)	0.28	(0.23)
Sector						
Manufacturing Sector	0.47	(0.50)	0.31	(0.46)	0.19	(0.40)
Retail Sector	0.08	(0.28)	0.14	(0.34)	0.13	(0.33)
Professional Sector	0.13	(0.34)	0.05	(0.22)	0.08	(0.27)
Observations	2383		20339174		25361184	

Note: This table compares the characteristics of workers in our survey (Columns 1–2) to the set of full-time workers ages 25–50 (Columns 3–4) and to full-time workers ages 25–50 who have made a job-to-job transition (Columns 5–6). We describe how we constructed these samples in Appendix B.1.

Table 2: Characteristics of Provided Firms

	All Firms		Connected Set	
	Unweighted	Weighted by Firm Size	Unweighted	Weighted by Firm Size
	(1)	(2)	(3)	(4)
Linkages				
To Wage Premia	95%	100%	97%	100%
To BHP	97%	100%	98%	100%
Size	29682	149565	34701	149891
(Std. Deviation)	(59653)	(93273)	(63227)	(93118)
Size (Imputed)	22733	140018	28385	140730
(Std. Deviation)	(51637)	(94320)	(56473)	(94036)
Size (Binned)				
1-10 Workers	3%	0%	2%	0%
11-50 Workers	4%	0%	3%	0%
51-200 Workers	5%	0%	3%	0%
201-1000 Workers	33%	1%	26%	1%
1001-10000 Workers	20%	3%	24%	3%
>10000 Workers	32%	96%	42%	96%
Firm Wage Premium	0.50	0.48	0.51	0.48
(Std. Deviation)	0.11	0.07	0.09	0.06
Firm Wage Premium (CHH)	0.16	0.06	0.16	0.05
(Std. Deviation)	0.22	0.16	0.21	0.16
Sector				
Manufacturing	40%	43%	42%	43%
Retail	9%	1%	8%	1%
Professional	24%	52%	25%	52%
Information	10%	2%	10%	2%
Finance	6%	0%	5%	0%
Number of Firms	3899	3899	949	949

Note: This table describes the firms that workers provided (Columns 1–2) and the subset of these firms in the connected set (remaining columns). Each column indicates how the statistics are weighted. We impute firm size when missing; Appendix B.3 describes this process. Appendix Table D1 presents results for alternative weighting schemes.

Table 3: Correlation Between Valuations of Different Demographic Groups

	Female (1)	Male (2)	College Degree (3)	No College Degree (4)
Female	1.000	0.437	0.609	0.522
Male	0.239	1.000	0.794	0.589
College Degree	0.512	0.716	1.000	0.389
No College Degree	0.518	0.574	0.336	1.000

Note: This table examines the pairwise correlation in valuations of amenities for different demographic groups. The numbers above the diagonal (shaded in grey) are rank correlations; the numbers below the diagonal are raw correlations. Firms are weighted by size. Results under alternative weighting schemes (including imputed firm size and number of mentions) and in alternative samples are in Appendix Table D2.

Table 4: External Validation: Amenities, Firm Size, and Employer Reviews

	Baseline Valuations		Valuations Excluding Wage Growth	
	(1)	(2)	(3)	(4)
A. Controlling for Observed Wage Premium				
Log Firm Size	0.340***	0.313***	0.424***	0.405***
	(0.084)	(0.083)	(0.142)	(0.148)
	552	552	305	305
Percent Recommend (0-100)	7.513**	5.412***	7.190**	3.280
	(2.938)	(1.877)	(3.146)	(2.137)
	420	420	242	242
B. Controlling for Log Median Pay				
Log Firm Size	0.392***	0.362***	0.552***	0.533***
	(0.091)	(0.090)	(0.142)	(0.143)
	577	577	318	318
Percent Recommend (0-100)	6.760**	4.636***	6.825**	1.970
	(2.732)	(1.730)	(3.186)	(2.111)
	434	434	251	251
C. Controlling for Log Mean Pay				
Log Firm Size	0.378***	0.347***	0.534***	0.516***
	(0.092)	(0.091)	(0.143)	(0.145)
	577	577	318	318
Percent Recommend (0-100)	6.886**	4.746***	6.982**	2.166
	(2.683)	(1.709)	(3.140)	(2.080)
	434	434	251	251
Fixed Effects	—	Sector	—	Sector

Note: This table examines the relationship between amenity valuations and two outcome variables, indicated in the left column: log firm size from the Betriebs-Historik Panel (BHP), and the percentage of employees who recommend the firm on Kununu. Each cell reports the coefficient on workers' valuation of firm-specific amenities from a regression of the outcome on this valuation, a measure of the firm's wage policy (indicated in the panel), and a constant. Columns 1–2 use our baseline valuations; Columns 3–4 use the valuations computed under the assumption that wage growth at the firm remains constant. The log firm size regressions are unweighted; the Percent Recommend (0–100) regressions are weighted by firm size. Levels of significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Classifying Job-to-Job Moves

	Change in Amenities			
	>2% Cut	<2% Change	>2% Raise	Total
	(1)	(2)	(3)	(4)
A. Change in Own Wage				
Decrease ($< -2\%$)	1.1%	19.4%	1.8%	22.3%
Similar ($\pm 2\%$)	1.5%	32.2%	2.1%	35.8%
Increase ($> 2\%$)	4.2%	32.4%	5.3%	41.9%
Total	6.8%	84.0%	9.2%	100.0%
B. Change in Firm Wage Premium				
Decrease ($< -2\%$)	4.6%	25.3%	6.5%	36.4%
Similar ($\pm 2\%$)	1.4%	27.2%	2.0%	30.6%
Increase ($> 2\%$)	4.0%	23.8%	5.3%	33.1%
Total	10.0%	76.3%	13.8%	100.1%
C. Change in Own Wage				
Decrease ($< -2\%$)	0.7%	21.2%	1.6%	23.5%
Similar ($\pm 2\%$)	1.1%	35.0%	1.7%	37.8%
Increase ($> 2\%$)	2.8%	32.1%	3.8%	38.7%
Total	4.6%	88.3%	7.1%	100.0%
D. Change in Firm Wage Premium				
Decrease ($< -2\%$)	3.0%	28.7%	5.8%	37.5%
Similar ($\pm 2\%$)	1.3%	26.1%	1.9%	29.3%
Increase ($> 2\%$)	3.2%	25.9%	4.0%	33.1%
Total	7.5%	80.7%	11.7%	99.9%

Note: This table reports the share of job-to-job transitions according to the change in wage and amenity values. The sample includes all full-time job movers between 2010 and 2019 who moved between firms for which we observe amenity values. Panels A and B use baseline amenity valuations; Panels C and D use valuations computed under the assumption that wage growth at the firm remains constant. Transitions above the greyed-out anti-diagonal are to lower-quality firms; transitions below the greyed-out anti-diagonal are to higher-quality firms.

Table 6: Amenities and the Gender Value Gap

	(1)	(2)	(3)	(4)	(5)	(6)
		All Workers			Sex-Specific	
	A. Firm Wage Premium (ψ_j)					
Female	-0.014^{*}	-0.010	-0.003	-0.022^{**}	-0.019^{**}	-0.013^{**}
	(0.008)	(0.007)	(0.003)	(0.009)	(0.008)	(0.005)
	B. Amenity Value ($a_j/\hat{\beta}$, log-wage equivalent)					
Female	-0.003	-0.003	0.001	0.010	0.010	0.014^{**}
	(0.004)	(0.004)	(0.002)	(0.007)	(0.006)	(0.006)
Mincer controls		Yes	Yes		Yes	Yes
Occupation FE			Yes			Yes
N	236,674	236,674	236,661	201,001	201,001	200,982

Note: This table examines the gender gap in firm wage premia (Panel A) and amenity values in money-metric (log-wage equivalent) units (Panel B). The firm wage premia stem from Bellmann et al. (2020). The amenity values are estimated using the full sample of workers (Columns 1–3) or using sex-specific models (Columns 4–6). The “Mincer controls” include age, a quadratic in experience, and a dummy for whether the worker has a college degree. Results are unweighted. Additional specifications are in Appendix Table A6. We cluster standard errors at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Figures

Figure 1: Key Survey Elicitations

"Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to?

Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. "PlaceHolder Inc"). These can be companies without current job vacancies."

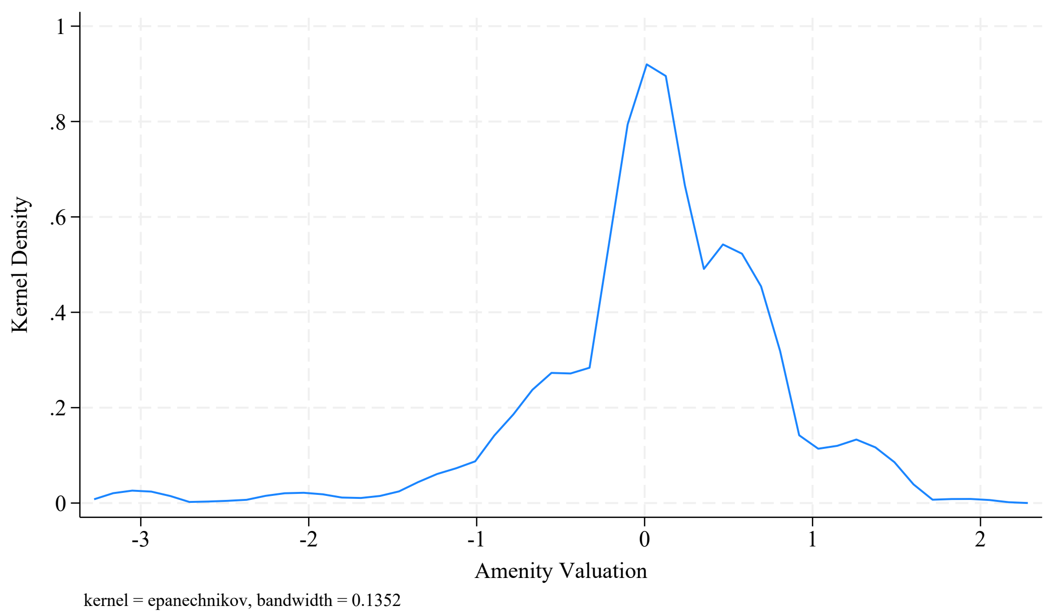
Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified.

Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

[Fill in Company 1] →	{Company 1} with a X% raise ______
[Fill in Company 2] →	{Company 2} with a Y% raise ______
[Fill in Company 3] →	{Company 3} with a Z% raise ______
❑ I do not want to answer this question	Remain at current firm at current pay _______

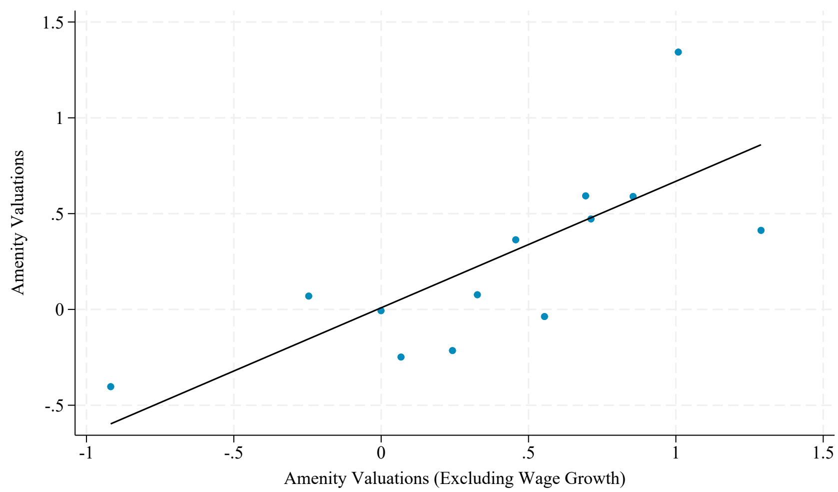
Note: This figure shows the two survey questions we use to estimate valuations. As noted in Section 2, we do not use information on the ranking of the current firm. Appendix E provides more information on the survey.

Figure 2: Distribution of Amenity Values



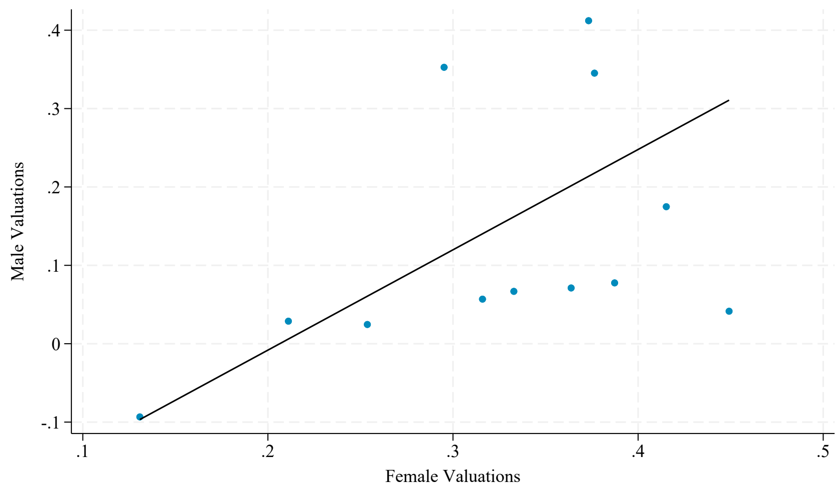
Note: This figure plots the distribution of the raw estimates of amenity values (in utility units), weighted by firm size. Section 3 and Appendix C provide details on estimation. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix D.1.

Figure 3: Relationship Between Valuations with and without Wage Growth



Note: This figure compares the baseline estimates of amenity valuations (in utility units) to those we obtain when we re-run the model using a discrete choice experiment where we told workers to assume that the pattern of promotions and pay growth would be the same at all outside firms. The dots come from a binned scatterplot and the line is a line of best fit. All estimates are relative to the same reference firm. To account for measurement error in the estimation of both types of amenities, we follow a split-sample approach and plot the fitted values of our baseline valuations (predicted values from regressing estimates from one split sample on the other) against the fitted values of our no-growth valuations. Results are weighted by firm size. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix D.1.

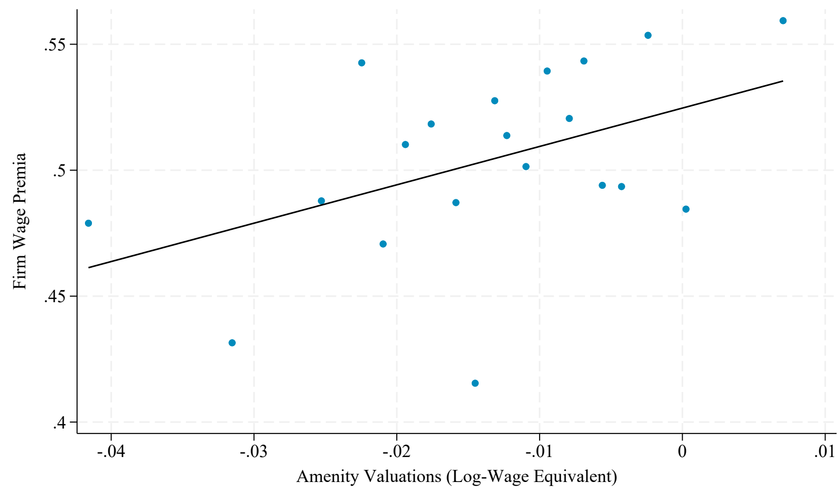
Figure 4: Relationship Between Male and Female Valuations



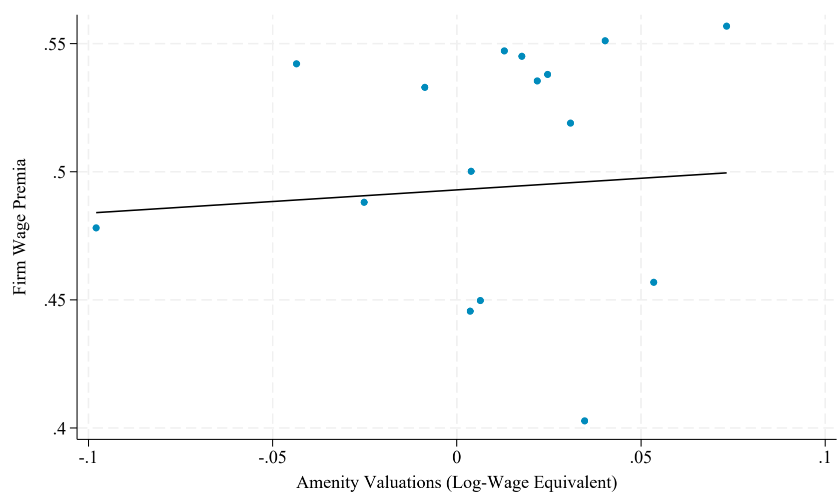
Note: This figure compares the baseline estimates of valuations (in utility units) for male and female workers. We account for measurement error using a split-sample approach. We first regress the valuations for one half of the workers in the group (male or female workers) on the valuations on the other half. We then plot the predicted values for male workers against those for female workers. Results are weighted by firm size. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix D.1.

Figure 5: Relationship Between Wage Premia and Amenity Valuations

A. Unweighted

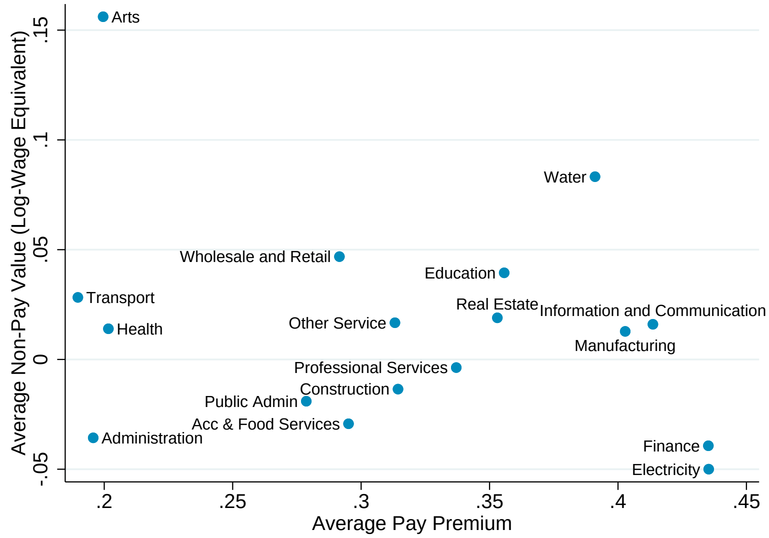


B. Weighted by Firm Size



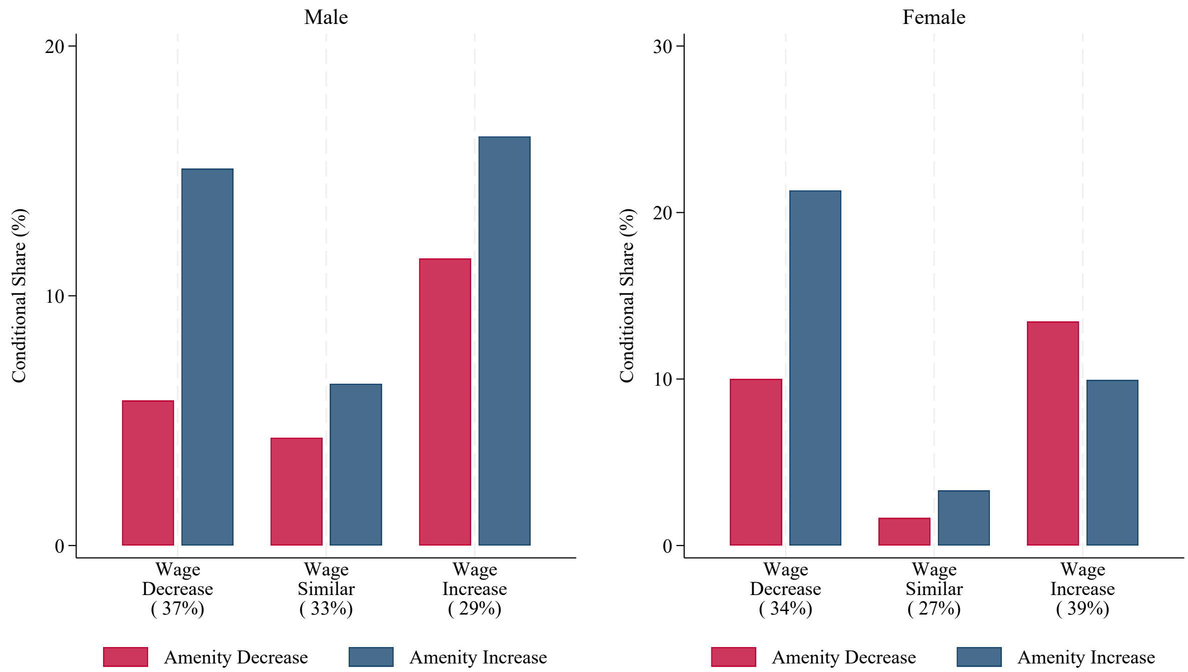
Note: This figure shows binned scatterplots of the relationship between firm wage premia (y-axis) and amenity values in log-wage equivalent units (x-axis) for firms in our connected set. We first predict amenity values estimated in one random half of the data using the values estimated in the other half and then create a binned scatterplot of the firm wage premia against these predicted values. Amenity values are converted to log-wage equivalents by dividing by the structural logit coefficient $\hat{\beta}$. The regression in Panel B is weighted by firm size. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix D.1.

Figure 6: Amenities, Wages, and Values by Sector



Note: This figure shows average amenity values in log-wage equivalent units (y-axis) and average wage premia (x-axis) by major sector. Amenity values are converted to log-wage equivalents by dividing by the structural logit coefficient $\hat{\beta}$. Within each sector we weight firms by firm size. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix Figure D4.

Figure 7: Classifying Job-to-Job Moves by Sex



Note: Each panel classifies job-to-job transitions by the direction of change in the firm wage premium (horizontal axis) and sex-specific amenity valuations. Bars show the conditional share of movers experiencing an amenity decrease (red) or increase (blue) within each wage-change category. Parenthetical values report the share of all movers in each wage category. The sample includes all full-time job-to-job movers between 2010 and 2019. Appendix Table A5 reports the full transition matrices.

A Appendix Tables and Figures

Table A1: Counts by Firm

		Connected Set	
	All	Both Waves	Wave 1
	(1)	(2)	(3)
1 Time	2861	251	228
2-5 Times	664	337	302
6-10 Times	210	197	173
11-14 Times	48	48	41
15-19 Times	29	29	23
20-50 Times	51	51	47
50+ Times	36	36	34
Total	3899	949	848

Note: This table tabulates the number of times each firm was provided by a worker and was therefore ranked against other outside firms.

Table A2: Reasons Workers Did Not Provide Firm Names

Know companies, but survey interface was too difficult	7%
Not comfortable sharing this information	8%
Could not think of specific companies	38%
No interest in switching	47%
Observations	829

Note: This table tabulates the reasons workers declined to provide the names of firms they would apply to if they wanted to switch firms. The sample includes workers who did not provide firms in the initial survey and did not provide three firms in the follow-up survey. All possible responses are listed.

Table A3: Randomization Assessment

Variable	p-value
Firm Wage Premium	0.405
Size	0.356
Median Wage	0.400
Mean Wage	0.435
Manufacturing Sector	0.910
Retail Sector	0.141
Professional Sector	0.816
Information Sector	0.144
Transport Sector	0.336
Finance Sector	0.524
HQ in East Germany	0.336
Year Incorporated	0.298
Total Assets/Employee	0.772
Fixed Assets/Employee	0.827
Indicator for Large Employer	0.663
Indicator for Popular Employer	0.871
Brand Recognition	0.322
Page Views (Kununu)	0.179
Pay Score (Kununu)	0.307
Image Score (Kununu)	0.508
Career Score (Kununu)	0.133
Atmosphere Score (Kununu)	0.197
Communication Score (Kununu)	0.506
Cohesion Score (Kununu)	0.132
Work-Life Balance (Kununu)	0.788
Supervisor Behavior (Kununu)	0.582
Interesting Tasks (Kununu)	0.066
Equality (Kununu)	0.558
Treatment of Older Workers (Kununu)	0.351

Note: We assess the randomization of raises to firms by regressing each firm characteristic on the randomized raise offered to that firm. The column presents the p-value from a test that the slope on the randomized raise is zero.

Table A4: Effect of Experimental Wage Difference on Pairwise Choice

	All	Male	Female	College	No College
	(1)	(2)	(3)	(4)	(5)
Panel A: Linear Probability Model					
Wage Difference	2.859***	2.888***	2.791***	2.913***	2.758***
	(0.107)	(0.129)	(0.190)	(0.132)	(0.181)
R-squared	0.062	0.063	0.060	0.065	0.058
p-value, Male = Female: 0.674					
p-value, College = No College: 0.488					
Panel B: Structural Logit					
Wage Difference	12.09***	12.22***	11.79***	12.34***	11.63***
	(0.50)	(0.60)	(0.89)	(0.62)	(0.84)
p-value, Male = Female: 0.687					
p-value, College = No College: 0.499					
Observations	20,740	14,485	6,255	13,550	7,190

Note: The dependent variable is a binary indicator for whether the worker prefers firm i over firm j. The regressor is the experimentally assigned difference in log wages between the two firms. Panel A reports OLS (linear probability) estimates; Panel B reports the coefficient on the wage difference from a structural logit, which we use as $\hat{\beta}$ in the utility function. Standard errors clustered at the worker level in parentheses. The p-values test whether the wage coefficient is equal across demographic groups using a pooled regression with an interaction term.

Table A5: Classifying Job-to-Job Moves: Sex-Specific Estimates of a_j

	Male				Female			
	Change in Amenities				Change in Amenities			
	Cut	Similar	Raise	Total	Cut	Similar	Raise	Total
A. Change in Own Wage								
Decrease ($< -2\%$)	0.8%	22.5%	1.4%	24.7%	0.9%	14.5%	2.4%	17.8%
Similar ($\pm 2\%$)	0.8%	36.8%	1.8%	39.5%	1.0%	26.7%	2.1%	29.8%
Increase ($> 2\%$)	2.6%	29.1%	4.1%	35.8%	4.9%	42.9%	4.5%	52.4%
N	2365				573			
B. Change in Firm Wage Premium								
Decrease ($< -2\%$)	2.2%	29.4%	5.6%	37.2%	3.4%	23.3%	7.2%	33.9%
Similar ($\pm 2\%$)	1.4%	29.7%	2.2%	33.3%	0.5%	26.0%	0.9%	27.4%
Increase ($> 2\%$)	3.4%	21.2%	4.8%	29.4%	5.2%	29.6%	3.8%	38.7%
N	1389				442			
C. Change in Log Median Wage								
Decrease ($< -2\%$)	2.2%	1.7%	2.6%	6.6%	3.5%	2.1%	5.0%	10.6%
Similar ($\pm 2\%$)	0.2%	85.4%	0.3%	85.9%	0.2%	80.4%	0.3%	80.9%
Increase ($> 2\%$)	1.8%	1.3%	4.4%	7.5%	3.3%	1.4%	3.8%	8.5%
N	2367				576			
D. Change in Log Mean Wage								
Decrease ($< -2\%$)	2.2%	1.8%	2.6%	6.6%	3.5%	2.1%	4.7%	10.2%
Similar ($\pm 2\%$)	0.3%	85.3%	0.4%	86.0%	0.0%	80.4%	0.7%	81.1%
Increase ($> 2\%$)	1.7%	1.3%	4.4%	7.4%	3.5%	1.4%	3.8%	8.7%
N	2367				576			

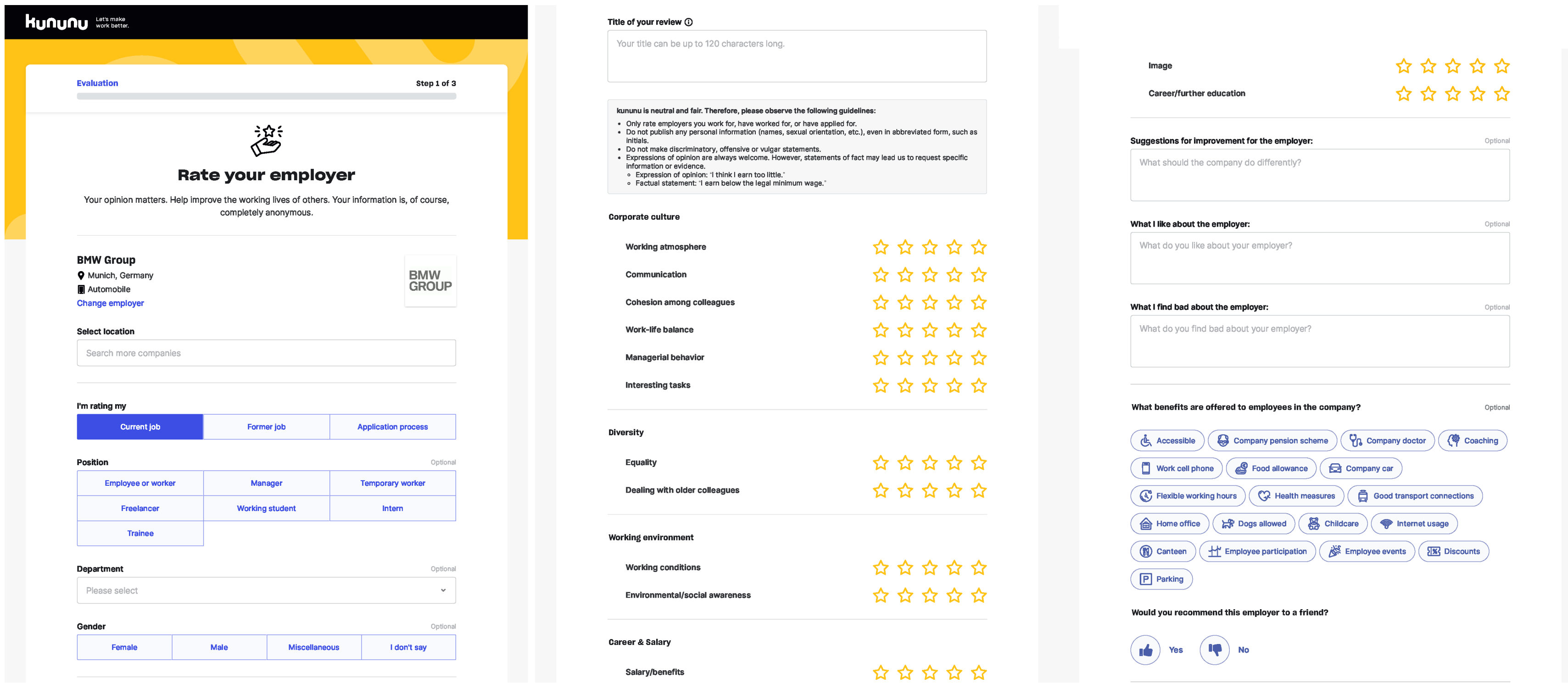
Note: This table reports the share of job-to-job transitions according to the change in wage and amenity values, separately by sex. The sample includes all full-time job movers between 2010 and 2019 who moved between firms for which we observe sex-specific amenity values. Within each panel, the first four columns report results for male workers and the next four for female workers. Panel A uses changes in workers' own wages, Panel B changes in the firm wage premium, Panel C changes in log median firm pay, and Panel D changes in log mean firm pay. Pooled (non-sex-specific) results for own wages and firm wage premia are in Table 5.

Table A6: Amenities and the Gender Value Gap: Robustness

	(1) Sector FE	(2) IPW Rewtd.	(3) No Growth Valuations	(4) Overall Mean Norm.	(5) Mfg. Norm.	(6) Median Norm.
	A. Firm Wage Premium (ψ_j)					
Female	-0.006	-0.024^{*}	-0.013^{**}	-0.013^{**}	-0.013^{**}	-0.013^{**}
	(0.004)	(0.013)	(0.005)	(0.005)	(0.005)	(0.005)
	B. Amenity Value ($a_j^g/\hat{\beta}$, log-wage equivalent)					
Female	0.011^{**}	0.008	0.040^{***}	0.003	0.006	0.004
	(0.005)	(0.009)	(0.010)	(0.006)	(0.006)	(0.006)
Mincer controls	Yes	Yes	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes

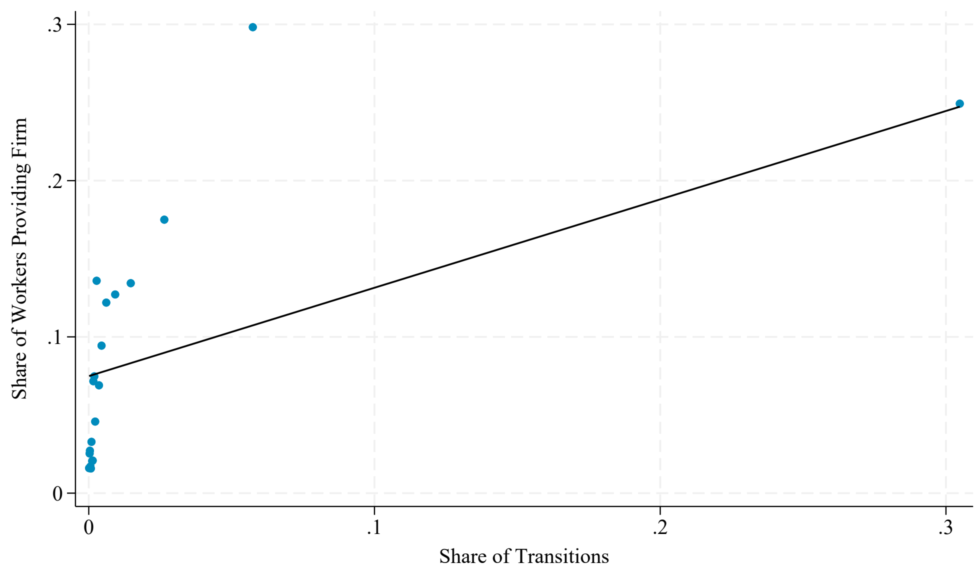
Note: This table examines the robustness of the baseline gender gap specification in Columns 3 and 6 of Table 6. All amenity values are in money-metric (log-wage equivalent) units. Column 1 adds sector fixed effects. Column 2 reweights observations using inverse probability weights to match the composition of the full 2% population sample, correcting for selection into the amenity estimation sample. Column 3 uses valuations elicited under the assumption of identical promotion and pay growth trajectories. Columns 4–6 vary the normalization anchor: overall employment-weighted mean, manufacturing sector, and median firm. All columns include Mincer controls (age, experience, education) and occupation fixed effects. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A1: Screenshots of Kununu's Review Interface



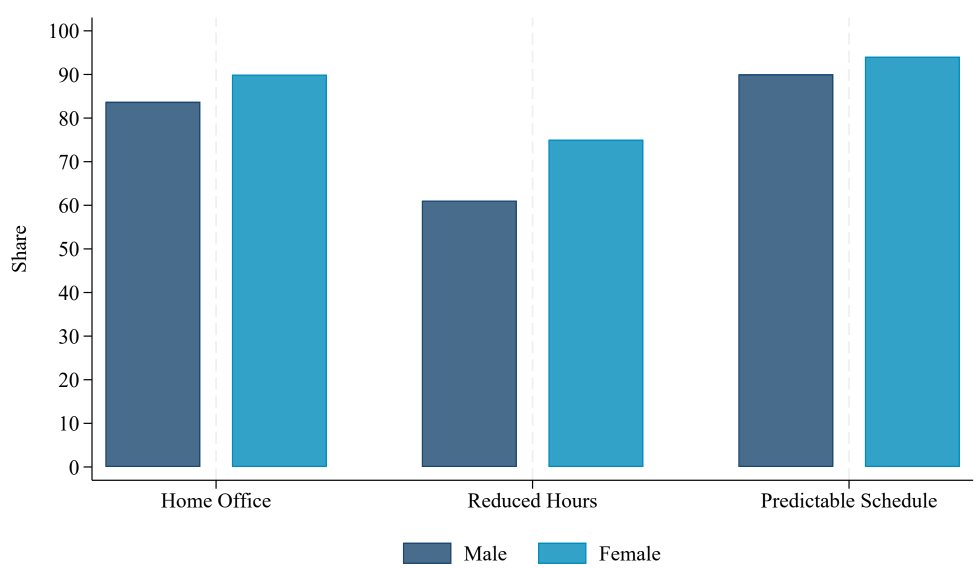
Note: This figure provides screenshots of the interface to submit reviews on Kununu, an online platform on which workers rate their current and former employers.

Figure A2: Relationship Between Stated and Revealed Destinations



Note: For each origin-destination pair we calculate the share of workers at the origin firm who move to a given destination firm. We then calculate the share of workers at this firm who name each destination firm as a place they would apply if they wanted to switch firms. We present a binned scatterplot of the share of workers providing each destination (y-axis) against the share moving to that firm (x-axis). The mobility data are from 2010–2019. The stated application data are from 2022 (initial survey) and 2024 (follow-up survey).

Figure A3: Gender Differences in Whether Specific Amenities Are Important



Note: This figure shows the percentage of male and female workers in our survey that rate specific amenities (remote work option, predictable hours, and reduced hours option) as "very important" or "extremely important" on a five-point Likert scale ranging from "not important" to "extremely important."

B Additional Data Sources

This section describes the additional data sources used in the paper.

B.1 IEB Social Security Records

We use the German Social Security records, which are assembled by the Institute for Employment Research (IAB) into the Integrated Employment Biographies (IEB) database, to select the sample of workers to whom we fielded the survey. These data capture all private-sector and public-sector employees with Social Security contributions. They contain information on employee demographics (e.g., gender, age, and education), employer information (e.g., sector and location), and job-spell-based information (e.g., start and end dates, full-time status, daily pay, and occupation). We impute daily pay for individuals whose pay is censored at the Social Security maximum.

We construct three primary samples:

- **Survey Sample:** The survey sample consists of workers in the "CHH Worker Survey" who provided the names of specific firms they would apply to if they wanted to switch firms. We pull employment histories for these workers. We use data from a single cross-section.
- **2% Sample:** We use a 2% sample of the IEB Social Security records. We use the subset of these workers who are in full-time, regular (i.e., not temporary) employment and who are between the ages of 25 and 50. These data cover the period 1975 to 2021; our primary analysis includes data from 2010 to 2019.
- **Movers Sample**: We use the employment histories of all workers in the IEB (100% sample) who switched firms between 2010 and 2019. We restrict to full-time, regular workers between 25 and 50. When we describe "job-to-job" movers, we use the subset of observations where workers moved directly from one firm to another. To identify job-to-job movers, we remove observations where the source is the Employment History (BeH)—a compilation of annual employment summaries required of firms by the government—and where the source of the previous observation is the Benefit Recipient History (LeH)—the IAB's database on unemployment assistance.

B.2 Linkage of Worker-Provided Firms to Administrative Data

To use data from the worker-provided firm module, we had to clean the firm names that workers provided. In the initial survey, respondents (overall, including those dropped from our main analysis sample) provided 5,828 unique strings. Of these, we could assign 5,016 (86%) to uniquely identifiable firms. We performed this assignment manually through online searches to ensure that the strings matched a unique firm.

Of the 812 strings that could not be uniquely identified, the majority are unspecific firm types (e.g., "City government", "police"); a small share (10%) represent non-response (e.g., "I do not want to answer") or unidentifiable content (e.g., "XXX"). Our manual assignment procedure, which corrects for differences in spelling and pools subdivisions of the same firm, assigns the 5,016 identifiable strings to 3,018 unique firms. Of these 3,018 firms, 2,013 can be matched to establishments in the IAB records. Among the firms that we cannot link to the IAB, 29% are foreign entities and 37% are in the public sector; these firms may not have any workers in Social Security-covered employment in Germany. We link 79% of firms named at least twice to the IAB records and obtain the municipality of headquarters for 99% of linked firms.

B.3 Imputation of Firm Size

For a subset of firms provided by workers, the number of full-time employees is not directly observable in the administrative data (BHP). Because our main results use employment weights to reflect the importance of firms in the aggregate labor market, excluding these firms would potentially bias our estimates. To address this, we perform a stochastic imputation of firm size using a nested hierarchy of Poisson Pseudo-Maximum Likelihood (PPML) regressions. Poisson models are suited for count-type data and ensure positive predicted sizes.

Methodology. We use a range of firm-level predictors from our administrative and survey data. Let S_j be the observed firm size for firm j and Z_j be a vector of candidate predictors. We estimate specifications of the following form:

$$\mathbb{E}[S_j|Z_j] = \exp(Z_j\gamma)$$

where the exponential form is estimated via PPML to account for the count-based nature of firm size.

For firms with missing size data, we predict the latent firm size $\hat{\mu}_j$ and apply stochastic imputation to preserve the variance of the size distribution:

$$\text{Firm Size Imputed}_j = \hat{\mu}_j + \eta_j, \quad \eta_j \sim N(0, \hat{\sigma}^2)$$

where $\hat{\sigma}$ is the root mean square error (RMSE) from the corresponding regression. This approach ensures that we do not artificially deflate the variance of the firm-size distribution in the imputed sample. We do not truncate the imputed values at zero; in practice, the predicted mean $\hat{\mu}_j$ is large enough relative to $\hat{\sigma}$ that negative draws are extremely rare. Our main results weight by actual firm size from the BHP; imputed sizes appear only in robustness exercises that include firms not linked to the administrative data.

Predictors and Model Hierarchy. We use the first available prediction from the following hierarchy:

1. Employees in Germany, global employees, survey mention frequency, and total Kununu reviews
2. Employees in Germany, global employees, and survey mention frequency
3. Employees in Germany, survey mention frequency, and total Kununu reviews
4. Survey mention frequency and total Kununu reviews
5. Employees in Germany, global employees, and total Kununu reviews
6. Survey mention frequency only
7. Employees in Germany only
8. Global employees only

The Kununu variables are only available for firms linked to Kununu. The number of global employees comes from Orbis. Following imputation, we replace the imputed value with the true administrative size whenever the latter is non-missing.

B.4 Estimates of AKM Wage Premia

Most of our analysis uses the AKM wage premia from Bellmann et al. (2020), which are computed from the full sample of workers, have been used by a variety of other researchers, and are provided directly by the IAB. However, in some specifications we use our own estimates so that we can use a split-sample approach to account for noise; this allows us to de-bias estimates of the variance and covariance.

Data Extraction and Initial Processing. From the population of Social Security records we construct a sample that consists of all full-time workers who switched firms between 2010 and 2019. For these workers we have data on wages before and after a transition.

Because wages are top-coded at the Social Security maximum, we impute wages above this threshold. We use this dataset to estimate separate Tobit regressions by year, sex, education, and age groups including as controls the following: age, mean log wage in other years, fraction of censored wages in other years, number of full-time employees at the current establishment and its square, dummy for 11 or more employees, mean years of schooling and fraction of university graduates at the current establishment, leave-one-out mean log coworker wage, the fraction of coworkers with censored wages, dummy for individuals observed only one year, dummy for employees of one-worker establishments, and an East Germany dummy. We set the mean log wage and fraction of censored wages in other years to the respective sample means for workers that are only observed in one year. We set the mean log wage and fraction of censored wages to the respective sample means for workers at firms with only one full-time employee. There are five education groups: missing, no qualification, apprenticeship, some post-secondary, and university graduate. We group workers into ten-year age ranges: 20–29, 30–39, 40–49, and 50–60.

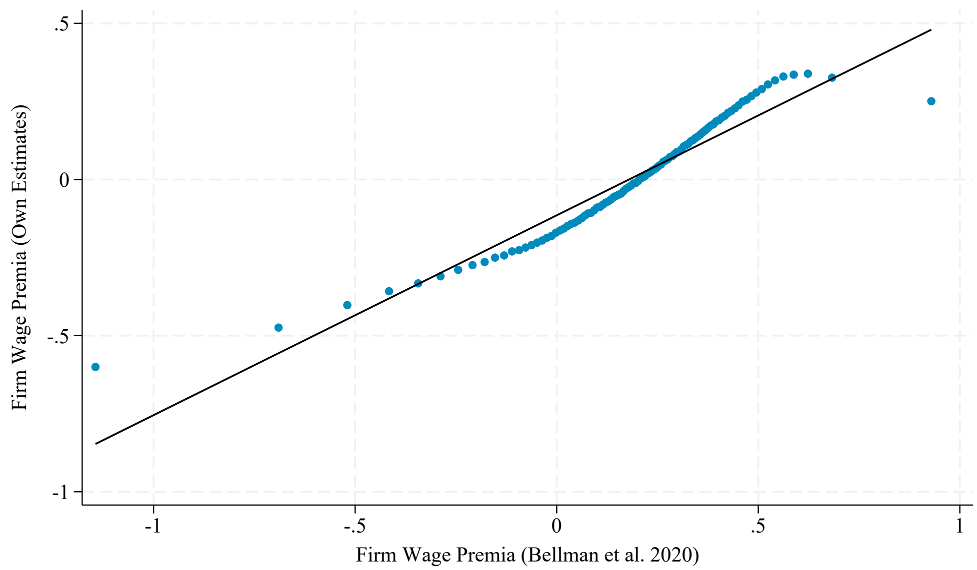
Estimation. We then fit the following model

$$y_{it} = \alpha_i + \psi_{j(i,t)} + x_{it}'\gamma + \varepsilon_{it} \tag{4}$$

where $j(i,t)$ denotes the employer of worker i in period t. The x_{it} include an unrestricted set of year dummies as well as quadratic and cubic terms in age fully interacted with educational attainment. We link the establishment-level wage premia back to the Social Security data. To obtain firm-level estimates we take the employment-weighted average across all establishments associated with a firm.

Comparison with Other Estimates. Figure B1 compares our firm effect estimates to those of Bellmann et al. (2020) in a binned scatterplot; the two sets of estimates are strongly correlated. However, there is substantial measurement error in the estimates of observed firm wage premia. A split-sample exercise reveals that the correlation between estimates from two samples is only 0.45, suggesting substantial noise.

Figure B1: Comparison of Firm Wage Premia



Note: This figure is a binned scatterplot that compares our estimates of firm-specific wage premia to those produced by Bellmann et al. (2020). The two sets of estimates use different reference firms for normalization, so the intercept is not zero. The slope of the regression line is 0.88.

B.5 Orbis

We collect additional information on the provided firms from the Orbis database, which is compiled by the Bureau van Dijk based on firm balance sheet information. The collected firm characteristics include year of incorporation, sector based on the 4-digit NACE industry code, whether the firm is headquartered in East Germany based on the zip code of its headquarters, and the number of employees.[24] To measure firm productivity, we also collect information on firms' fixed assets and total assets. Fixed assets refer to the total amount (after depreciation) of non-current assets (intangible assets, tangible assets, other fixed assets) and thus represent long-term assets that are unlikely to be converted into cash in the near term. Total assets are the sum of fixed assets and current assets (e.g., cash and any assets that will be converted into cash within the year).

For each variable from Orbis, we use the most recent available year. For over 90% of our firms, the most recent information is not older than three years. For fixed and total assets, we CPI-adjust our variables. To find the worker-provided firms in Orbis, we manually match each firm by name and address. We are able to match 79% of worker-provided firms that were named at least twice in the initial survey; we did not attempt this match for firms that were named only once.

[24]Since Orbis draws on firms' balance sheet information, the number of employees may include employees outside of Germany. We therefore also hand-collect information on the number of employees firms have in Germany from the firms' websites.

B.6 Kununu

To better understand the types of amenities offered by different employers, we use data from Kununu, a leading employer rating platform in Germany. Kununu allows current and former employees to rate their workplace across multiple dimensions and to indicate whether their employer offers a specific set of amenities. These data provide an externally validated, worker-reported source of firm characteristics that complement our survey and administrative data. Because the information is public and generated by employees, it captures workers' perceptions of actual workplace offerings, rather than relying on employer-provided descriptions that may be shaped by marketing or compliance incentives.

B.6.1 Data Collection and Linkages

This section describes the firm-level information we collected from Kununu. We linked firms in the Kununu data to the firms provided by workers based on firm name. Among the worker-provided firms that were named at least twice in the initial survey, this is true for 88% of firms. As of February 2023 (when we collected the Kununu data), Kununu contained reviews for almost 200,000 German firms. In total, these firms received over 2.5 million ratings and had around 1.5 billion page views.

Kununu provides information on the total number of reviews a company received and the share of reviewers who would recommend the firm to a friend. Each company has an overall rating as well as ratings in different categories, such as pay or company culture. Ratings range from one to five, with higher numbers representing better ratings. When submitting these ratings, reviewers are prompted with the full list of subcategories and can choose between one to five stars for each.

Appendix Table B1 describes the variables we collect from Kununu. To measure how well-known provided firms are, we use the number of page views and number of reviews of each firm. As a measure of firm popularity, we use the share of reviewers who would recommend a firm as an employer to a friend and the overall rating of the firm. Recommendations are not provided for all firms on Kununu. To further characterize provided firms (e.g., high- versus low-paying), we use the numerical ratings of the different subcategories.

B.6.2 Accuracy of Information

Several features of the Kununu data support their validity. First, there is variation in the provided information: only 6.4% of firms receive a maximum rating of 5 out of 5. Second, firm-specific measures are internally consistent: a firm's overall rating and the share of workers who would recommend the firm as an employer are highly correlated (Panel B of Appendix Figure B2).

B.6.3 Variation in Employer-Provided Amenities

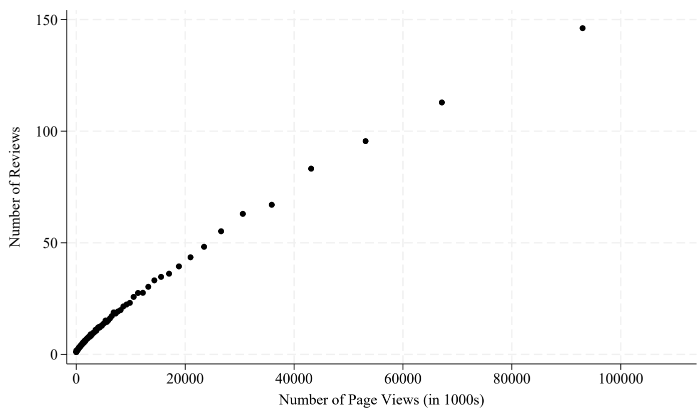
The platform includes indicators for a wide range of specific amenities, such as flexible working hours, remote work options, childcare services, transportation and food benefits, coaching opportunities, and participation in decision-making. Figure B3 reports the average share of reviewers who indicate each amenity, weighted by the number of reviews per firm. The most commonly reported amenities are flexible working hours, parking, employee events, and retirement benefits.

Table B1: Variables Collected from Kununu

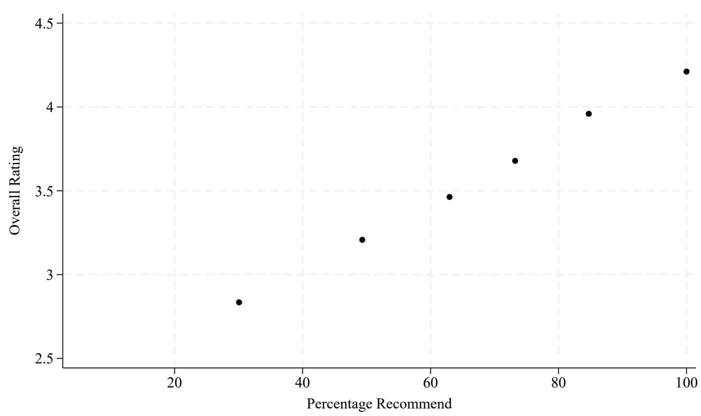
Note: This table presents information on the variables we collected from Kununu on all German firms that are covered by Kununu as of February 2023.

Figure B2: Assessment of Internal Consistency

Panel A. Page Views (in 1000s) and Reviews



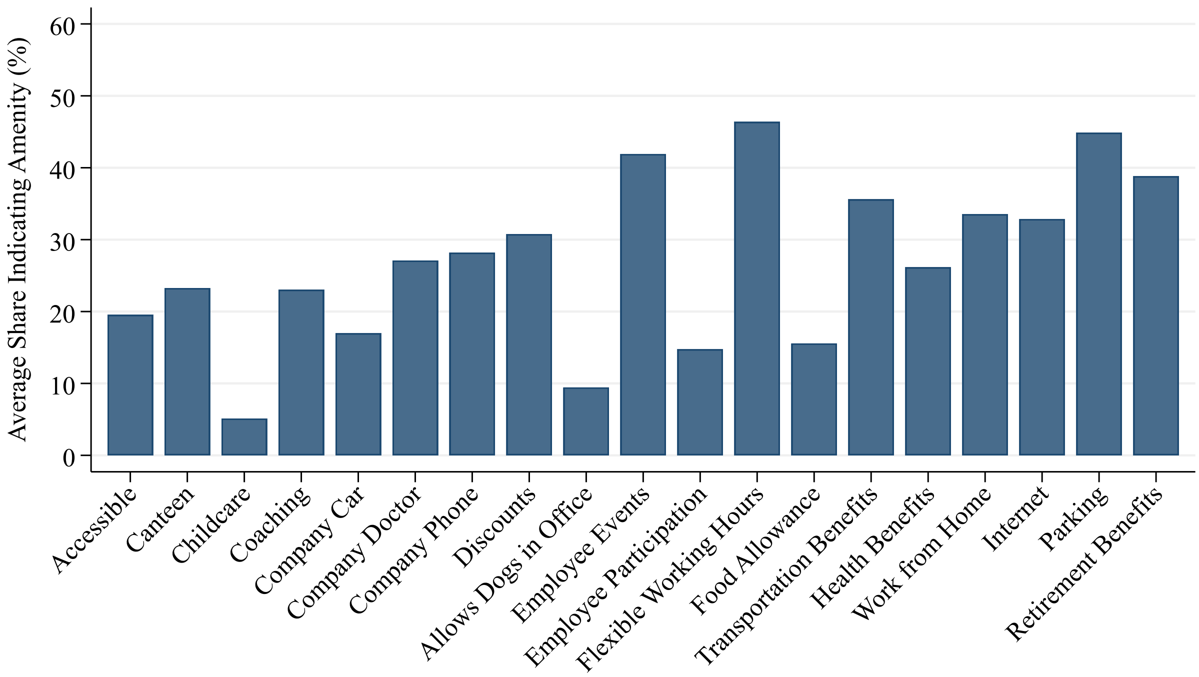
Panel B. Overall Rating and Percent Recommend



Note: This figure examines the within-Kununu consistency of the measures we collect. Panel A presents a binned scatterplot of a firm's number of page views (in 1000s) against the number of reviews it receives. Firms in the top 1% of page views are dropped to make more points visible. Panel B presents a binned scatterplot of the overall Kununu score and the percent of reviewers that would recommend the firm as an employer. Each figure includes firms that were mentioned at least twice by workers in the survey. The binned scatterplots are created using the program provided by Cattaneo et al. (2024).

Other amenities—such as access to a company doctor, food allowances, and coaching—are less common but still present in a nontrivial share of firms.[25]

Figure B3: Averages of Kununu Amenity Responses, Weighted by Number of Reviews



Note: Kununu reports the percentage of reviewers at each firm who indicate the firm offers each amenity. Each bar gives the mean percentage for the indicated amenity, weighted by the number of employee reviews of each firm.

Figure B4 illustrates the share of employees at each firm who report that a given amenity is offered by their employer. We focus on four of the amenities included in the previous graph: flexible working hours, work from home, childcare, and coaching. The distributions show mass in intermediate ranges—not concentrated at 0% or 100%. For many firms, employees disagree about whether an amenity is offered, reflecting differences in access or perception.

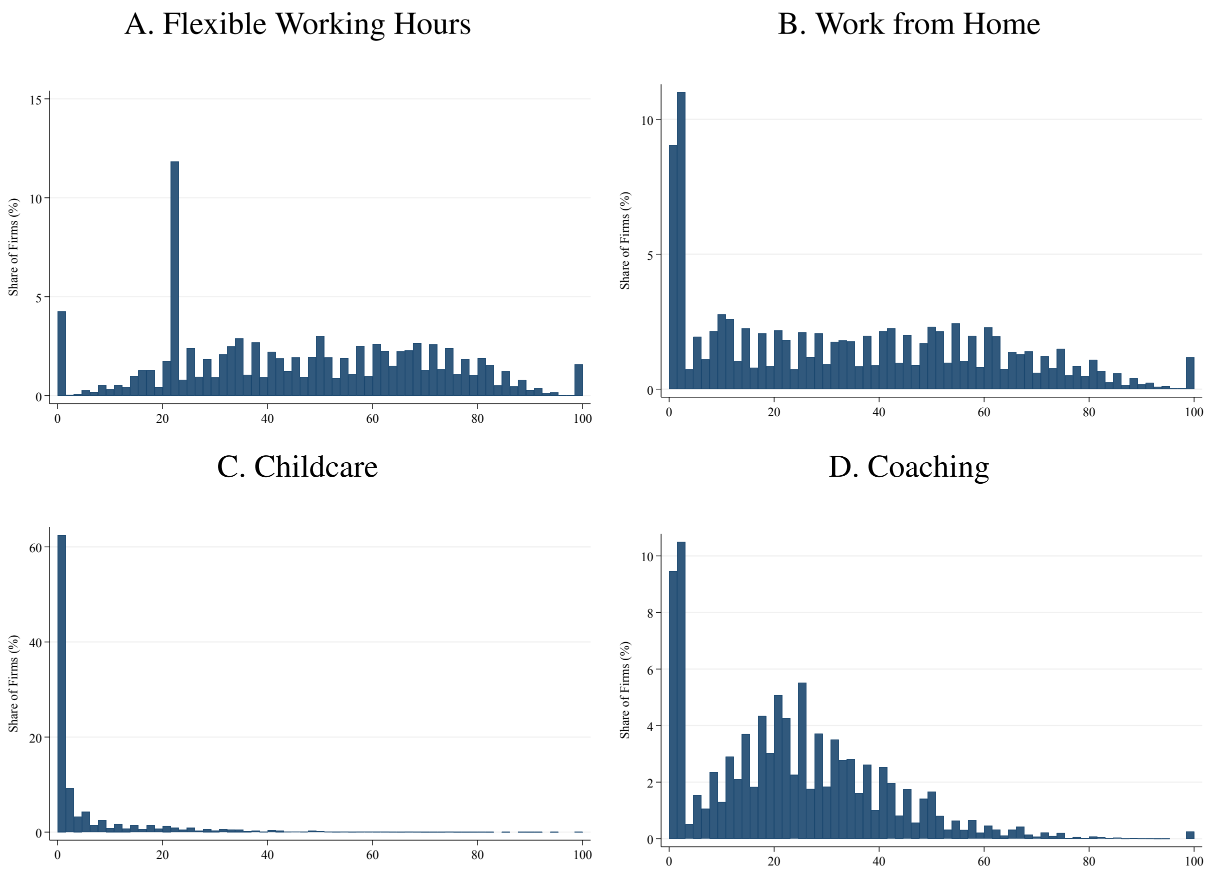
For example, Panel C shows the distribution of childcare availability. The vast majority of firms have near-zero confirmation rates, consistent with childcare being a relatively rare benefit. This pattern likely reflects limited provision rather than disagreement. Panel D shows a similar pattern for coaching: few firms have unanimous endorsement, but many have partial confirmation across employees.

Even for more common amenities such as flexible working hours and home office options (Panels A and B), we observe a wide spread of confirmation rates across firms. While many firms

[25]In cases where no employees indicated any amenities in their reviews, it is unclear whether the firm truly has none of the amenities or the employees skipped the amenities question despite the firm offering amenities. Because most employees indicate at least one amenity, we assume nonresponse to the amenities question means zero amenities if a firm has 3 or more reviews. Firms with fewer reviews are excluded if no employee responded to the amenities question.

have a majority of employees reporting availability, relatively few reach full consensus. These patterns suggest that Kununu data capture real within-firm variation in the perceived or experienced availability of amenities.

Figure B4: Distribution of Kununu Amenity Responses



Note: Each panel reports the distribution of the amenity indicated. Each bar reports the percent of firms for which the percent of reviewers on the x-axis indicated the firm offers the benefit, weighted by the number of employee reviews of each firm.

C Theoretical Appendix

This appendix describes the estimation strategy and shows that it recovers amenity values.

C.1 Setup

The value worker i obtains from working at firm j is:

$$u_{ij} = \beta \log w_{ij} + a_j + \epsilon_{ij} \tag{5}$$

where β is the marginal utility of log wages, w_{ij} is the individual's wage if they work at firm j, a_j is the value of the firm's amenities (in utility units), and ϵ_{ij} is worker i's idiosyncratic preference for firm j. We assume that the ϵ_{ij} are drawn iid from a type I extreme value distribution.

Our discrete choice experiments specify proportional changes in the wage, relative to the worker's current wage ω_i. They specify that the log wage worker i would earn at firm j is:

$$\log w_{ij} = \omega_i + r_{ij},$$

where r_{ij} is the randomized raise associated with firm j and ω_i is the worker's current wage.

Given the offered wages, worker i prefers k to firm j if

$$\begin{aligned} \beta r_{ij} + a_j + \epsilon_{ij} &< \beta r_{ik} + a_k + \epsilon_{ik} & (6)\\ \epsilon_{ij} - \epsilon_{ik} &< \beta(r_{ik} - r_{ij}) + (a_k - a_j). & (7) \end{aligned}$$

We define the probability a worker prefers firm k to j as:

$$\begin{aligned} P_{kj} &= \Pr[\epsilon_{ij} - \epsilon_{ik} < \beta(r_{ik} - r_{ij}) + (a_k - a_j)] & (8)\\ &= \mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ik} + a_k)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right], & (9) \end{aligned}$$

where the second line comes from the distributional assumptions on ϵ_{ij}. In a stationary equilibrium, the flow of workers from firm k to firm j must equal the reverse flow. Letting x_k denote the mass of workers at firm k, this detailed balance condition requires

$$x_k \, P_{jk} = x_j \, P_{kj} \quad \forall\, j, k.$$

Rearranging, we find the solutions $\{x_k\}$ to the linear problem

$$x_k = \frac{\sum_j P_{kj} x_j}{\sum_j P_{jk}}, \tag{10}$$

which holds for all j when

$$\mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ij} + a_j)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right] x_k = \mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ik} + a_k)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right] x_j. \tag{11}$$

Intuitively, the share of workers who prefer j to k (relative to the share of workers who prefer k to

j) is larger whenever the gap in utilities is larger. We obtain these $\{x_j\}$ by finding the fixed point of the above linear program.

C.2 Randomization Allows Direct Identification of a_j

The rankings implied by our estimation identify amenity values (not overall values as in Sorkin, 2018).

We observe workers' choices based on comparisons of $\beta r_{ij} + a_j$, not a_j. Let

$$x_k \propto \exp(a_k + \xi_k) \tag{12}$$

where ξ_k is a term that captures potential deviations in what we estimate from a_k. Then

$$\mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ij} + a_j)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right]\exp(a_k + \xi_k) \tag{13}$$

$$= \mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ik} + a_k)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right]\exp(a_j + \xi_j)$$
$$\Longrightarrow$$

$$\mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ij})}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right]\exp(\xi_k)$$
$$= \mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ik})}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)}\right]\exp(\xi_j)$$

$$\frac{\exp(\xi_k)}{\exp(\xi_j)} = \frac{\mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ik})}{\exp(\beta r_{ik}+a_k)+\exp(\beta r_{ij}+a_j)}\right]}{\mathbb{E}_{j,k}\left[\frac{\exp(\beta r_{ij})}{\exp(\beta r_{ik}+a_k)+\exp(\beta r_{ij}+a_j)}\right]}$$

The first-order expansion of $\mathbb{E}[X/Y]$ around $\mathbb{E}[X]$ and $\mathbb{E}[Y]$ is $\mathbb{E}[\frac{X}{Y}] \approx \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$. This approximation is valid when the variance of the randomized raises is small relative to differences in amenity values, which holds in our experimental design. Under this approximation,

$$\frac{\exp(\xi_k)}{\exp(\xi_j)} \approx \frac{\mathbb{E}_k\left[\exp(\beta r_{ik})\right]}{\mathbb{E}_j\left[\exp(\beta r_{ij})\right]} = 1 \tag{14}$$

The final equality holds because r_{ik} and r_{ij} are drawn from the same distribution—by design—so $\mathbb{E}[\exp(\beta r_{ik})] = \mathbb{E}[\exp(\beta r_{ij})]$ for any β. Therefore

$$x_k \propto \exp(a_k). \tag{15}$$

The eigenvector recovers a_k in utility units regardless of β.

While this proof relies on the delta method approximation, unreported simulations confirm that the randomization of raises yields accurate estimates of $\{a_j\}$ across the range of wage values used in our surveys.

C.3 Obtaining Money-Metric Values

Given that

$$x_k \propto \exp(a_k), \tag{16}$$

$c \cdot x_k = \exp(a_k)$ for some constant c. This implies

$$\log(x_k) = a_k - \log(c). \tag{17}$$

The log of the eigenvector entries recovers a_k in the units of the utility function. The money-metric (log-wage equivalent) amenity value is a_k/β. We estimate β from the structural logit coefficient on the experimental wage difference (Table A4) and use it to convert to money-metric units in Section 5.1.

However, there is no way to recover the subtracted term from discrete choice experiments: values are only identified relative to a reference firm. To see this, note that workers' decision rule satisfies

$$\beta(r_{ij} + \log w_i) + a_j + \epsilon_{ij} < \beta(r_{ik} + \log w_i) + a_k + \epsilon_{ik}. \tag{18}$$

Observe that all decisions will be the same as those implied by the decision rule

$$\beta(r_{ij} + \log w_i) + a_j - \log(c) + \epsilon_{ij} < \beta(r_{ik} + \log w_i) + a_k - \log(c) + \epsilon_{ik}$$

for any value of c. We therefore normalize to a base firm selected to have amenity estimates near the median, held constant across models. This ensures that amenity values are measured in the same wage-equivalent units across groups (given equal β), but cross-group comparisons of levels still depend on the normalization assumption. For example, in sex-specific models, the estimated gender gap for firm k reflects the difference $(a_k^F - a_k^M) - (a_{\text{base}}^F - a_{\text{base}}^M)$ where the latter term is not identified.

C.4 Normalization in Cross-Group Comparisons

The Identification Problem. The discrete choice experiment identifies group-specific amenity valuations $\hat{a}_k^g$ only up to an additive constant within each group $g \in \{M, F\}$:

$$\hat{a}_k^g = a_k^g - c^g, \tag{19}$$

where a_k^g is the true amenity value and c^g is a group-specific constant that pairwise comparisons cannot identify.

1. *Within-group objects are fully identified.* $\hat{a}_k^g - \hat{a}_\ell^g = a_k^g - a_\ell^g$ for any two firms k and ℓ; the constant cancels. Rank orderings, within-group variances, and regressions of $\hat{a}_k^g$ on firm-level covariates are normalization-free.
2. *Cross-group level comparisons require a normalization.* The estimated gender gap at firm k is
$$\hat{a}_k^F - \hat{a}_k^M = \left(a_k^F - a_k^M\right) - \left(c^F - c^M\right). \tag{20}$$
Any claim about the *level* of women's amenities relative to men's therefore depends on the chosen value of $c^F - c^M$.

Normalization Strategies. A normalization pins down c^g. We consider five alternatives:

1. **Services-sector anchor (baseline).** Set the employment-weighted mean of $\hat{a}_k^g$ among professional services firms to zero within each group:

$$c^g = \bar{a}^g_{\text{serv}} \equiv \sum_{j \in \text{serv}} s_j \, a_j^g,$$

   where s_j is firm j's employment share within services. This assumes $\bar{a}^F_{\text{serv}} = \bar{a}^M_{\text{serv}}$: men and women derive, on average, the same amenity value from services firms.
2. **Overall employment-weighted mean.** Set the employment-weighted mean over *all* firms to zero within each group: $c^g = \bar{a}^g_{\text{all}}$. This assumes men and women derive the same average amenity value across the full economy.
3. **Manufacturing anchor.** Set the employment-weighted mean in manufacturing to zero: $c^g = \bar{a}^g_{\text{manuf}}$.
4. **Median-firm anchor.** Set the amenity value of the median firm (by employment) to zero within each group.
5. **Overlap-set restriction.** Restrict the sample to firms evaluated by at least one worker of each sex, retaining the services-sector normalization from alternative 1. This is a sample restriction rather than a separate normalization; it tests whether the results are driven by firms evaluated by only one sex.

Direction of Bias under the Services Anchor. The services anchor assumes $\bar{a}^F_{\text{serv}} = \bar{a}^M_{\text{serv}}$. If this assumption is violated, substituting $c^g = \bar{a}^g_{\text{serv}}$ into equation (20) gives

$$\hat{a}_k^F - \hat{a}_k^M = \left(a_k^F - a_k^M\right) - \left(\bar{a}^F_{\text{serv}} - \bar{a}^M_{\text{serv}}\right). \tag{21}$$

Within-sex gradients (e.g., the relationship between a woman's wage premium and her amenity valuation) are normalization-free: the constant c^g cancels within a single group.

Cross-sex level comparisons—including the coefficient on a Female indicator in a regression of $\hat{a}_j^{g(i)}$ on worker demographics—depend on the normalization. The Female coefficient captures

$$\theta_1 = \left(\bar{a}^F_{\text{women's firms}} - \bar{a}^F_{\text{serv}}\right) - \left(\bar{a}^M_{\text{men's firms}} - \bar{a}^M_{\text{serv}}\right),$$

which shifts by exactly $-(\bar{a}^F_{\text{serv}} - \bar{a}^M_{\text{serv}})$ under the services anchor. The positive coefficients in Panel B of Table 6 therefore depend on the normalization; we verify their robustness across alternatives in Table A6.

Suppose services firms are truly more attractive to women than to men on non-wage dimensions, so that $\bar{a}^F_{\text{serv}} > \bar{a}^M_{\text{serv}}$. Then $\bar{a}^F_{\text{serv}} - \bar{a}^M_{\text{serv}} > 0$, and equation (21) implies

$$\hat{a}_k^F - \hat{a}_k^M = \left(a_k^F - a_k^M\right) - \left|\bar{a}^F_{\text{serv}} - \bar{a}^M_{\text{serv}}\right|. \tag{22}$$

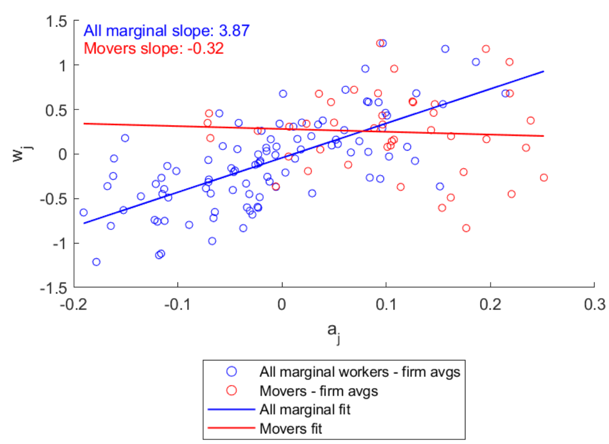
The estimated gender gap at every firm is shifted *downward*: anchoring to a sector where women may derive higher amenity value sets a higher floor for women's normalized values, deflating their amenity estimates relative to men's across all other sectors.

C.5 Estimating Valuations Based on Movers

We field the survey to a broad sample of workers rather than to job-to-job movers because the theoretical literature shows that—even when the correlation between wages and amenities is positive—focusing on movers can generate spurious negative correlations (Bonhomme and Jolivet, 2009).

In simulations that assume noiseless observation of workers' firm-specific valuations, this bias is large.[26] In reasonable specifications this flipped the sign of the correlation from (by assumption) large and positive to (when focusing only on movers) large and negative. Figure C1 illustrates the bias for a representative parameterization: across all firms (blue), the slope of the firm wage premium on the firm amenity value is strongly positive; when the same data are restricted to realized job-to-job movers (red), the slope is slightly negative. The results in Section 5.2 confirm that many workers make trade-offs between wages and amenities when switching firms.

Figure C1: Bias from Estimating Wage–Amenity Correlations Using Movers Only



Note: This figure presents results from a simulation in which firms differ in both wage premia w_j and amenity values a_j, with a positive population-level correlation between the two. Each circle is a firm. Blue circles plot the firm-level pair (a_j, w_j) for the population of marginal workers; red circles plot the same pair after restricting to firms reached through realized job-to-job moves. The blue line is the OLS fit across all marginal workers; the red line is the OLS fit among movers. Conditioning on movers flips the sign of the slope, illustrating the selection mechanism formalized in Bonhomme and Jolivet (2009).

[26] The assumption that we can observe point estimates of workers' firm-specific valuations without error is strong. Stated valuation approaches are known to yield estimates that are off by a factor of ten (Rodemeier, 2023). Validated approaches, such as discrete choice experiments, do not yield point estimates for each worker (see, e.g., Mas and Pallais, 2017; Wiswall and Zafar, 2018).

D Additional Results and Analysis

D.1 Robustness to Alternative Weighting Schemes

This section examines the robustness of our baseline results to alternative weighting schemes. The results in the main text weight by firm size (observed, not imputed). Here we also present specifications that weight by imputed firm size (which allows us to include firms not linked to the BHP establishment database; see Appendix B.3 for details on the imputation procedure) and by the number of times a firm is mentioned in the survey; weighting observations by their sample size is a common approach to dealing with measurement error.

D.1.1 Descriptive Statistics Reported in Section 2

Table D1: Characteristics of Provided Firms

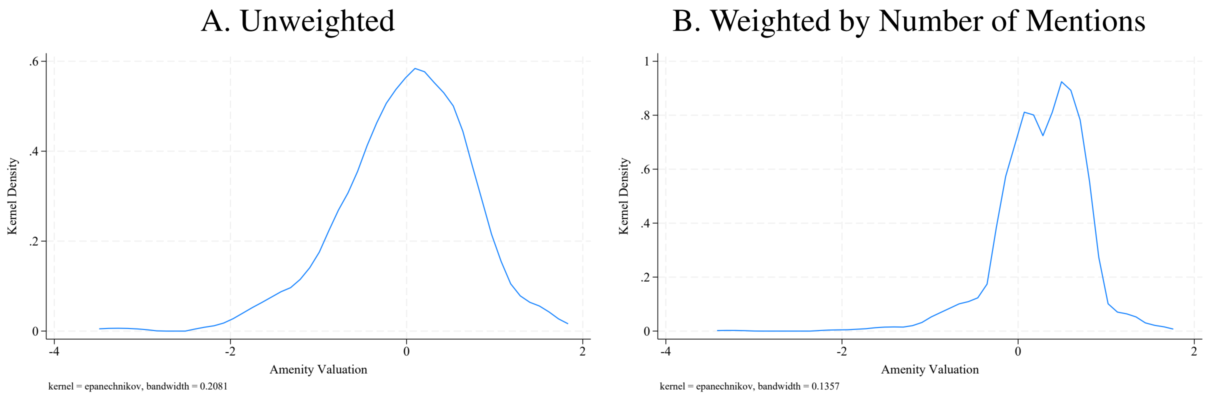
	All Firms				Connected Set			
	Unweighted	Firm Size	Mentions	Firm Size (Imputed)	Unweighted	Firm Size	Mentions	Firm Size (Imputed)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Linkages								
To Wage Premia	95%	100%	100%	100%	97%	100%	100%	100%
To BHP	97%	100%	100%	100%	98%	100%	100%	100%
Size	29682	149557	76253	149565	34701	149891	76316	149891
(Std. Deviation)	(59653)	(93276)	(78671)	(93273)	(63227)	(93118)	(78674)	(93118)
Size (Imputed)	22733	149565	69887	140018	28385	149891	69987	140730
(Std. Deviation)	(51637)	(93273)	(75381)	(94320)	(56473)	(93118)	(75390)	(94036)
Size (Binned)								
1-10 Workers	3%	0%	0%	0%	2%	0%	0%	0%
11-50 Workers	4%	0%	1%	0%	3%	0%	1%	0%
51-200 Workers	5%	0%	0%	0%	3%	0%	0%	0%
201-1000 Workers	33%	0%	2%	1%	26%	0%	2%	1%
1001-10000 Workers	20%	3%	8%	3%	24%	3%	8%	3%
>10000 Workers	32%	96%	88%	96%	42%	97%	88%	96%
Firm Wage Premium	0.50	0.48	0.51	0.48	0.51	0.48	0.51	0.48
(Std. Deviation)	0.11	0.07	0.08	0.07	0.09	0.06	0.08	0.06
Firm Wage Premium (CHH)	0.16	0.06	0.15	0.06	0.16	0.05	0.15	0.05
(Std. Deviation)	0.22	0.16	0.16	0.16	0.21	0.16	0.16	0.16
Sector								
Manufacturing	40%	44%	58%	43%	42%	44%	58%	43%
Retail	9%	0%	2%	1%	8%	0%	2%	1%
Professional	24%	53%	29%	52%	25%	53%	29%	52%
Information	10%	1%	7%	2%	10%	1%	7%	2%
Finance	6%	0%	1%	0%	5%	0%	1%	0%
Number of Firms	3899	1941	3587	3899	949	603	949	949

Note: This table describes the firms that workers provided (Columns 1–4) and the subset of these firms in the connected set (remaining columns). Each column indicates how the statistics are weighted. Appendix B.3 describes how we imputed firm size.

D.1.2 Results Reported in Section 4

In Section 4 we describe the distribution of amenity values, as well as how these amenity values agree (or do not agree) across groups. Here we show that these core findings are invariant to weighting scheme. In particular, amenity valuations are highly dispersed; there is disagreement between groups in how to value firms; and despite this, the single-index model has predictive power. We replicate Figure 2 (distribution of amenity values), Table 3 (between-group correlations), and Figure 3 (relationship between valuations with and without wage growth). We do not replicate the log firm size regressions in Table 4, as those are naturally unweighted. The Kununu regressions in that table are weighted by firm size and could in principle be replicated; we omit them for brevity. Figure 4 (relationship between male and female valuations) is visually similar under all weighting schemes.

Figure D1: Distribution of Amenity Values: Alternative Weighting Schemes



Note: This figure replicates Figure 2 (amenity values in utility units) using alternative weighting schemes. Panel A is unweighted; Panel B weights by the number of times the firm is mentioned by workers in the survey. The baseline figure weights by firm size.

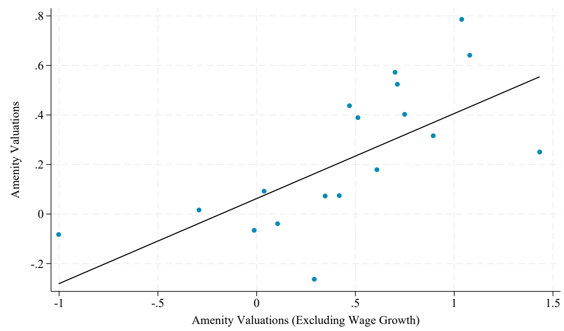
Table D2: Correlation Between Valuations of Different Demographic Groups: Weighting by Number of Times Mentioned

	Female (1)	Male (2)	College Degree (3)	No College Degree (4)
Female	1.000	0.526	0.696	0.504
Male	0.360	1.000	0.868	0.665
College Degree	0.624	0.819	1.000	0.469
No College Degree	0.478	0.537	0.341	1.000

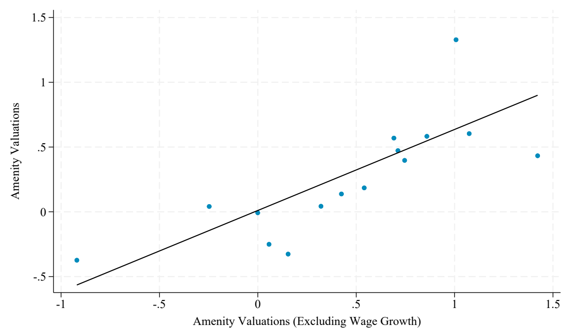
Note: This table examines the pairwise correlation in valuations of amenities for different demographic groups. The numbers above the diagonal (shaded in grey) are rank correlations; the numbers below the diagonal are raw correlations. Firms are weighted by the number of times they are mentioned by workers in the survey. The baseline estimates are in Table 3.

Figure D2: Relationship Between Valuations with and without Wage Growth: Alternative Weighting Schemes

A. Weighted by Number of Mentions



B. Weighted by Imputed Firm Size

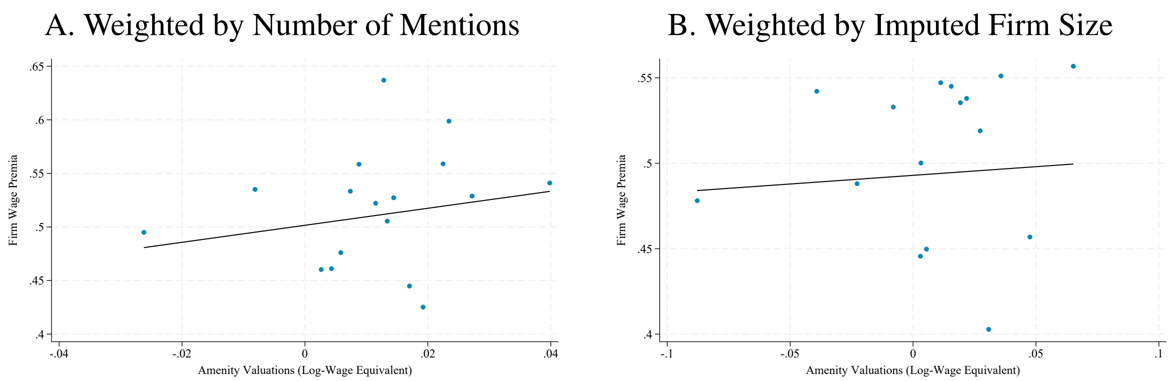


Note: This figure compares the baseline estimates of amenity valuations (in utility units) to those we obtain when we re-run the model using a discrete choice experiment where we told workers to assume that the pattern of promotions and pay growth would be the same at all outside firms. Panel A weights by the number of times firms are mentioned. Panel B weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 3.

D.1.3 Results Reported in Section 5

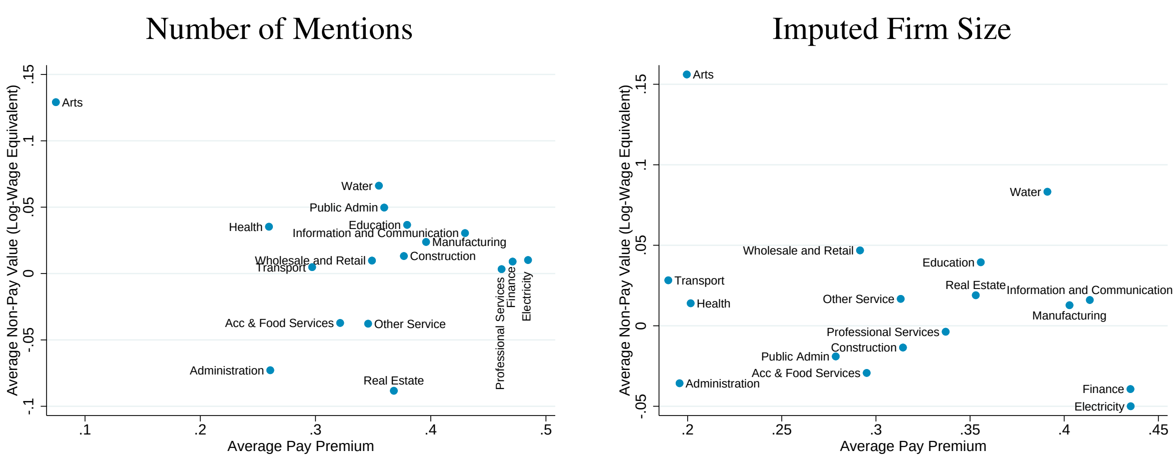
In Section 5 we examine the relationship between amenities, wages, and inequality. Here we replicate Figure 5 (relationship between wage premia and amenity valuations) and Figure 6 (amenities, wages, and values by sector), showing that the core result—that there is a non-negative correlation between valuations and wage premia so amenity provision does not dampen inequality—is robust to alternative weighting schemes. We do not replicate Table 5 (classifying job-to-job moves) or Table 6 (amenities and the gender value gap), as those regressions are conducted at the worker level and are not weighted by firm size.

Figure D3: Relationship Between Wage Premia and Amenity Valuations: Alternative Weighting Schemes



Note: This figure shows binned scatterplots of the relationship between firm wage premia (y-axis) and amenity values in log-wage equivalent units (x-axis) for firms in our connected set. Panel A weights by the number of times firms are mentioned. Panel B weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 5.

Figure D4: Amenities, Wages, and Values by Sector: Alternative Weighting Schemes

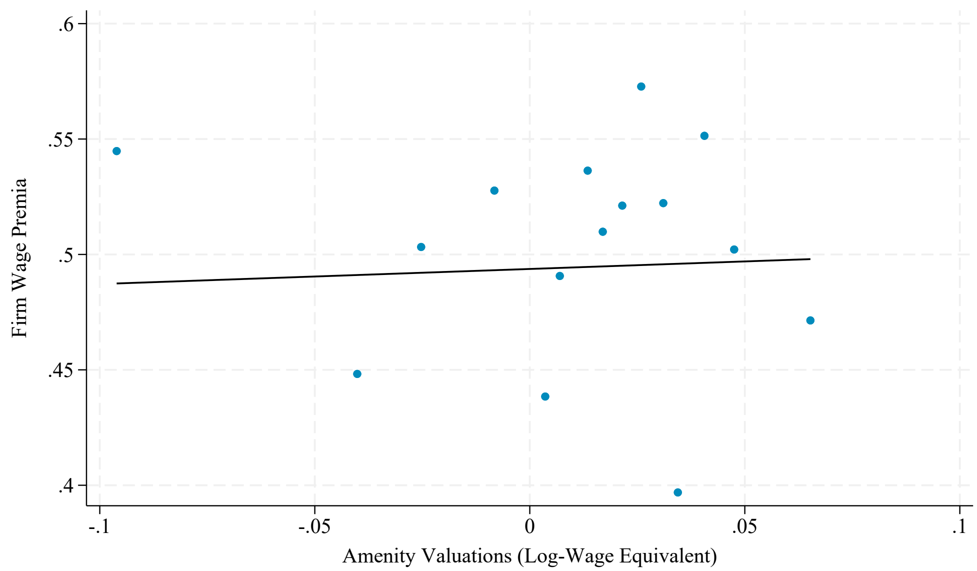


Note: This figure shows the average wage premium (x-axis) and average non-wage amenity value in log-wage equivalent units (y-axis) by major sector. The left panel weights by the number of times each firm was mentioned. The right panel weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 6.

D.1.4 Wave 1 Robustness

Figure D5 replicates the baseline wage–amenity binscatter (Figure 5) using only amenity estimates from Wave 1 of the survey, in which randomized raises ranged from 5% to 20% with no pay cuts offered. The near-zero slope confirms that the orthogonality result does not depend on the introduction of pay cuts in the follow-up wave and that valuations are stable across waves.

Figure D5: Relationship Between Wage Premia and Amenity Valuations: Wave 1 Only

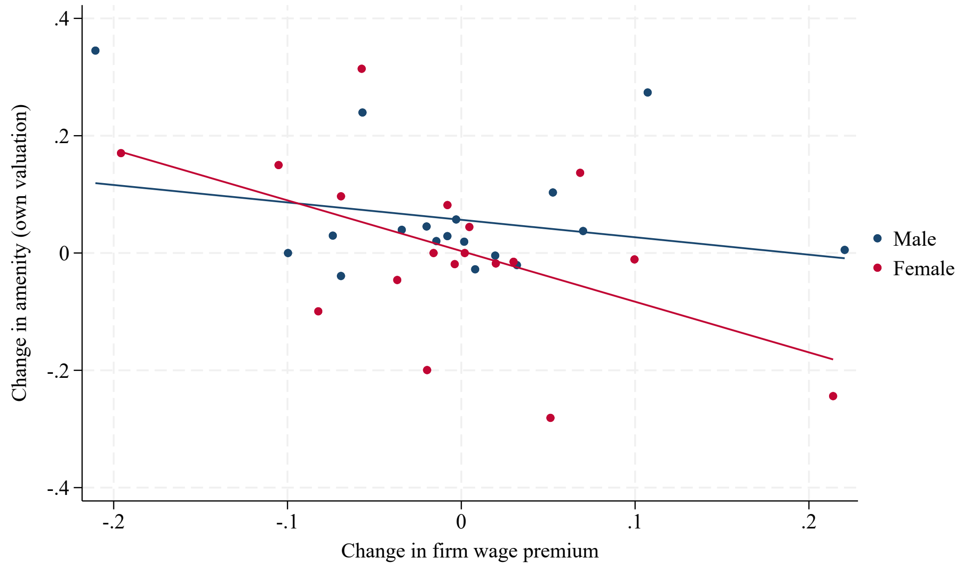


Note: This figure replicates the split-sample binscatter in Figure 5, Panel B, using only amenity estimates from Wave 1 of the survey. In Wave 1, randomized raises ranged from 5% to 20%; no pay cuts were offered. Firms are weighted by employment.

D.1.5 Mobility Results Reported in Section 5.2

Figure D6 presents binscatters of the wage–amenity trade-off among job-to-job movers, separately by sex. We instrument the change in amenity valuations from one random split-half of respondents with the change from the other, and plot the change in the firm wage premium against the instrumented amenity change. Workers who move to firms with higher predicted amenities accept lower firm premia, and the slope is steeper for women.

Figure D6: Wage–Amenity Trade-off Among Movers



Note: Binned scatterplot of the change in sex-specific amenity valuations (y-axis) against the change in firm wage premium (x-axis) among job-to-job movers, separately for men and women. Amenity valuations are instrumented using the split-sample approach: the change from one random half of respondents is instrumented with the change from the other.

Appendix Tables D3 and D4 present robustness checks on the transition matrix in Table 5. Table D3 restricts to likely voluntary movers (direct job-to-job transitions with no intervening unemployment spell). Table D4 uses a 10% threshold (rather than 2%) to define large changes in wages and amenities. The patterns are similar across both specifications.

Table D3: Classifying Job-to-Job Moves: Voluntary Movers

	Change in Amenities			
	>2% Cut	<2% Change	>2% Raise	Total
	(1)	(2)	(3)	(4)
A. Change in Own Wage				
Decrease ($< -2\%$)	1.0%	19.6%	1.6%	22.2%
Similar (±2%)	1.5%	32.6%	2.1%	36.2%
Increase ($> 2\%$)	4.0%	32.6%	5.2%	41.8%
Total	6.5%	84.8%	8.9%	100.2%
B. Change in Firm Wage Premium				
Decrease ($< -2\%$)	4.3%	25.6%	6.2%	36.1%
Similar (±2%)	1.3%	27.6%	1.8%	30.7%
Increase ($> 2\%$)	3.9%	24.0%	5.2%	33.1%
Total	9.5%	77.2%	13.2%	99.9%
C. Change in Log Median Wage				
Decrease ($< -2\%$)	2.9%	1.9%	4.0%	8.8%
Similar (±2%)	0.1%	81.2%	0.2%	81.5%
Increase ($> 2\%$)	3.4%	1.7%	4.6%	9.7%
Total	6.4%	84.8%	8.8%	100.0%
D. Change in Log Mean Wage				
Decrease ($< -2\%$)	2.8%	1.8%	4.0%	8.6%
Similar (±2%)	0.1%	81.2%	0.4%	81.7%
Increase ($> 2\%$)	3.4%	1.7%	4.5%	9.6%
Total	6.3%	84.7%	8.9%	99.9%

Note: This table replicates Table 5 restricting to likely voluntary movers—workers who transition directly from one employer to another without an intervening unemployment spell. Panels A and B use baseline amenity valuations; Panels C and D use no-growth valuations. Changes are classified as large when they exceed 2%.

Table D4: Classifying Job-to-Job Moves: 10% Threshold

	Change in Amenities			
	>10% Cut	<10% Change	>10% Raise	Total
	(1)	(2)	(3)	(4)
A. Change in Own Wage				
Decrease ($< -10\%$)	0.1%	5.3%	0.1%	5.5%
Similar ($\pm 10\%$)	0.6%	68.4%	1.4%	70.4%
Increase ($> 10\%$)	0.8%	22.6%	0.7%	24.1%
Total	1.5%	96.3%	2.2%	100.0%
B. Change in Firm Wage Premium				
Decrease ($< -10\%$)	0.3%	8.4%	0.7%	9.4%
Similar ($\pm 10\%$)	1.4%	78.4%	2.0%	81.8%
Increase ($> 10\%$)	0.5%	7.7%	0.6%	8.8%
Total	2.2%	94.5%	3.3%	100.0%
C. Change in Log Median Wage				
Decrease ($< -10\%$)	0.4%	5.9%	0.9%	7.2%
Similar ($\pm 10\%$)	0.2%	84.4%	0.4%	85.0%
Increase ($> 10\%$)	0.9%	6.1%	0.9%	7.9%
Total	1.5%	96.4%	2.2%	100.1%
D. Change in Log Mean Wage				
Decrease ($< -10\%$)	0.4%	5.8%	1.0%	7.2%
Similar ($\pm 10\%$)	0.1%	84.5%	0.4%	85.0%
Increase ($> 10\%$)	0.9%	6.0%	0.8%	7.7%
Total	1.4%	96.3%	2.2%	99.9%

Note: This table replicates Table 5 using a 10% threshold (rather than 2%) to define large changes in wages and amenities. Panels A and B use baseline amenity valuations; Panels C and D use no-growth valuations.

D.2 Results from the Researcher-Provided Firm Module

The results in the main text use data from a module in which we asked workers to provide the names of specific firms, and then to rank hypothetical offers from these firms under researcher-randomized raises. For the analysis in Caldwell, Haegele and Heining (2025), we designed a parallel module focusing on researcher-selected firms. We can use these data to probe the robustness of our core findings.

We estimate firm-specific amenity valuations via rank-ordered logit, separately by demographic group. All valuations are normalized relative to the same base firm (a manufacturing firm), which is the omitted category in the rank-ordered logit. Because the same base firm is used for all subsamples, differences in valuations across groups are identified. We restrict attention to firms that workers indicated they would consider applying to.

Table D5 reports correlations and formal tests of equality in valuations from the researcher-provided module. Because many workers evaluated the same researcher-selected firms, these estimates are substantially more precise than those from worker-provided firms. Panel A uses all ranked firms; Panel B restricts to firms workers indicated they would consider applying to ("marginal workers"). Columns (1)–(2) report pairwise correlations between group-specific valuations. Columns (3)–(5) report p-values from paired t-tests, Kolmogorov-Smirnov tests, and a joint χ^2 test accounting for estimation error (delta method).

Table D5: Between-Group Heterogeneity in Valuations: Researcher-Provided Module

Panel A: All Workers	Correlations		p-values		
	Pearson	Spearman	Paired t	K-S	Joint χ^2
	(1)	(2)	(3)	(4)	(5)
Male vs. Female	0.424	0.430	0.000^{***}	0.007^{***}	0.000^{***}
College vs. No College	0.770	0.792	0.001^{***}	0.236	0.501
Firms	30	30			

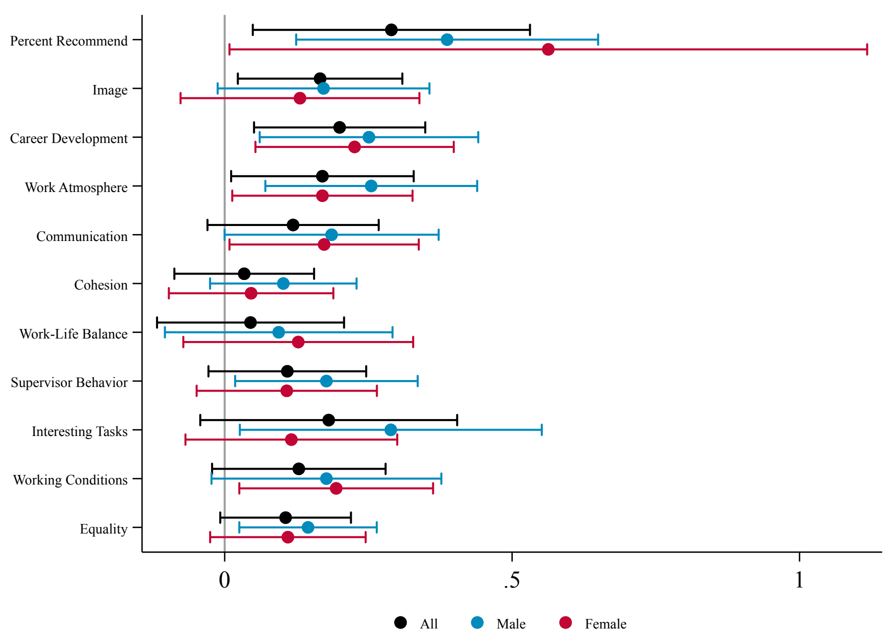
Panel B: Marginal Workers	Correlations		p-values		
	Pearson	Spearman	Paired t	K-S	Joint χ^2
	(1)	(2)	(3)	(4)	(5)
Male vs. Female	-0.069	-0.087	0.741	0.388	0.758
College vs. No College	0.036	0.206	0.610	0.035^{**}	0.717
Firms	30	30			

D.3 Correlation Between Firm Valuations and Amenity Provision

Our primary analysis estimates the overall value of the bundle of non-wage amenities provided by each firm. In this section, we correlate our firm-level amenity valuations with external measures of job quality from Kununu, a leading employer review platform in Germany (described in Appendix Section B.6).

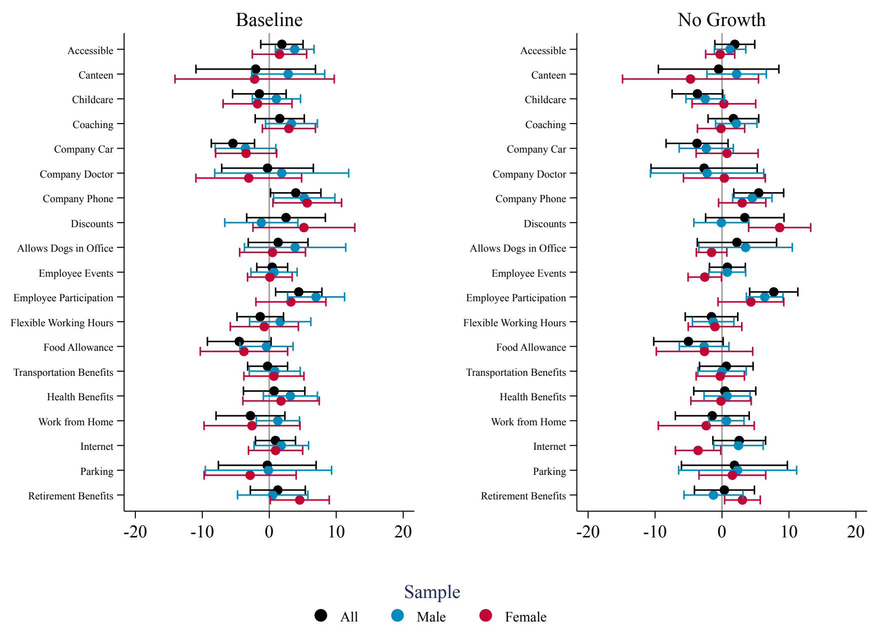
Figure D7 plots coefficients from regressions of standardized Kununu scores on our baseline amenity valuations for all workers, as well as separately for men and women. Firms with higher amenity valuations are significantly more likely to be recommended by their employees, and across nearly all subcategories—including Work Atmosphere, Career Development, and Supervisor Behavior—there is a robust positive correlation with our valuations. Figure D8 correlates our valuations with amenity provision percentages for specific amenities reported on Kununu. Our index is positively associated with several specific benefits, such as Employee Participation, Coaching, and Company Phones. However, as noted in Section 3.3, because we did not cross-randomize individual amenities, these correlations should be viewed as validation that our estimates capture real workplace differences rather than as a decomposition of the value of specific attributes.

Figure D7: Correlates of Firm Amenity Values



Note: This figure reports results from regressions of Kununu scores on our baseline amenity values for all workers (black), male workers (blue), and female workers (red). The dots are the coefficient on the amenity valuation (in utility units); whiskers show 95% confidence intervals. Kununu scores are standardized to have unit standard deviations. Regressions are weighted by firm size (see Appendix B.3).

Figure D8: Correlates of Firm Amenity Values: Kununu Amenities



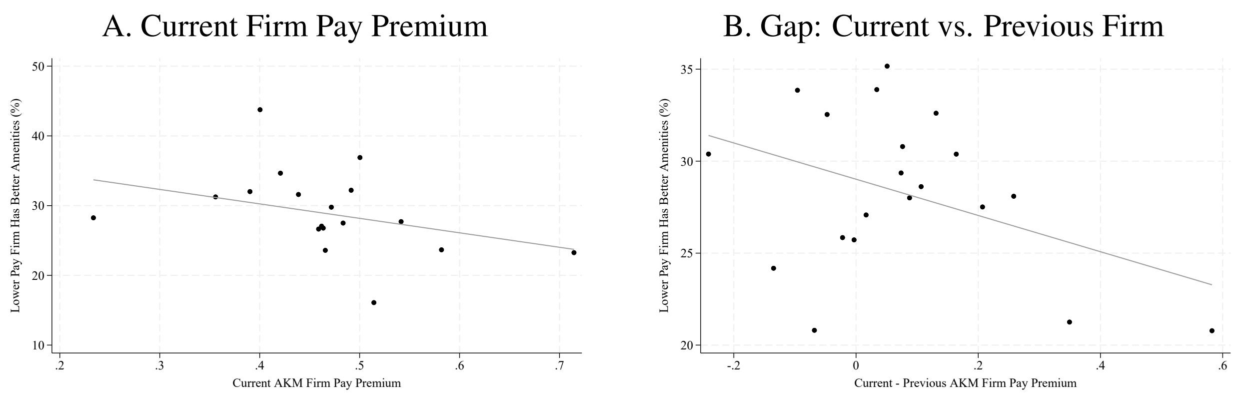
Note: This figure reports results from regressions of Kununu amenity provision percentages on our amenity values for all workers (black), male workers (blue), and female workers (red). The left panel uses baseline amenity values; the right panel uses alternative valuations that hold wage-growth expectations constant (see Section 4.2). The dots are the coefficient on the amenity valuation (in utility units); whiskers show 95% confidence intervals. Regressions are weighted by firm size (see Appendix B.3).

D.4 Beliefs About the Correlation Between Wages and Amenities

Caldwell, Haegele and Heining (2025) document that, in the follow-up survey, we asked workers whether they thought a firm paying 30% above market offered amenities that were better, the same, or worse than those offered by a firm paying 10% above market. We found that at most 29% of workers believe the higher-paid firm offered worse non-wage amenities. Here we describe the predictors of this belief. Because we did not ask whether the gap in non-wage amenities was enough to offset the gap in wages, we interpret this as a descriptive investigation of the share of workers who believe there is a negative correlation between wages and amenities.

Panel A of Appendix Figure D9 shows that, within sector, there is a negative correlation between these beliefs and the pay premium of the respondent's current firm. Panel B shows a similar negative correlation with the gap between the pay premia of a worker's current and previous firm, consistent with the idea that workers who move to lower-pay firms are more likely to believe that high wages come with lower amenities.

Figure D9: Heterogeneity in Beliefs About $\mathrm{Cov}(a_j, \psi_j)$ by Firm



Note: These figures present binned scatterplots plotting the share of respondents who say the lower-pay firm has better amenities against the pay premium of the respondent's current firm (Panel A) or the gap between the pay premia of their current and previous firm (Panel B). Both control for sector fixed effects.

D.5 Additional Validation Exercises

Table D6 shows that there is a positive relationship between the sector of provided firms and those associated with previous transitions. For instance, among workers in the manufacturing sector, 39% stay in that sector when they make a transition (Panel A); 38% of manufacturing workers in our sample provided a manufacturing firm. There is a positive correlation between the share of workers making a specific sector-to-sector transition and the share of workers providing firms that would imply the same sector-to-sector move.

Table D6: Stated and Revealed Transitions: Sector

A. Revealed Transitions

		Destination Firm						
		Manufacturing	Retail	Accommodations and Food Services	Information	Financial	Professional	Other
		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Origin Firm	Manufacturing	39%	12%	2%	2%	1%	7%	37%
	Retail	15%	37%	3%	3%	1%	6%	36%
	Accommodations and Food Services	10%	11%	33%	2%	1%	4%	39%
	Information	9%	9%	1%	39%	2%	14%	25%
	Financial	6%	7%	1%	6%	44%	11%	25%
	Professional	14%	10%	2%	8%	3%	33%	31%
	Other	12%	9%	2%	2%	1%	5%	70%

B. Stated Transitions

		Stated Firm						
		Manufacturing	Retail	Accommodations and Food Services	Information	Financial	Professional	Other
		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Current Firm	Manufacturing	38%	10%	-	12%	8%	17%	16%
	Retail	19%	24%	-	18%	3%	17%	20%
	Accommodations and Food Services	-	-	-	-	-	-	-
	Information	24%	17%	-	22%	16%	8%	13%
	Financial	25%	12%	-	7%	30%	17%	9%
	Professional	43%	11%	-	13%	8%	13%	12%
	Other	33%	12%	-	8%	4%	12%	31%

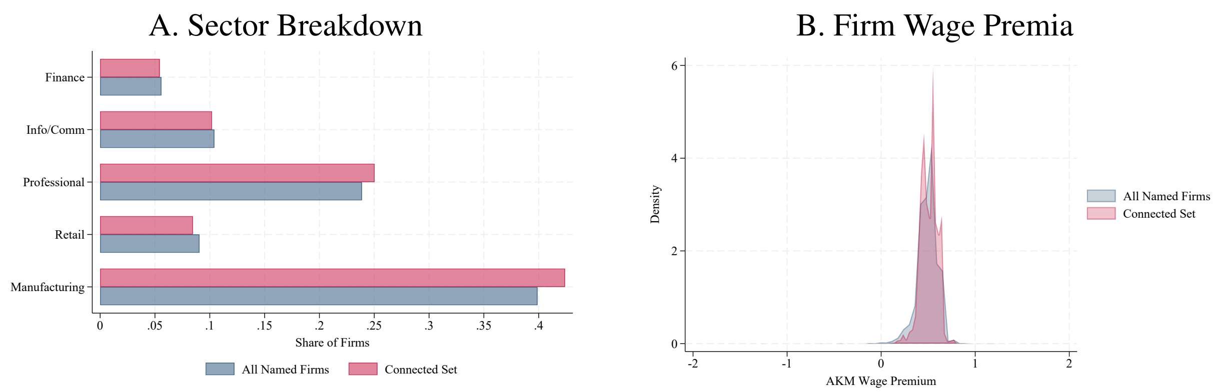
Note: This table examines the characteristics of workers' destination firms, by the characteristics of their origin firms. Panel A uses data on observed mobility and computes statistics at the establishment level; Panel B uses data on workers' provided firms and computes statistics at the firm level. Apart from the worker-provided firms, we do not have a mapping from establishments to firms in the administrative data, so the observed mobility data in Panel A are necessarily reported at the establishment level. Small cells are suppressed for privacy.

D.6 Analysis of the Connected Set

We can only provide estimates of valuations for firms in the connected set. While the previous section showed that workers provided firms which match (in characteristic space and sector space) the firms that workers from their firm have previously moved to, the connected set is naturally smaller. In particular, it excludes many firms that are named by only a single worker. A natural concern is that selection into the connected set could be correlated with firm characteristics, potentially biasing our estimates of the relationship between wages and amenities.

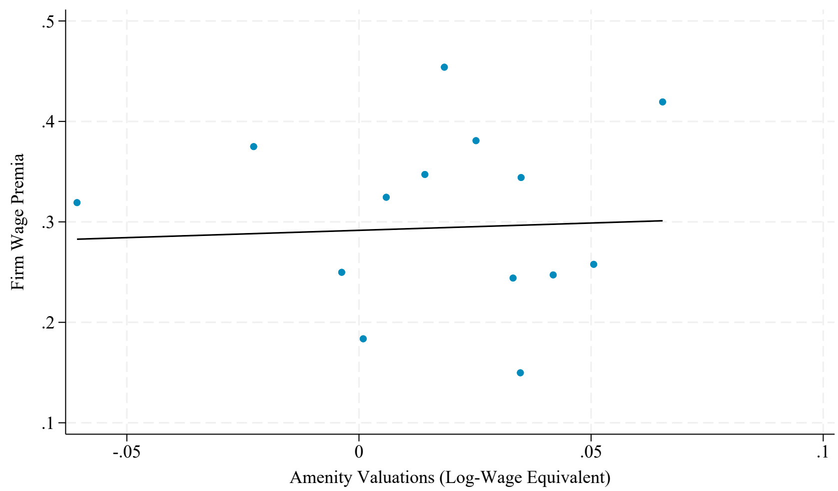
Appendix Figure D10 shows that the sector distribution and wage premium distribution of firms in the connected set are similar to those among all named firms; in particular, connected-set firms are not concentrated among high- or low-wage employers.

Figure D10: Provided Firms vs. Connected Set



Note: This figure compares the sector distribution (Panel A) and firm wage premia distribution (Panel B) of workers' provided firms to those of the subset in the connected set.

Figure D11: Relationship Between Wage Premia and Amenity Valuations: Excluding Wage Growth



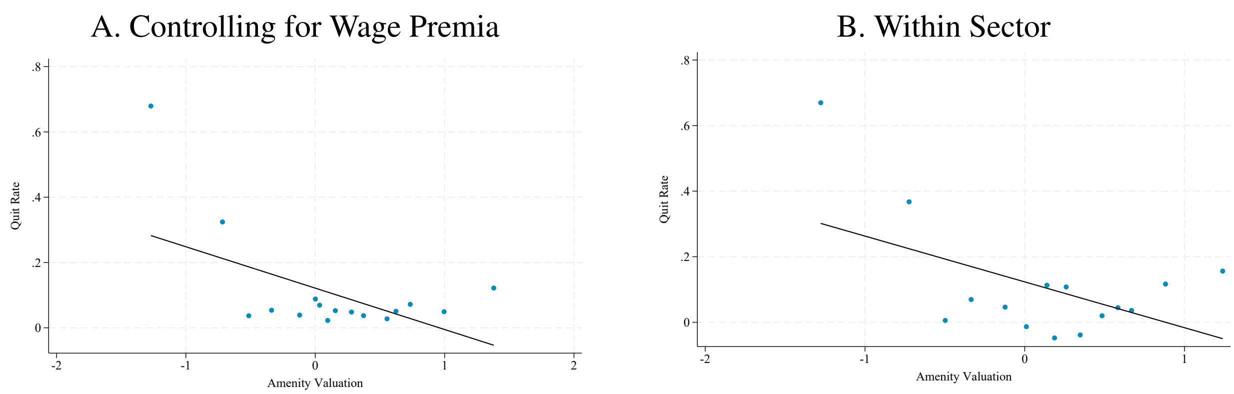
Note: This figure replicates the weighted panel of Figure 5 using amenity valuations (in log-wage equivalent units) estimated under the assumption that wage growth at the firm remains constant. Firms are weighted by employment. Wage premia are estimated from the CHH mobility sample (Appendix B.4) rather than from Bellmann et al. (2020); results are similar in either case. The relationship between firm wage premia and amenity valuations remains approximately zero, indicating that differences in dynamic expectations do not drive the orthogonality result.

D.7 Amenities and Quit Rates

A key prediction of compensating differentials models is that firms offering better amenities should, all else equal, experience lower separation rates. We provide descriptive evidence on this prediction using administrative data on job-to-job transitions from the IEB. We construct a firm-level measure of "quit rates" by counting the number of workers who leave each firm and move directly to another employer (i.e., without an intervening unemployment spell), divided by firm size. This is an imperfect proxy for voluntary separations: not all direct job-to-job moves are voluntary, and some separations involving short unemployment spells may be. Nonetheless, the measure captures the intensive margin of worker retention that models of amenity provision emphasize.

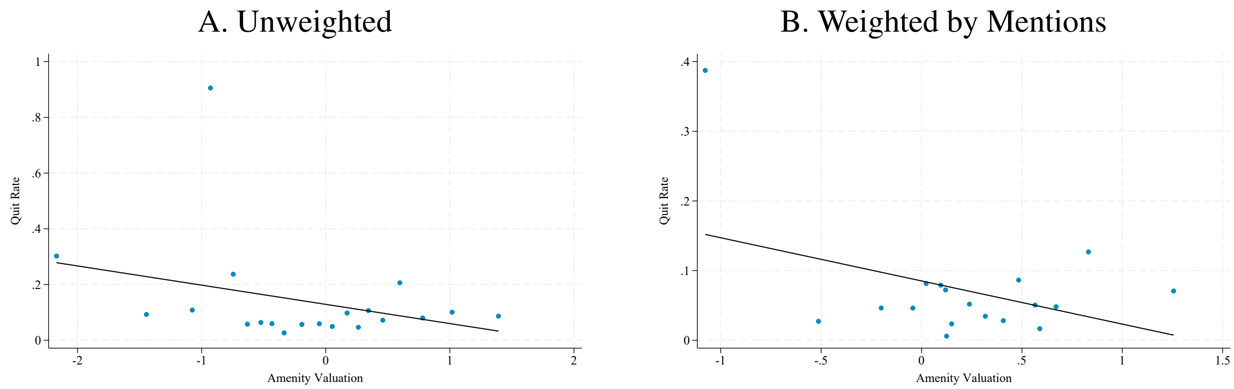
Figure D12 presents binned scatterplots of quit rates against amenity valuations, residualized on firm wage premia. Firms with higher amenity valuations have substantially lower quit rates, conditional on wages. This pattern holds both unconditionally (Panel A) and within sectors (Panel B), suggesting that it is not driven by sectoral composition. The relationship is also robust to alternative weighting schemes (Figure D13). While the cross-sectional nature of this exercise precludes a causal interpretation, the pattern confirms that firms with higher amenity values retain workers at higher rates. Given the noisy proxy, we present these patterns as descriptive.

Figure D12: Amenity Valuations and Quit Rates



Note: Binned scatterplots of firm-level quit rates against amenity valuations (in utility units), controlling for firm wage premia. Quit rates are the average annual number of direct job-to-job separations divided by firm size (IEB, 2010–2019). Panel B additionally controls for sector fixed effects. Firms are weighted by size.

Figure D13: Amenity Valuations and Quit Rates: Alternative Weighting Schemes



Note: This figure replicates Panel A of Figure D12 (amenity valuations in utility units) under alternative weighting schemes. Both panels control for firm wage premia.

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E Survey Logistics

This appendix provides additional details on how we selected and invited survey participants. We also describe how we randomly assigned incentives to participate and discuss survey response patterns. This section replicates some of the results and discussion presented in the appendix to Caldwell, Haegele and Heining (2025), which analyzes the implementation and response rates to the initial survey. This document also provides the questionnaires.

E.1 Implementation of the Initial Survey

E.1.1 Sample Construction

We used German Social Security records to identify participants for the survey. Our eligible pool consisted of workers who were—as of December 30, 2020—between the ages of 25 and 50, employed full-time, and at their current establishment for fewer than eight years. To manage the large number of letters, we mailed the survey in batches. For the first batch, we drew 82,500 workers (75% of the batch) from the set of eligible workers at firms that participated in the firm survey linked to IAB records in Caldwell, Haegele and Heining (2025). We oversampled from surveyed firms to achieve adequate power for the analysis in Caldwell, Haegele and Heining (2025). We selected the remaining 25% (N=27,500) at random from (a random 5% sample of) workers at non-surveyed firms. The entire second batch (N=25,000) was selected from the random 5% sample of eligible workers at non-surveyed firms. We mailed reminders to a random 25% subset of individuals in the first batch who had not responded to the initial invitation when we prepared the second mailing. In spring 2024, we invited all respondents from the initial survey who provided panel consent to participate in a follow-up survey.

E.1.2 Invitations

After we identified workers for inclusion in the survey, a specialized department at the IAB pulled their addresses. The IAB fielded the survey and the director of the IAB signed the invitation to participate. We mailed invitations to the initial survey between June 2022 and December 2022. We described the survey to potential respondents as a scientific study on pay progression in Germany. To manage the large number of letters, we staggered the mailings.

We invited respondents via mail instead of e-mail or phone because the Federal Employment Agency in Germany (*Bundesagentur für Arbeit*) only has e-mail and phone numbers of individuals who have recently been unemployed or have participated in re-employment measures. Postal addresses are available for all workers. In the invitation, we informed respondents that the survey would take approximately 10 minutes to complete. Appendix Figure G1 shows (a translation of) the wording of the invitation; the sample we include is for a worker randomized into one of the gift card treatments.

In one batch of the invitations, a printing issue led to some cases where endorsement and cover letters were mixed up. This meant that two different addresses (endorsement person a, cover person b) ended up in one envelope. Letters were sent to the addresses provided on the endorsement letter, but the cover letter information included the name and password of a different individual. If an individual who received an incorrect mailing participated in the survey, they would have been linked to the wrong respondent's record. Based on inspection of the frequency of this error in letters that were returned due to incorrect addresses (which is an independent issue, but allows us to analyze the frequency of the error), we calculate that at most 431 people among the set of 17,772 recipients in this batch were likely affected (2.4%), assuming the share among returns is the same as in the overall sample. Recipients may have realized the mix-up and declined to participate. Since this only affected people with endorsement letters (which were randomly assigned), this may have reduced the response rate for this group.

E.1.3 Exogenous Variation in Incentives to Participate

To characterize patterns of non-response, we introduced random variation in individuals' incentives to participate. We did this through (1) randomized financial incentives, (2) randomized endorsement letters, and (3) randomized follow-up.

Figure G1: Original Invitation (English Translation)

INSTITUT FÜR ARBEITSMARKT- UND BERUFSFORSCHUNG
Die Forschungseinrichtung der Bundesagentur für Arbeit

Bei Rückfragen wenden Sie sich bitte an:
Dr. Jörg Heining
Regensburger Str. 104
90478 Nürnberg
E-Mail: lf-befragung@iab.de

Institut für Arbeitsmarkt- und Berufsforschung
Regensburger Str. 104 · Re100 407 · 90478 Nürnberg

«First name» «Last name»
«Street address»
«Zip code» «City»

Anschreiben-ID: «anschreiben_id»
Nürnberg, «date»

Scientific Study on Wage Flexibility in Germany

Hello «First name» «Last name»,

The Institute for Employment Research (IAB) is conducting a scientific study to understand how the salary progression of employees in Germany is changing. We are therefore interested in your experience in the labor market and would like to invite you to a survey. By participating, you support the IAB in advising political decision-makers and thus help to improve economic and social policy in Germany.

In a nutshell - we will interview you via the Internet
The survey will not take more than **10 minutes** of your time. Your participation is of course voluntary and anonymous. To get to the survey, you can use the display **QR code** or click on the following link:
https://umfragen.iab.de/goto/LF-befragung

Your personal password for participation is: **«survey_password»**

Participating pays off
As a thank you for your participation, we are raffling off 1,500 vouchers, each with a value of 10 euros.

Your information is confidential
The safety of your personal data is important to us. We assure you that your information will be treated with strict confidentiality in accordance with the statutory data protection regulations and will only be used for scientific purposes. Your answers cannot be linked to your person. You will find additional data protection explanations attached.

Thank you for your cooperation and for your trust!

Kind regards

B. Fitzenberger

Prof. Bernd Fitzenberger Ph.D.
Direktor des Instituts für Arbeitsmarkt- und Berufsforschung (IAB)

Note: This is a screenshot of the translated cover letter sent to workers randomized to the high-gift-card

Gift Card Lottery. Information on the gift card lottery was randomly assigned to random subsets of the first 110,000 workers selected for batch 1 and was included with the initial invitation. 10,000 workers were selected to be in a raffle for a 5 euro gift card and 20,000 workers were selected to be in a raffle for a 10 euro gift card. Workers were informed about the lottery in the cover letter, which provided the gift card value, as well as the number of gift cards that we would raffle off. After the survey closed in January 2023, the IAB conducted the gift card lottery by randomizing among participants who started the survey. Winners were informed either via e-mail or mail, according to the preferences indicated in the survey.

Endorsement Letters. The endorsement letter was assigned to a subset of 82,500 of the first 110,000 workers in batch 1 and was included with the initial invitation to participate. The letter was signed by one of the 2021 Nobel Prize winners in economics, identified both as one of the 2021 Laureates and as a previous collaborator of the IAB. The letter highlighted the importance of scientific labor market research and urged recipients to complete the survey.

Reminder Mailings. There were 99,698 initial non-responders. We randomly selected 25,000 to receive a follow-up letter. The reminder letters had nearly identical wording to the initial invitations, but reminded individuals that they had previously been invited to participate in the survey. The letters included the same information on data protection as before. Individuals who had been included in the gift card raffle in the initial invitation received reminders of their offer to participate. Individuals who had been randomized to receive endorsement letters did not receive a second endorsement letter.

E.1.4 Balance Check

Appendix Table G1 describes the workers we invited for the survey. As Columns 2 and 3 indicate, conditional on the strata used for selection (whether an individual is at a surveyed firm), there is no difference in the characteristics of eligible workers and those invited to participate in our survey. Columns 4 and 5 compare those selected for the treatment and control groups for the endorsement letters and gift card treatments. Conditional on the strata used to assign these treatments, we find no difference in the characteristics of those selected and those not selected.[27] Column 6 shows that among those eligible to receive a reminder (initial non-responders in batch 1), there is no difference between those selected to receive a reminder and those not selected.

E.1.5 Response Rate and Consent

The survey was open until January 15, 2023. We received 13,680 total responses. Subtracting the number of letters that could not be delivered, the survey had an effective response rate of 11.4%. This response rate is much higher than those of other surveys at the IAB that invite respondents for the first time (Haas et al., 2021). Among the 13,680 individuals who started the survey, 11,868 completed it; this represents a completion rate of 87%.[28] The median response time among individuals who completed the survey was 9 minutes.

We asked participants for their consent to link their answers to the employer-employee data at the IAB. We have 10,134 complete responses with linkage consent, which we link to the IAB records. While this direct consent is necessary under German privacy laws to link the survey data to other data sources, we analyze the raw, unlinked data for both consenters and non-consenters. We also asked participants for their consent to participate in follow-up surveys. 8,416 respondents who provided consent for this linkage also provided consent to be contacted for future survey waves. Among the 10,134 linkage consenters, this represents a panel consent rate of 83%.

E.1.6 Impact of Randomized Incentives on Response Rates

Appendix Table G2 shows that neither the gift cards nor the endorsement letter had a statistically significant (or economically meaningful) impact on response rates in the initial survey. By contrast, the reminder message increased response rates by 4 percentage points among workers who did not

[27] We grouped individuals into two groups based on their federal state of residence and randomly assigned gift cards within these strata. We randomly assigned endorsement letters without regard to state.

[28] We define a response as complete if a respondent clicked through to the (second to last) question eliciting consent for participating in a follow-up survey. We do not require respondents to have answered every question to be counted as complete responses. The survey did not require individuals to respond to particular questions.

Table G1: Randomization Assessment

		Selection		Wave 1 Randomization		
	Eligible Mean	Wave 1 - Eligible	Wave 2 - Eligible	Lottery - No Lottery	Letter - No Letter	Reminder - No Reminder
	(1)	(2)	(3)	(4)	(5)	(6)
Demographics						
Female	0.32	0.63	0.33	0.48	0.41	0.32
Age	33.02	0.29	0.94	0.75	0.73	0.82
German Citizen	0.77	0.22	0.65	0.75	0.67	0.22
College Education	0.34	0.59	0.47	0.06	0.35	0.39
Apprenticeship	0.45	0.90	0.67	0.64	0.51	0.69
Daily Earnings	129.12	0.22	0.39	0.23	0.47	0.09
Occupation Group						
Manager	0.04	0.96	0.45	0.95	0.71	0.57
Recent Entrant	0.34	0.86	0.84	0.95	0.06	0.01
Sector						
Manufacturing	0.32	0.86	0.72	0.01	0.74	0.44
Retail	0.10	0.47	0.40	0.53	0.47	0.41
Professional	0.10	0.95	0.42	0.08	0.44	0.70
Establishments		24928	21248	7253	19600	7204
Workers		110000	25000	30000	82500	25000
F-statistic		0.693	0.444	1.558	0.982	1.039
p-value		0.747	0.937	0.104	0.460	0.408

Note: This table documents that the random assignment of survey invitations and incentives was successful. Column 1 describes workers eligible for inclusion in our worker survey. Each entry in Columns 2 to 6 presents the p-value for a two-sided test of the coefficient from a regression of the characteristic in each row on the treatment in each column, controlling for the strata used for random assignment. P-values are calculated using robust standard errors. At the bottom of the table we present the F-statistic and associated p-value from a test of whether all of the coefficients are jointly zero. Columns 2 and 3 show that, conditional on the strata used for selection (whether an individual worked at a surveyed firm in 2020), selected individuals are not statistically distinguishable from non-selected individuals. Columns 4 to 6 show that, conditional on the strata used for random assignment, individuals selected to receive each of the three types of incentives are not distinguishable from those who were not selected. We only randomized these incentives to workers in batch 1.

Table G2: Impact of Randomized Incentives on Response Rates in the Initial Survey

	Endorsement Letter	Gift Card Level	Gift Card Binary	Reminder
	(1)	(2)	(3)	(4)
Treatment	0.000	-0.000	-0.002	0.040***
	(0.002)	(0.000)	(0.002)	(0.001)
Observations	109995	109995	109995	99698

Note: This table analyzes the effect of the randomized incentives on the likelihood that invited individuals completed the survey and provided linkage consent. Each coefficient stems from a separate regression of survey completion on an indicator for the respective incentive, conditional on the strata used for random assignment. Column 1 focuses on endorsement letters. Columns 2 and 3 focus on gift cards. Column 4 focuses on the survey reminder. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

initially respond to the survey. Both incentives were visible only to individuals who opened the mailer; the null effects suggest that much of the initial non-response reflects individuals who never opened the invitation.

E.1.7 Selection into Non-Response

We follow Dutz et al. (2021) in analyzing the characteristics of compliers. We restrict attention to individuals in batch 1 of the initial survey, because this is the batch in which we implemented the randomization of incentives. We do not characterize endorsement letter or gift card compliers because these incentives did not have a significant impact on response rates. Appendix Table G3 describes three populations of workers. Column 1 describes "early always takers": those who responded to the survey before we mailed the reminder. Column 2 describes the "late always takers": those who we randomized into not receiving a reminder, but who nonetheless responded after we mailed the reminders. Column 3 describes the reminder compliers. As Column 2 indicates, virtually all of the always takers responded before we mailed the reminder. Column 3 indicates that the reminder compliers are, relative to the early always takers, somewhat more likely to be male or covered by a collective bargaining agreement.

We also examine selection into providing consent to link survey responses to administrative data and to be invited to participate in future surveys. Appendix Table G4 compares the character-

Table G3: Characteristics of Reminder Compliers and Always Takers in the Initial Survey

	Always Takers		Reminder
	Early	Late	Compliers
	(1)	(2)	(3)
Shares	0.05	0.000	0.03
Female	0.27	0.50	0.24
Age	33.85	34.36	34.02
German	0.94	0.93	0.91
College Degree	0.63	0.64	0.60
Apprenticeship	0.31	0.21	0.34
Daily Pay	186.04	202.40	184.97
Censored Pay	0.26	0.29	0.29
Hours Worked	40.63	41.39	40.37
CBA	0.61	0.71	0.65
Sector			
Manufacturing	0.56	0.64	0.56
Retail	0.07	0.00	0.05
Professional	0.17	0.14	0.20
AKM Fixed Effects			
Person Effect	4.47	4.33	4.44
Firm Effect	0.52	0.58	0.54
Risk Preferences	6.22	5.57	6.19
Outside Options	1.41	1.14	1.41
Did Not Provide Names	0.45	0.50	0.50

Note: This table compares the characteristics of early and late always takers to reminder compliers. Early always takers are workers in the control group who responded before we mailed the (randomized) reminders. Late always takers are workers in the control group who responded after we mailed the reminders. Reminder compliers are workers who responded after being mailed a reminder. Following Dutz et al. (2021), we estimate the reminder compliers' average characteristics via an instrumental variables regression with $Y_i(1-R_{i1})R_{i2}$ as the outcome variable, $(1-R_{i1})R_{i2}$ as the endogenous variable, and Z_i as the instrument. Y_i is the characteristic of interest, R_{i1} is an indicator for responding before we mailed the reminders, R_{i2} is an indicator for responding after we mailed the reminders, and Z_i is an indicator for having received a reminder. Shares reported in Row 1 are out of all workers invited to respond to the initial survey (Column 1), those who neither responded earlier nor received a reminder (Column 2), and those who did not respond earlier but did receive a reminder (Column 3).

istics of invited individuals (Column 1) to those who completed the survey and provided linkage consent (Column 2) and to those who additionally provided consent to participate in future surveys (Column 4). Columns 3 and 5 compare the samples in Columns 2 and 4 to the samples in Columns 1 and 2, respectively. We find modest differences in respondent characteristics with respect to gender, age, occupation type, and sector. For instance, while the female share is 30% among invited individuals, it is 32% among those who responded and provided consent. German citizens were much more likely to respond than non-citizens, likely because they are more comfortable with the survey language and may feel a greater obligation to contribute to research on the German labor market. We also find that more educated, higher-earning workers were more likely to respond, likely because they are more familiar with the IAB. Columns 6 and 7 show that workers who responded to the follow-up survey were again positively selected on citizenship, education, and earnings.

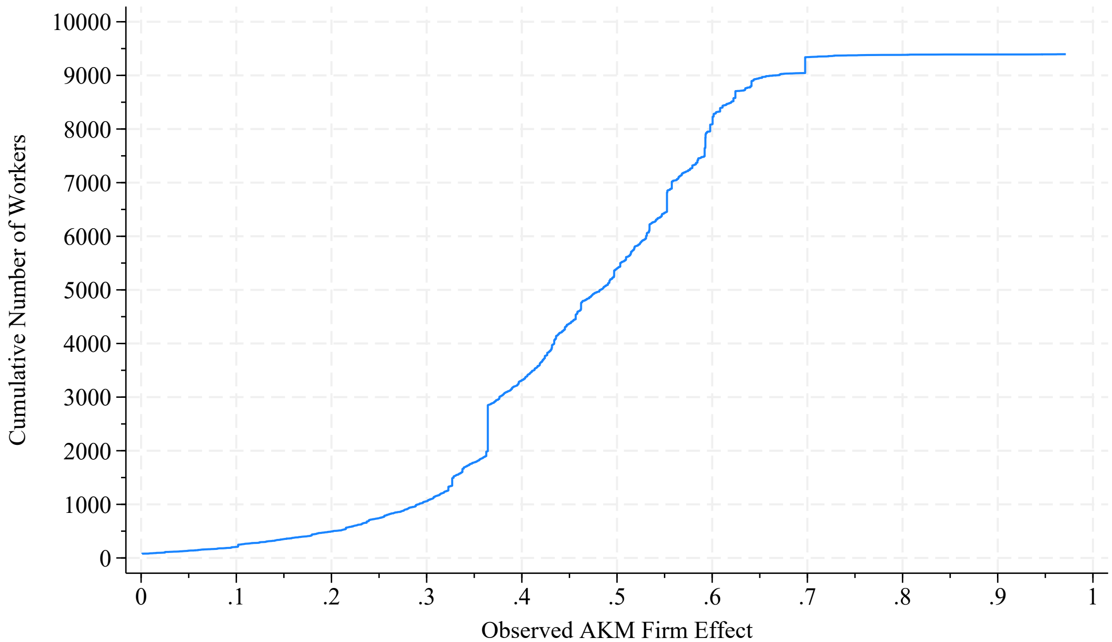
While there is some positive selection into the survey, our final sample includes workers at a broad set of firms, including low-pay firms (Figure G2).

Table G4: Non-Response and Consent

	Invited	Linkage Consent		Panel and Linkage		Responded to Follow-Up	
	Mean	Mean	Difference Rel. Invited	Mean	Difference Rel. Linked	Mean	Difference Rel. Invited
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Demographics							
Female	0.30	0.32	0.02 ***	0.32	0.01	0.31	-0.01
	(0.46)	(0.46)	(0.00)	(0.47)	(0.01)	(0.46)	(0.01)
Age	33.63	33.33	-0.32 ***	33.33	-0.02	33.41	0.14
	(6.59)	(6.23)	(0.06)	(6.14)	(0.17)	(6.16)	(0.14)
German Citizen	0.81	0.92	0.12 ***	0.92	0.03 ***	0.94	0.02 ***
	(0.39)	(0.27)	(0.00)	(0.26)	(0.01)	(0.24)	(0.01)
College Education	0.39	0.59	0.22 ***	0.60	0.07 ***	0.65	0.07 ***
	(0.49)	(0.49)	(0.01)	(0.49)	(0.01)	(0.48)	(0.01)
Apprenticeship	0.45	0.33	-0.12	0.32	-0.05	0.29	-0.06
	(0.50)	(0.47)	(0.00)	(0.47)	(0.01)	(0.45)	(0.01)
Daily Earnings	146.03	169.79	25.69 ***	170.92	6.61 ***	175.02	7.27 ***
	(60.77)	(56.71)	(0.59)	(56.67)	(1.50)	(55.27)	(1.24)
Occupation Group							
Manager	0.05	0.06	0.02 ***	0.06	0.01 *	0.07	0.01 **
	(0.21)	(0.24)	(0.00)	(0.24)	(0.01)	(0.25)	(0.01)
Recent Entrant	0.26	0.20	-0.06 ***	0.20	-0.02 *	0.19	-0.02 **
	(0.44)	(0.40)	(0.00)	(0.40)	(0.01)	(0.39)	(0.01)
Sector							
Manufacturing	0.44	0.46	0.03 ***	0.46	0.00	0.47	0.01
	(0.50)	(0.50)	(0.01)	(0.50)	(0.01)	(0.50)	(0.01)
Retail	0.10	0.09	-0.01 ***	0.09	0.01	0.09	0.00
	(0.30)	(0.28)	(0.00)	(0.28)	(0.01)	(0.28)	(0.01)
Professional	0.10	0.13	0.04 ***	0.13	0.01	0.13	0.00
	(0.29)	(0.34)	(0.00)	(0.34)	(0.01)	(0.34)	(0.01)
Establishments	42705	3556		2983		1457	
Observations	134995	10134		8416		3664	

Note: This table compares the characteristics of workers we invited to complete the survey (Column 1) to workers who completed the survey and provided consent to link their responses to the administrative data (Column 2), to the subset of these workers who additionally provided consent to participate in follow-up surveys (Column 4), and those who participated in the follow-up survey (Column 6). Columns 3, 5, and 7 present the strata-adjusted differences between the samples indicated in the header and sub-header. For instance, Column 3 reports the difference between workers who completed the survey and provided linkage consent to the set of invited workers. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

Figure G2: Sample Size by AKM Firm Effect



Note: This figure shows the cumulative number of workers in our sample (y-axis) who, at the time of sampling, worked at an establishment whose AKM firm effect was at or below the value indicated on the x-axis. Workers whose firms do not have associated estimates are not included in this graph.

E.2 Implementation of the Follow-Up Survey

In spring 2024, we invited all respondents from the initial survey who provided panel consent to participate in a follow-up survey. Participants who did not provide an e-mail address in the initial survey were contacted via mail by the IAB. The invitation letter closely followed the initial survey letter but reminded respondents of their prior participation and introduced the follow-up. Among the participants who did provide an e-mail address in the initial survey, we randomized the initial mode of contact: 25% received an invitation sent by the IAB via e-mail and 75% first received an invitation letter via mail (identical to those participants who did not provide an e-mail) and then received another invitation via e-mail. Within each group of respondents, we randomly selected 75% to receive a further reminder (either by mail or e-mail).

The survey was open for 14 weeks and received a 50% response rate. The completion rate was 92%. The median response time among individuals who completed the survey was 9 minutes. Among those who completed the follow-up survey, 90% provided linkage consent.

E.2.1 Impact of Randomized Incentives on Response Rates

We asked individuals in the initial wave of the survey whether we could contact them for a follow-up survey. We also asked these individuals whether we could contact them via e-mail. Most workers indicated they would prefer to be contacted by e-mail in the follow-up survey. In the follow-up survey, we grouped individuals by whether they provided an e-mail address and targeted incentives accordingly.

Overall, about 50% of workers we invited to complete the follow-up survey successfully did so. Appendix Table G5 shows that each of the reminders in this follow-up had a significant impact on response rates. Column 1 examines the response rates of individuals in the letter (non-email) sample. For these workers, we randomized a single incentive: the provision of a reminder letter (as in the initial survey wave). As Column 1 indicates, those who did not receive the reminder (25% of the sample) had a response rate of just under 40%; the reminder boosted response rates by 8 percentage points.

Among workers who provided an e-mail address, we cross-randomized two incentives. First, we mailed a random subset (75%) of these individuals a letter before we e-mailed them regarding the survey. Second, we sent an e-mail reminder to a random subset (75%, cross-randomized) of these individuals. Both incentives were highly effective. The initial letter boosted response rates by 23 percentage points (Column 2). The e-mail reminder boosted response rates by 7 percentage points (Column 3).

E.2.2 Selection Into Non-Response

Analogous to the initial survey, we analyze the characteristics of compliers for the follow-up survey following Dutz et al. (2021). Appendix Table G6 describes three groups of individuals: early always takers (Column 1), late always takers (Column 2), and reminder compliers (Column 3).

Table G5: Impact of Randomized Incentives on Response Rates

	Provided an E-Mail			
	No	Yes		
	(1)	(2)	(3)	(4)
Reminder Letter	0.079***			
	(0.019)			
Initial Letter		0.230***		0.232***
		(0.015)		(0.015)
Reminder E-mail			0.070***	0.077***
			(0.016)	(0.016)
Constant	0.360***	0.270***	0.393***	0.210***
	(0.017)	(0.013)	(0.014)	(0.018)
Observations	3405	5011	5011	5011

Note: This table analyzes the effect of the randomized incentives on the likelihood that invited individuals completed the follow-up survey and provided linkage consent. In the initial survey, we asked individuals whether we could contact them for a follow-up survey and whether we could contact them via e-mail (which they then provided). Column 1 focuses on workers who did not provide an e-mail address with which to contact them for the follow-up survey. Columns 2 to 4 focus on workers who did provide an e-mail address for this purpose. Each column presents the coefficients from a regression among workers in the indicated sample; the dependent variable is an indicator for whether the individual responded to the follow-up survey and provided consent to link this new response to the administrative records; the independent variables are an indicator for the survey incentive(s) indicated in the rows and a constant. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

As in the initial survey, almost all of the always takers responded before the reminder was sent. Column 3 indicates that the reminder compliers are very similar to the early always takers.

Table G6: Characteristics of Reminder Compliers and Always Takers in the Follow-Up Survey

	Always Takers		Reminder
	Early	Late	Compliers
	(1)	(2)	(3)
Shares	0.35	0.01	0.13
Female	0.32	0.50	0.31
Age	33.37	32.55	33.36
German	0.94	0.91	0.93
College Degree	0.65	0.77	0.61
Apprenticeship	0.29	0.14	0.33
Daily Pay	175.12	157.38	171.34
Censored Pay	0.22	0.18	0.19
Hours Worked	40.45	40.48	40.52
CBA	0.56	0.33	0.61
Sector			
Manufacturing	0.47	0.36	0.48
Retail	0.09	0.09	0.09
Professional	0.13	0.18	0.13
AKM Fixed Effects			
Person Effect	4.45	4.41	4.42
Firm Effect	0.46	0.44	0.45
Risk Preferences	6.09	5.68	6.22
Outside Options	1.40	1.45	1.41
Did Not Provide Firm Names	0.44	0.41	0.43

Note: This table compares the characteristics of early and late always takers to reminder compliers for the follow-up survey. Early always takers are workers in the control group who responded before we mailed the (randomized) reminders. Late always takers are workers in the control group who responded after we mailed the reminders. Reminder compliers are workers who responded after being mailed a reminder. Following Dutz et al. (2021), we estimate the reminder compliers' average characteristics via an instrumental variables regression with $Y_i(1\text{-}R_{i1})R_{i2}$ as the outcome variable, $(1\text{-}R_{i1})R_{i2}$ as the endogenous variable, and Z_i as the instrument. Y_i is the characteristic of interest, R_{i1} is an indicator for responding before we mailed the reminders, R_{i2} is an indicator for responding after we mailed the reminders, and Z_i is an indicator for having received a reminder. The shares reported in Row 1 are unconditional shares of all workers invited to respond to the follow-up survey.

References

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F Questionnaire

Although we included questions for workers who indicated they were self-employed or non-employed at the time of the survey, the survey targeted employed workers. We exclude self-employed and non-employed individuals from our analysis (in all papers) as the number of workers in these categories was low.

F.1 English Translation of Questionnaire

This section provides the questionnaires of both the initial and follow-up survey. From both surveys, we omit questions about negotiation that were designed for and described in detail in Caldwell, Haegele and Heining (2025). The appendix of that paper provides more information on those questions. We first provide the English translation of both questionnaires, followed by the original wording in German.

F.1.1 Main Survey

1. In order to keep the survey short, we would like to include data in the evaluation that are available at the Federal Employment Agency in Nuremberg. This is, for example, information about your previous employment.

   We strictly comply with all data protection regulations or these are strictly controlled by the IAB. All information and answers are evaluated and presented anonymously, i.e. without name, address and contact information. Nobody can therefore recognize which answers you have given. There is also no transfer of data that would identify you. Further information on data protection can be found https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx.

   The evaluation is carried out for purely scientific purposes, i.e., not for commercial purposes such as advertising or marketing. Your consent is of course voluntary.

   Do you agree to your data being linked?

   a. Yes, I agree to the link

   b. No, I do not agree to the link

2. What best describes your current main employment?
   (a) Employed with social security contributions
   (b) Self-employed
   (c) Not employed
   (d) Other

Module: Employment and Bargaining
Submodule: Employed or missing

3. When did you first join your current company?

(a) In the past 6 months
(b) 6-12 months ago
(c) 1-2 years ago
(d) 2-3 years ago
(e) >3 years ago

4. Is your current position covered by a CBA agreement (i.e., are you paid according to a CBA)?
   (a) Yes
   (b) No
   (c) I don't know
5. How many hours do you work in a typical week? {fill in}
6. At the time that you first applied to your current company, did you know anyone who worked at the company?
   (a) Yes
   (b) No, but I knew someone who used to work at the company
   (c) No, I did not know anyone
7. At the time that you applied, did you know what salary you would earn?
   (a) I had no or very little idea
   (b) I only had a rough idea what is paid in my region or sector
   (c) I had at least a rough idea what this company pays for the position
   (d) I knew exactly what this company pays for the position
8. During the past six months, have you done any of the following? Please select all that apply.
   (a) Looked at job postings
   (b) Updated public resume or employment information (e.g., Xing, LinkedIn)
   (c) Reached out to people in my network for information about potential job opportunities
   (d) Applied to a job at another company
9. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g., sent you a job opening or offered to provide a referral)?
   (a) Yes
   (b) No
10. During the past six months, did you receive any job offers from other companies?
    (a) Yes
    (b) No
11. During the past six months, did your company offer you a salary increase without you asking?
    (a) Yes
    (b) No
12. During the past six months, did you actively ask for an increase in salary?
    (a) Yes
    (b) No

Submodule: Self-Employed or Non-Employed

13. How many hours do you work in a typical week?{fill in}
14. During the past six months, have you done any of the following? Please select all that apply.

o Looked at job postings

o Updated public resume or employment information (e.g., Xing, LinkedIn)

o Reached out to people in my network for information about potential job opportunities

o Applied for jobs

15. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g. sent you a job opening or offered to provide a referral)?
    (a) Yes
    (b) No
16. During the past six months, did you receive any job offers?
    (a) Yes
    (b) No

Module: Worker-Provided Firms

17. Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to? Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. "Place-Holder Inc"). These can be companies without current job vacancies.
    (a) {fill in Company 1}
    (b) {fill in Company 2}
    (c) {fill in Company 3}
    (d) I do not want to answer this question [exclusive option]
18. **[Employed]** What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?
    (a) {fill in Company 1}
    (b) {fill in Company 2}
    (c) {fill in Company 3}
19. What best describes how certain you feel about your pay estimates?

Company Name	Very Uncertain	Uncertain	Somewhat Certain	Certain	Very Certain
{fill in Company 1}					
{fill in Company 2}					
{fill in Company 3}					

20. Some of these companies may have subsidiaries in multiple locations. Which location did you have in mind?

Company Name	Location
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

21. Do you know any employees at these companies (e.g. former coworkers, friends and family)?

Company Name	Yes, I know current employees	Yes, I know former employees	No, I do not know any who ever worked at this co
{fill in Company 1}			
{fill in Company 2}			
{fill in Company 3}			

22. (table above)

23. Suppose one of your colleagues got offers from the following companies for a position like yours. Which company do you think offers the highest gross annual pay? Please rank the offers from 1 (highest pay) to 3 (lowest pay).

Company Name	Ranking
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

24. Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 3 (least likely to take).

Company Name	Ranking
{fill in Company 1}	
fill in Company 2}	
{fill in Company 3}	

25. **[Employed]** Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {15%/15%} raise	
Remain at Current Firm at Current Pay	

26. **[Employed]** Now we would like to ask you to re-rank the firms under 2 scenarios. First, suppose that you have the same three offers, but would be able to have the same **commute/route** to work. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain at Current Firm at Current Pay	

27. **[Employed]** Now suppose that you have the same set of offers, but instead of the commute staying the same your **career growth** (future raises/promotions) are identical at each option. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain at Current Firm at Current Pay	

Module: Researcher-Provided Firms

28. Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.
    (a) Company 1
    (b) Company 2
    (c) Company 3
    (d) Company 4
    (e) Company 5
    (f) Company 6
    (g) Company 7
29. **[Employed/Self-Employed]** We are now interested in your perception of job opportunities available at different companies. What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company Name	Pay in Euros
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

30. **[Non-Employed]** We are now interested in your perception of job opportunities available at different companies. What do you think your gross annual pay would be if you worked at these companies in a position similar to your most recent one?

Company Name	Pay in Euros
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

31. **[Employed]** Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain at Current Firm at Current Pay	

32. **[Self-employed]** Now suppose you can remain self-employed or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Option	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain Self-employed at Current Pay	

33. **[Non-employed]** Now suppose you can remain non-employed or join any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Option	Ranking
{fill in Company 1} with a {10%/15%} higher income	
{fill in Company 2} with a {10%/15%} higher income	
{fill in Company 3} with a {5%/20%} higher income	
Remain Non-employed	

34. In your opinion, how easy would it be for you to obtain a job offer from a different company that you would prefer to your current position?
    (a) Very Easy
    (b) Easy
    (c) Difficult
    (d) Very difficult
35. If you had to move to a new company within the next {one/three/six} months, what best describes where you would search for a job? This only applies to positions **without** a home office option.
    (a) I would search for jobs which are not further away from my place of residence than {fill-in} Kilometers
36. Finally, we would like to ask you to assess yourself. Are you generally a person who is willing to take risks or do you try to avoid taking risks? Please choose a value on the scale below, where the value 0 means “not at all willing to take risks” and the value 10 means “very willing to take risks”.
    (a) 0 (Not at all willing to take risks) 1 2 3 4 5 6 7 8 9 10 (Very willing to take risks)

Module: Consent

37. To thank participants for their time, we are raffling off {1500}/{1000} gift cards with a value of {10}/{5} Euros. If you would like to participate in the raffle, please provide your email address below. ________________ {open-field that accepts e-mail addresses only}
38. Your information supports IAB’s research on employees’ salary progression in Germany. Since current labor market dynamics are rapidly changing, we would like to survey you again in a few months. Would you like to continue to contribute to providing well-founded information on employees’ circumstances in Germany? Then we ask you to provide us with your email address so that we can question you again. Of course, we will only use the e-mail address to write to you for our survey. You can of course unsubscribe from the survey at any time. To which email address can we send our invitation?
    (a) {open field that accepts e-mail addresses only}

(b) I do not want to be contacted by email.

39. May we then invited you again by mail?
    (a) Yes
    (b) No

F.1.2 Follow-up Survey

1. In order to keep the survey short, we would like to include data in the evaluation that are available at the Federal Employment Agency in Nuremberg. This is, for example, information about your previous employment. We strictly comply with all data protection regulations or these are strictly controlled by the IAB. All information and answers are evaluated and presented anonymously, i.e., without name, address and contact information. Nobody can therefore recognize which answers you have given. There is also no transfer of data that would identify you. Further information on data protection can be found https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx. The evaluation is carried out for purely scientific purposes, i.e. not for commercial purposes such as advertising or marketing. Your consent is of course voluntary. Do you agree to your data being linked?
    (a) Yes, I agree to the linkage
    (b) No, I do not agree the linkage
2. Have you started a new job at a different company since {month_survey1 2022/2023}?
    (a) Yes
    (b) No
3. **[Q2=Yes]** What best describes your current main employment?
    (a) Employed with social-security contributions
    (b) Self-employed
    (c) Not employed
    (d) Other
4. During the past six months, have you done any of the following? Please select all that apply.
    (a) Looked at job postings
    (b) Updated public resume or employment information (e.g., Xing, LinkedIn)
    (c) Reached out to people in my network for information about potential job opportunities
    (d) Applied to a job at another company
5. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g., sent you a job opening or offered to provide a referral)?
    (a) Yes
    (b) No
6. During the past six months, how many job offers from other companies did you receive?
    (a) 0
    (b) 1
    (c) 2
    (d) 3 or more

Module: Worker-Provided Firms

7. Next, we are interested in how you perceive employment opportunities in Germany. Even if you don't know something exactly, we would like to ask you to share your best guess with us. Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to? Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g., "Place-Holder Inc"). These can be companies without current job vacancies.
   (a) {fill in Company 1}
   (b) {fill in Company 2}
   (c) {fill in Company 3}
   (d) I do not want to answer this question [exclusive option]
8. **[Q7=d]** What best describes why you did not provide any company names in the previous question?
   (a) I am not comfortable sharing this information
   (b) There are companies I would consider, but the survey interface was too difficult
   (c) I could not think of specific companies
   (d) I have no interest in switching companies and have not thought about it at all
9. **[Q7 has only 1 or 2 firms provided]** What best describes why you did not provide three company names in the previous question?
   (a) I am not comfortable sharing this information
   (b) There are more companies I would consider, but the survey interface was too difficult
   (c) I could not think of any other specific companies
   (d) I have no interest in switching companies and have not thought much about it
10. **[Q7 is not missing]** What do you think your **gross annual pay** would be if you worked at these companies in a position similar to your current position?

Company	**Gross Annual Pay in Euros**
{Company 1}	
{Company 2}	
{Company 3}	

11. **[Q7 is not missing]** Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the pay specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{Company 1} with a pay of{}	
{Company 2} with a pay of{}	
{Company 3} with a pay of{}	
Remain at current company with a pay of{}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Researcher-Provided Firms

12. Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.
    (a) Company 1
    (b) Company 2
    (c) Company 3
    (d) Company 4
    (e) Company 5
    (f) Company 6
    (g) Company 7
13. What do you think your **gross annual pay** would be if you worked at these companies in a position similar to your current position?

Company	Gross Annual Pay in Euros
{Company 1}	
{Company 2}	
{Company 3}	

14. **[Q13 is missing]** Even if you are unsure, we are interested in your guess of how these companies compare with respect to what they pay. Please rank the companies from 1 (highest pay) to 3 (lowest pay).

Company	Ranking
{Company 1}	
{Company 2}	
{Company 3}	

15. Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the pay specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{Company 1} with a pay of {}	
{Company 2} with a pay of {}	
{Company 3} with a pay of {}	
Remain at current company with a pay of {}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

16. Now we would like to ask you to rerank the companies under two scenarios. First suppose that you have the same three offers but would be able to have the same **commute** to work as before. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{Company 1} with a pay of {}	
{Company 2} with a pay of {}	
{Company 3} with a pay of {}	
Remain at current company with a pay of {}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

17. Now suppose that you have the same set of offers, but instead of the commute staying the same you knew that the **training and learning opportunities** at each of the three firms was the same as at your current company. Please rank the offers from 1 (most likely to take) to 4 (least likely to take)

Job Offer	Ranking
{Company 1} with a pay of{}	
{Company 2} with a pay of {}	
{Company 3} with a pay of{}	
Remain at current company with a pay of {}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Applications

18. Suppose you planned to move to a new company in the next {one/three/six} months, how many different companies do you think would you apply to?
    (a) The number of companies I would apply to is: _____
19. **[Randomize who sees Q19 (50%) and who sees Q20/Q21 (50%)]** In your opinion, what are the main reasons that make employees hesitant to switch jobs? Please choose the two most important reasons.
    (a) Personal ties to coworkers
    (b) Location
    (c) Pay
    (d) Benefits/ company culture
    (e) Reluctance to undergo changes
    (f) Lack of opportunities at other companies
20. Suppose you are comparing job opportunities at two different companies: Company 1 pays 10% above the market average and Company 2 pays 30% above the market average.Which company do you think attracts more qualified applicants per opening?
    (a) Company 1
    (b) Company 2
    (c) Both attract the same number of applicants
21. Which company do you think provides better non-wage amenities (e.g., home office, child-

care subsidy)?

   (a) Company 1
   (b) Company 2
   (c) Both provide the same non-wage amenities

22. Imagine you were to discover that other companies in your area pay {5%/10%/20%} more than your current employer. How likely is it that you would start applying for jobs at other companies?

   (a) Please type in a number between 0% (I would definitely not apply) and 100% (I would start applying immediately): []%

23. How sure would you have to be that you would get a job at one of the higher-paying companies for you to start applying for a new job?

   (a) At least 90% sure
   (b) At least 75% sure
   (c) At least 50% sure
   (d) At least 25% sure
   (e) I would start applying in any case

24. When considering a potential job opportunity, how important are each of the following features to you? [Dropdown scale with 1 - not important at all, 2 - slightly important, 3 - moderately important, 4 - very important, 5 - extremely important]

   (a) Opportunity for reduced hours
   (b) Home office option
   (c) Predictable schedule (including overtime)

25. To what extent do you agree with the following statements? [Dropdown scale 1 (totally disagree) 2 3 4 5 6 7 (totally agree)]

   (a) "I have confidence in my capabilities."
   (b) "If I were comparing companies to apply to, I would factor in the amount of competition from other applicants."
   (c) "I would be reluctant to apply for a job if the probability I would get an offer is low."

Module: Person-Specific Information

26. At the end of the survey, we are interested in your personal background. Are you married?

   (a) Yes
   (b) No

27. Do you have children?

   (a) Yes
   (b) No

F.2 Original German Questionnaire

F.2.1 Initial Survey

Module: Consent

1. Um die Befragung kurz zu halten, würden wir gerne bei der Auswertung Daten einbeziehen,

die bei der Bundesagentur für Arbeit in Nürnberg vorliegen. Dabei handelt es sich zum Beispiel um Informationen zu Ihrer bisherigen Berufstätigkeit. Wir halten alle datenschutzrechtlichen Bestimmungen streng ein bzw. diese werden vom IAB streng kontrolliert. Alle Angaben und Antworten werden in anonymisierter Form, also ohne Namen, Anschrift und Kontaktinformationen, ausgewertet und dargestellt. Niemand kann daher erkennen, welche Antworten Sie gegeben haben. Es erfolgt auch keine Weitergabe von Daten, die Ihre Person erkennen lassen. Weitere Informationen zum Datenschutz finden Sie unter https://www.iab.de/de/befragungen/lohnfle in-deutschland.aspx). Die Auswertung erfolgt zu rein wissenschaftlichen Zwecken, das heißt nicht für kommerzielle Zwecke wie Werbung oder Marketing. Ihre Zustimmung ist selbstverständlich freiwillig. Sind Sie mit der Zuspielung Ihrer Daten einverstanden?

   (a) Ja, ich bin mit einer Zuspielung einverstanden.
   (b) Nein, ich bin mit einer Zuspielung nicht einverstanden.

2. Was trifft am besten auf Ihre aktuelle Haupterwerbstätigkeit zu?
   (a) Sozialversicherungspflichtig beschäftigt
   (b) Selbstständig
   (c) Nicht erwerbstätig
   (d) Sonstiges

   Module: Employment and Bargaining

   Submodule: Employed or missing

3. Bitte beantworten Sie folgende Fragen in Bezug auf Ihren Hauptarbeitgeber. Seit wann sind Sie in Ihrem jetzigen Unternehmen beschäftigt?
   (a) Seit weniger als 6 Monaten
   (b) Seit 6-12 Monaten
   (c) Seit 1-2 Jahren
   (d) Seit 2-3 Jahren
   (e) Seit mehr als 3 Jahren
4. Ist Ihre Stelle tarifgebunden (d.h. werden Sie nach Tarifvertrag bezahlt)?
   (a) Ja
   (b) Nein
   (c) Ich weiß nicht
5. Wie viele Stunden arbeiten Sie in einer typischen Woche?{fill in}
6. Als Sie sich zum ersten Mal bei Ihrem jetzigen Unternehmen beworben haben, kannten Sie jemanden, der bei diesem Unternehmen beschäftigt war?
   (a) Ja.
   (b) Nein, aber ich kannte jemanden, der in der Vergangenheit bei diesem Unternehmen beschäftigt war.
   (c) Nein, ich kannte niemanden.
7. Wussten Sie zum Zeitpunkt Ihrer Bewerbung welches Gehalt Sie verdienen würden?
   (a) Ich hatte keine oder nur sehr wenig Ahnung.
   (b) Ich hatte nur eine ungefähre Vorstellung davon, was in meiner Region oder Branche bezahlt wird.
   (c) Ich hatte zumindest eine ungefähre Vorstellung, was dieses Unternehmen für die Stelle

bezahlt.

(d) Ich wusste genau, was dieses Unternehmen für die Stelle bezahlt.

8. In den vergangenen sechs Monaten ...... haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.
   (a) Stellenausschreibungen angesehen
   (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
   (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
   (d) Sich auf eine Stelle in einem anderen Unternehmen beworben
9. In den vergangenen sechs Monaten ...... hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, eine Empfehlung für Sie auszusprechen)?
   (a) Ja
   (b) Nein
10. In den vergangenen sechs Monaten ...... haben Sie Stellenangebote von anderen Unternehmen erhalten?
   (a) Ja
   (b) Nein
11. In den vergangenen sechs Monaten...... hat Ihr Unternehmen Ihnen eine Gehaltserhöhung angeboten, ohne dass Sie danach gefragt haben?
   (a) Ja
   (b) Nein
12. In den vergangenen sechs Monaten....... haben Sie proaktiv nach einer Gehaltserhöhung gefragt?
   (a) Ja
   (b) Nein

13. Wie viele Stunden arbeiten Sie in einer typischen Woche{fill in}
14. In den vergangenen sechs Monaten ...... haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.
   (a) Stellenausschreibungen angesehen
   (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
   (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
   (d) Auf eine Stelle beworben
15. In den vergangenen sechs Monaten ...... hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, Empfehlung für Sie auszusprechen)?
   (a) Ja
   (b) Nein

16. In den vergangenen sechs Monaten ...... haben Sie Stellenangebote erhalten?
    (a) Ja
    (b) Nein

Module: Worker-Provided Firms

17. Als Nächstes interessiert uns, wie Sie Beschäftigungsmöglichkeiten in Deutschland wahrnehmen. Auch wenn Sie etwas nicht genau wissen, möchten wir Sie bitten, Ihre Einschätzung mit uns zu teilen. Angenommen, Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln. Bei welchen Unternehmen würden Sie erwägen, sich zu bewerben? Bitte geben Sie drei Unternehmen an, bei denen Sie sich bewerben würden und die Beschäftigte in Stellen wie Ihrer einstellen (z.B. „Musterunternehmen GmbH“). Das können auch Unternehmen ohne aktuelle Stellenangebote sein.
    (a) {Unternehmen 1 eintragen}
    (b) {Unternehmen 2 eintragen}
    (c) {Unternehmen 3 eintragen}
    (d) Ich möchte diese Frage nicht beantworten [exklusive Option]
18. **[Employed]** Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

19. Was beschreibt am besten, wie sicher Sie sich bei Ihren Gehaltsschätzungen sind?

Unternehmen	Sehr unsicher	Unsicher	Etwas sicher	Sicher
{Unternehmen 1}				
{Unternehmen 2}				
{Unternehmen 3}				

20. Einige dieser Unternehmen haben möglicherweise Niederlassungen in mehreren Orten. An welchen Ort hatten Sie jeweils gedacht?

Unternehmen	Ort
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

21. Kennen Sie Mitarbeiter bei diesen Unternehmen (z.B. ehemalige Kollegen, Freunde oder Familienmitglieder)?

Unternehmen	Ja, ich kenne derzeitige Mitarbeiter	Ja, ich kenne ehemalige Mitarbe
{Unternehmen 1}		
{Unternehmen 2}		
{Unternehmen 3}		

22. Nehmen Sie nun an, einer Ihrer Kollegen bekommt Jobangebote von folgenden Unternehmen für eine Stelle wie Ihre. Was glauben Sie, welches Unternehmen bietet das höchste Bruttojahresgehalt? Bitte ordnen Sie die Angebote von 1 (höchstes Gehalt) bis 3 (niedrigstes

Gehalt).

Unternehmen	Rangfolge
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

23. **[Employed]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

24. **[Self-Employed]** Angenommen, Sie könnten selbstständig bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Mit aktuellem Gehalt selbstständig bleiben	

25. **[Non-employed]** Angenommen, Sie könnten weiterhin nicht erwerbstätig sein oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Einkommenserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Einkommen	
{Fill-in 2} mit {10%/15%} höherem Einkommen	
{Fill-in 3} mit {15%/15%} höherem Einkommen	
Weiterhin keine Erwerbstätigkeit	

26. **[Employed]** Nun möchten wir Sie bitten, die Rangfolge der Unternehmen unter zwei Szenarien neu einzustufen. Nehmen Sie zuerst an, Sie hätten wieder die drei Jobangebote zur Auswahl, aber Sie hätten den gleichen Arbeitsweg wie bisher. Bitte ordnen Sie die Angebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

27. **[Employed]** Statt dem Arbeitsweg, nehmen Sie nun an, dass Ihre Karriereentwicklung (zukünftige Gehaltserhöhungen/Beförderungen) für alle Optionen identisch ist. Bitte ordnen Sie die

Angebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

Module: Researcher-Provided Firms

28. Falls Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, würden Sie erwägen, sich bei folgenden Unternehmen zu bewerben? Bitte kreuzen Sie alle zutreffenden Antworten an.
    (a) Unternehmen 1
    (b) Unternehmen 2
    (c) Unternehmen 3
    (d) Unternehmen 4
    (e) Unternehmen 5
    (f) Unternehmen 6
    (g) Unternehmen 7
29. **[Employed or Self-Employed]** Uns interessiert nun, wie Sie Stellenangebote in verschiedenen Unternehmen wahrnehmen. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

30. **[Non-Employed]** Uns interessiert nun, wie Sie Stellenangebote in verschiedenen Unternehmen wahrnehmen. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie Ihrer letzten Stelle arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

31. **[Employed]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {15%/10%} höherem Gehalt	
{Unternehmen 12} mit {15%/10%} höherem Gehalt	
{Unternehmen 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

32. **[Self-Employed]** Angenommen, Sie könnten selbstständig bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung.

Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {15%/10%} höherem Gehalt	
{Unternehmen 2} mit {15%/10%} höherem Gehalt	
{Unternehmen 13} mit {15%/15%} höherem Gehalt	
Mit aktuellem Gehalt selbstständig bleiben	

33. **[Non-Employed]** Angenommen, Sie könnten weiterhin nicht erwerbstätig sein oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Einkommenserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

34.

Jobangebot	Rangfolge
{Unternehmen 1} mit {15%/10%} höherem Einkommen	
{Unternehmen 2} mit {15%/10%} höherem Einkommen	
{Unternehmen 3} mit {15%/10%} höherem Einkommen	
Weiterhin keine Erwerbstätigkeit.	

35. Was glauben Sie, wie einfach wäre es für Sie, ein Stellenangebot von einem anderen Unternehmen zu erhalten, das Sie Ihrer jetzigen Stelle vorziehen würden?
    (a) Sehr einfach
    (b) Einfach
    (c) Schwierig
    (d) Sehr schwierig
36. Wenn Sie {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} das Unternehmen wechseln müssten, was beschreibt am besten, wo Sie nach einer Stelle suchen würden? Es geht dabei ausschließlich um Stellen ohne Home-Office Option.
    (a) Ich würde nach Stellen suchen, die nicht weiter entfernt von meinem Wohnort sind als. {eintragen} Kilometer
37. Abschließend interessiert uns Ihre Selbsteinschätzung. Sind Sie generell ein risikobereiter Mensch oder versuchen Sie Risiken zu vermeiden? Verwenden Sie dazu bitte eine Skala von 0 bis 10. Der Wert 0 bedeutet „gar nicht risikobereit“ und der Wert 10 „sehr risikobereit“. Mit den Werten dazwischen können Sie Ihre Einschätzung abstufen.
    (a) 0 (gar nicht risikobereit) 1 2 3 4 5 6 7 8 9 10 (sehr risikobereit)

Module: Consent

38. Als Dankeschön für Ihre Teilnahme verlosen wir {1500}/{1000} Geschenkkarten im Wert von {10}/{5} Euro. Wenn Sie an der Verlosung teilnehmen möchten, geben Sie bitte Ihre E-Mail-Adresse an. {open field that accepts e-mail addresses only}
39. Ihre Angaben unterstützen das IAB bei seiner Forschung zu Gehaltsentwicklung von Beschäftigten in Deutschland. Da sich die aktuellen Arbeitsmarktdynamiken stetig verändern, möchten wir Sie gerne in einigen Monaten erneut befragen. Wollen Sie auch weiterhin dazu beitragen, fundierte Informationen zur Gehaltsentwicklung in Deutschland bereitzustellen? Dann bitten wir Sie, uns Ihre E-Mail-Adresse zur Verfügung zu stellen, damit wir Sie erneut zu einer Befragung einladen können. Die Teilnahme daran ist freiwillig. Natürlich werden wir die E-Mail-Adresse ausschließlich nutzen, um Sie für unsere Befragung anzuschreiben. Sie können sich selbstverständlich jederzeit wieder aus der Befragung abmelden. An welche

E-Mail-Adresse dürfen wir unsere Einladung schicken?

(a) {open field that accepts e-mail addresses only}
(b) Ich möchte nicht per E-Mail kontaktiert werden.

40. Dürfen wir Sie dann postalisch erneut einladen?

    (a) Ja
    (b) Nein

F.2.2 Follow-up Survey

1. Um die Befragung kurz zu halten, würden wir gerne bei der Auswertung Daten einbeziehen, die bei der Bundesagentur für Arbeit in Nürnberg vorliegen. Dabei handelt es sich zum Beispiel um Informationen zu Ihrer bisherigen Berufstätigkeit. Wir halten alle datenschutzrechtlichen Bestimmungen streng ein bzw. diese werden vom IAB streng kontrolliert. Alle Angaben und Antworten werden in anonymisierter Form, also ohne Namen, Anschrift und Kontaktinformationen, ausgewertet und dargestellt. Niemand kann daher erkennen, welche Antworten Sie gegeben haben. Es erfolgt auch keine Weitergabe von Daten, die Ihre Person erkennen lassen. Weitere Informationen zum Datenschutz finden Sie unter https://www.iab.de/de/befragungen/lohnfle in-deutschland.aspx). Die Auswertung erfolgt zu rein wissenschaftlichen Zwecken, das heißt nicht für kommerzielle Zwecke wie Werbung oder Marketing. Ihre Zustimmung ist selbstverständlich freiwillig. Sind Sie mit der Zuspielung Ihrer Daten einverstanden?

    (a) Ja, ich bin mit einer Zuspielung einverstanden.
    (b) Nein, ich bin mit einer Zuspielung nicht einverstanden.

2. Haben Sie seit {month_survey1 2022/2023} eine neue Stelle bei einem anderen Unternehmen angetreten?

    (a) Ja
    (b) Nein

3. [Q2=1]Was trifft am besten auf Ihre aktuelle Haupterwerbstätigkeit zu?

    (a) Sozialversicherungspflichtig beschäftigt
    (b) Selbstständig
    (c) Nicht erwerbstätig
    (d) Sonstiges

4. In den vergangenen sechs Monaten... ...haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.

    (a) Stellenausschreibungen angesehen
    (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
    (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
    (d) Sich auf eine Stelle in einem anderen Unternehmen beworben

5. In den vergangenen sechs Monaten......hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, eine Empfehlung für Sie auszusprechen)?

   (a) Ja
   (b) Nein
6. In den vergangenen sechs Monaten......wie viele Stellenangebote von anderen Unternehmen haben Sie erhalten?
   (a) 0
   (b) 1
   (c) 2
   (d) 3 oder mehr

   Module: Worker-Provided Firms
7. Als Nächstes interessiert uns, wie Sie Beschäftigungsmöglichkeiten in Deutschland wahrnehmen. Auch wenn Sie etwas nicht genau wissen, möchten wir Sie bitten, Ihre Einschätzung mit uns zu teilen. Angenommen, Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln. Bei welchen Unternehmen würden Sie erwägen, sich zu bewerben? Bitte geben Sie drei Unternehmen an, bei denen Sie sich bewerben würden und die Beschäftigte in Stellen wie Ihrer einstellen (z.B. „Musterunternehmen GmbH“). Das können auch Unternehmen ohne aktuelle Stellenangebote sein.
   (a) {Unternehmen 1 eintragen}
   (b) {Unternehmen 2 eintragen}
   (c) {Unternehmen 3 eintragen}
   (d) Ich möchte diese Frage nicht beantworten [exklusive option]
8. **[Q7=d]** Was beschreibt am besten, warum Sie in der vorherigen Frage keine Unternehmen angegeben haben?
   (a) Ich möchte diese Informationen nicht teilen
   (b) Es gibt Unternehmen, die ich in Betracht ziehen würde, aber die Eingabe in der Umfrage war zu kompliziert
   (c) Mir fallen keine konkreten Unternehmen ein
   (d) Ich habe kein Interesse daran, das Unternehmen zu wechseln und habe darüber bis jetzt nicht nachgedacht
9. **[Q7 has only 1 or 2 provided firms]** Was beschreibt am besten, warum Sie in der vorherigen Frage nicht drei Unternehmen angegeben haben?
   (a) Ich möchte diese Informationen nicht teilen
   (b) Es gibt Unternehmen, die ich in Betracht ziehen würde, aber die Eingabe in der Umfrage war zu kompliziert
   (c) Mir fallen keine weiteren, konkreten Unternehmen ein
   (d) Ich habe kein Interesse daran, das Unternehmen zu wechseln und habe darüber bis jetzt nicht nachgedacht
10. **[Q7 is not missing]** Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Fill-in 1}	
{Fill-in 2}	
{Fill-in 3}	

11. **Q7 is not missing]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort das unten angegebene Gehalt. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Researcher-Provided Firms

12. Falls Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, würden Sie erwägen, sich bei folgenden Unternehmen zu bewerben? Bitte kreuzen Sie alle zutreffenden Antworten an.

    (a) Firm 1
    (b) Firm 2
    (c) Firm 3
    (d) Firm 4
    (e) Firm 5
    (f) Firm 6
    (g) Firm 7

13. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

14. **[Q13 is missing]** Auch wenn Sie sich nicht sicher sind, interessieren wir uns sehr für Ihre Einschätzung, wie diese Unternehmen im Hinblick auf das Gehalt abschneiden. Bitte ordnen Sie die Unternehmen von 1 (höchstes Gehalt) bis 3 (niedrigstes Gehalt) ein.

Unternehmen	Rangfolge
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

15. Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der

genannten Unternehmen wechseln und erhalten sofort das unten angegebene Gehalt. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

16. Nun möchten wir Sie bitten, die Rangfolge der Unternehmen unter zwei Szenarien neu einzustufen. Nehmen Sie zuerst an, Sie hätten wieder die drei Jobangebote zur Auswahl, aber Sie hätten den gleichen Arbeitsweg wie bisher. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

17. Statt dem Arbeitsweg, nehmen Sie nun an, dass die Training- und Weiterbildungsmöglichkeiten der drei Unternehmen die gleichen sind wie in Ihrem aktuellen Unternehmen. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Applications

18. Falls Sie planen würden {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, bei wie vielen verschiedenen Unternehmen würden Sie sich bewerben?

    (a) Die Anzahl der Unternehmen, bei denen ich mich bewerben würde, ist: _____

19. **[Randomize who sees Q19 (50%) and who sees Q20/Q21 (50%)]** Was sind Ihrer Meinung

nach die Hauptgründe, die Mitarbeitende zögern lassen, den Arbeitsplatz zu wechseln? Bitte wählen Sie die zwei wichtigsten Gründe aus.

(a) Persönliche Bindung zu den Arbeitskollegen
(b) Standort
(c) Gehalt
(d) Nebenleistungen/ Unternehmenskultur
(e) Abneigung gegen Veränderungen
(f) Fehlende Möglichkeiten bei anderen Unternehmen

20. Angenommen, Sie vergleichen die Stellenangebote bei zwei verschiedenen Unternehmen: Unternehmen 1 zahlt 10% über dem Durchschnitt im Arbeitsmarkt und Unternehmen 2 zahlt 30% über dem Durchschnitt im Arbeitsmarkt. Welches Unternehmen zieht Ihrer Meinung nach mehr qualifizierte Bewerbende pro Stelle an?

    (a) Unternehmen 1
    (b) Unternehmen 2
    (c) Beide ziehen die gleiche Anzahl von Bewerbenden an

21. Welches Unternehmen bietet Ihrer Meinung nach die besseren Nebenleistungen (z. B. Home Office, Kita-Zuschuss)?

    (a) Unternehmen 1
    (b) Unternehmen 2
    (c) Beide bieten die gleichen Nebenleistungen

22. Nehmen sie an, Sie stellen fest, dass andere Unternehmen in Ihrer Gegend {5%/10%/20%} mehr bezahlen als Ihr aktueller Arbeitgeber. Wie wahrscheinlich ist es, dass Sie sich auf eine neue Stelle bei einem anderen Unternehmen bewerben würden?

    (a) Bitte geben Sie eine Zahl zwischen 0% (ich würde mich auf keinen Fall bewerben) und 100% (ich würde mich sofort bewerben) an: ___%.

23. Wie sicher müssten Sie sein, dass Sie eine Stelle bei einem der besser bezahlenden Unternehmen bekommen würden, um sich für einen neuen Job zu bewerben?

    (a) Mindestens 90% sicher
    (b) Mindestens 75% sicher
    (c) Mindestens 50% sicher
    (d) Mindestens 25% sicher
    (e) Ich würde mich in jedem Fall bewerben

24. Falls Sie einen neuen Job in Betracht ziehen, wie wichtig ist jede der folgenden Eigenschaften für Sie? [Dropdown scale with 1 - Überhaupt nicht wichtig, 2 - Etwas wichtig, 3 - Mäßig wichtig, 4 - Sehr wichtig, 5 - Äußerst wichtig]

    (a) Möglichkeit für reduzierte Arbeitszeiten
    (b) Homeoffice-Option
    (c) Planbare Arbeitszeiten (einschließlich Überstunden)

Module: Person-Specific Information

25. Am Ende der Umfrage interessieren wir uns für Ihren persönlichen Hintergrund.Sind Sie verheiratet?

    (a) Ja

(b) Nein

26. Haben Sie Kinder?
    (a) Ja
    (b) Nein